

Abstract Nonsense

III

Contents

Basics of Commutative Algebra	7
The Jacobson ideal	7
Definition 1.1.1 (Jacobson Ideal)	7
Lemma 1.1.2 (Characterization of Jacobson ideal)	7
Proposition 1.1.3 (Properties of the Jacobson ideal)	7
Proposition 1.1.4 (Basics of divisibility for ideals)	8
The Zariski topology	8
Definition 1.2.1 (Zariski topology)	8
Lemma 1.2.2 (Ideals contained in union of prime ideals)	8
Proposition 1.2.3 (Basic properties of Zariski topology)	8
Contraction and extension of ideals	9
Definition 1.3.1 (Contraction and extension)	9
Lemma 1.3.2 (Properties of contraction and extension)	9
Lemma 1.3.3 (Embedding with contraction)	9
Proposition 1.3.4 (Contraction is continuous)	9
Lemma 1.3.5 (Contraction and quotient ring)	9
Lemma 1.3.6 (Contraction and localization, I)	10
Proposition 1.3.7 (Contraction and localization, II)	10
Theorem 1.3.8 (Fiber of contraction)	10
Nakayama's Lemma	11
Proposition 1.4.1 (Endomorphisms trapped in an ideal)	11
Corollary 1.4.2 (Modules trapped in an ideal)	12
Corollary 1.4.3 (Nakayama's Lemma)	12
Primary Decomposition	13
Primary ideals	13
Definition 2.1.1 (Primary ideals)	13
Lemma 2.1.2 (Characterization of primary ideals)	13
Lemma 2.1.3 (Radical and intersection of prime divisors)	13
Corollary 2.1.4 (Radical of primary ideals)	13
Definition 2.1.5 ($\mathfrak{p}$ -primary ideals)	13
Lemma 2.1.6 ($\mathfrak{m}$ -primary ideals)	14
Lemma 2.1.7 (Primary ideals in a PID)	14
Lemma 2.1.8 (Characterization of $(\mathfrak{q} : x)$)	14
Primary decomposition	14
Definition 2.2.1 (Decomposable ideals)	14
Lemma 2.2.2 (Intersection of $\mathfrak{p}$ -primary ideals)	15
Theorem 2.2.3 (Prime ideals associated with a decomposable ideal)	15
Proposition 2.2.4 (Associated primes and prime divisors)	15
Proposition 2.2.5 (Union of associated primes)	16
Corollary 2.2.6 (Zero-divisors and nilpotent elements)	16
Localization and primary decomposition	17
Lemma 2.3.1 (Characterization of $\mathfrak{a}^{\text{ec}}$)	17
Proposition 2.3.2 (Primary ideals in a localized ring)	17
Lemma 2.3.3 (Extension of primary ideals)	17
The second uniqueness theorem	18
Definition 2.4.1 (Isolated family of associated primes)	18
Theorem 2.4.2 (Second uniqueness)	18
Integral domain with Krull dimension one	18
Definition 2.5.1 (Krull dimension)	18
Lemma 2.5.2 (Integral domain with special dimensions)	18
Proposition 2.5.3 (Primary decomposition and $\dim D = 1$)	19

Primary decomposition in Noetherian rings	19
Definition 2.6.1 (Irreducible ideals)	19
Lemma 2.6.2 (Irreducible decomposition in Noetherian rings)	19
Proposition 2.6.3 (Noetherian rings and primary decomposition)	19
Lemma 2.6.5 (Every prime ideal contains a minimal prime)	20
Corollary 2.6.6 (Minimal prime ideals in a Noetherian ring)	20
Integral Extension of Rings	21
Integral closure	21
Definition 3.1.1 (Integral extension)	21
Proposition 3.1.2 (Characterization of integral elements)	21
Corollary 3.1.3 (Extension by integral elements)	22
Corollary 3.1.4 (Integral closure is a subring)	22
Definition 3.1.5 (Integrally closed integral domain)	22
Lemma 3.1.6 (Tower of integral extensions)	22
Corollary 3.1.7 (Integral closure is integrally closed)	22
Integral extension, factor ring, and localization	23
Lemma 3.2.1 (Integrality in localized or factored rings)	23
Proposition 3.2.2 (Integral closure and localization)	23
Theorem 3.2.3 (Integral closedness is a local property)	23
Contraction and integral embedding	24
Lemma 3.3.1 (Integral extension with fields)	24
Corollary 3.3.2 (Contraction and maximal ideals)	24
Proposition 3.3.3 (Lying-over in integral extensions)	24
Corollary 3.3.4 (Contraction is a closed map)	25
Theorem 3.3.5 (Going-up)	25
*Lemma 3.3.6 (Incomparability in integral extensions)	25
Theorem 3.3.7 (Dimension of rings in an integral extension)	26
Openness of contraction	26
Definition 3.4.1 (Integrality over an ideal)	26
Lemma 3.4.2 (Characterization of integrality over an ideal)	26
Lemma 3.4.3 (Minimal polynomial and integrality over an ideal)	26
Theorem 3.4.4 (Going-down)	27
Theorem 3.4.5 (Openness of contraction)	27
Noetherianity and integral extension	28
Theorem 3.5.1 (Integral extension and quotient fields)	28
Corollary 3.5.2 (Noetherianity and integral extension)	28
Lemma 3.5.2 (Finite separable extension and basis)	29
Corollary 3.5.3 (Trace and norm)	29
Definition 3.5.4 (Non-degenerate bilinear form)	29
Lemma 3.5.5 (Characterization of non-degeneracy)	30
Lemma 3.5.6 (Non-degeneracy and reflexivity)	30
Lemma 3.5.7 (Dual basis)	30
Lemma 3.5.8 (Trace and non-degeneracy)	30
Lemma 3.5.9 (Field of fraction and integral closure)	31
Corollary 3.5.10 ($\mathcal{O}_k$ as an additive group)	31
Valuation Rings	32
Valuation ring and Nullstellensatz	32
Definition 4.1.1 (Valuation ring)	32
Proposition 4.1.2 (Properties of valuation ring)	32
Theorem 4.1.3 (Integral domain and valuation ring)	32
Proposition 4.1.4 (Characterization of integral closure)	34
Theorem 4.1.5 (Extension of morphism)	34

Theorem 4.1.6 (Nullstellensatz, I)	34
Theorem 4.1.7 (Nullstellensatz, II)	35
Theorem 4.1.8 (Nullstellensatz, III)	35
Theorem 4.1.9 (Nullstellensatz, IV)	35
Proposition 4.1.10 (Equivalency of four Nullstellensatz)	36
Finitely generated k -algebra and normalization	36
Definition 4.2.1 (Jacobson ring)	36
Theorem 4.2.2 (Finitely generated k -algebra and Jacobson ring)	36
Corollary 4.2.3 (Jacobson ring and Jacobson ideal)	36
Lemma 4.2.4 (Some automorphisms of $k[x_1, \dots, x_n]$)	37
Corollary 4.2.5 (Normalization of polynomial)	37
Theorem 4.2.6 (Noether's normalization lemma)	37
Corollary 4.2.7 (Dimensions of finitely generated k -algebras)	38
Noetherian Rings of Special Dimensions	39
Composition series and Artinian modules	39
Lemma 5.1.1 (Noetherian rings of dimension 0)	39
Corollary 5.1.2 (Composition series and Noetherian rings)	39
Definition 5.1.3 (Composition series of modules)	39
Proposition 5.1.4 (Length of composition series)	39
Definition 5.1.5 (Artinian modules)	40
Lemma 5.1.6 (Basic properties of Artinian modules)	40
Lemma 5.1.7 (Noetherian vector space)	40
Proposition 5.1.8 (Modules with composition series)	40
Artinian rings	40
Lemma 5.2.1 (Noetherian rings of dimension 0, again)	40
Lemma 5.2.2 (Dimension of Artinian rings)	41
Lemma 5.2.3 (Spectrum of Artinian rings)	41
Proposition 5.2.4 (Jacobson ideal of Artinian rings)	41
Theorem 5.2.5 (Artinian and Noetherian rings)	41
Theorem 5.2.6 (Structure of Artinian rings)	41
Proposition 5.2.7 (Properties of local Artinian rings)	42
Dedekind domains	42
Lemma 5.3.1 (Properties of local Noetherian domains)	42
Theorem 5.3.2 (Characterization of discrete valuation rings)	43
Theorem/Definition 5.3.3 (Characterization of Dedekind domains)	44
Corollary 5.3.4 (Ideals in Dedekind domains)	44
Corollary 5.3.5 ($\mathcal{O}_k$ is an Dedekind domain)	44
Definition 5.3.6 ($\text{Frac } A$ and $\text{Prin } A$)	44
Lemma 5.3.7 ($\text{Frac } A$ and $\text{Prin } A$ as algebraic objects)	44
Lemma 5.3.8 (Invertible elements in $\text{Frac } A$)	45
Proposition 5.3.9 (Invertibility is a local property)	45
Corollary 5.3.10 (Invertibility and Dedekind domains)	45
Definition/Corollary 5.3.11 (Class group of a Dedekind domain)	45
Height of Ideals	46
Krull's height theorem and principal ideal theorem	46
Definition 6.1.1 (Height of an ideal)	46
Proposition 6.1.2 (Associated primes in a Noetherian ring)	46
Lemma 6.1.3 (Ideals with height zero)	46
Theorem 6.1.4 (Krull's height theorem (HT))	46
Corollary 6.1.5 (Dimension of $k[x_1, \dots, x_n]$)	47
Theorem 6.1.6 (Krull's principal ideal theorem (PIT))	47
Theorem 6.1.7 (Converse to Krull's HT)	48

Corollary 6.1.8 (Noetherian domain and UFD)	48
Theorem 6.1.9 (Dimension of Noetherian local rings)	48
Cohen-Macaulay rings	49
Theorem 6.2.1 (Dimension of quotient Noetherian rings)	49
Definition 6.2.2 (A -regular sequence)	49
Corollary 6.2.3 (A -regular sequences and dimension of quotient rings)	49
Definition 6.2.4 (Depth and Cohen-Macaulay rings)	49
Theorem 6.2.5 (Characterization of Cohen-Macaulay ring)	49
Finitely generated k -domains	50
Theorem 6.3.1 (Noether's normalization lemma, stronger version)	50
Theorem 6.3.2 (Dimension of a finitely generated k -domain)	51
Appendix	52
Integral extension	52
Nullstellensatz and algebraic geometry	53
Dedekind domains and DVR	53
Miscellaneous	54

Basics of Commutative Algebra

The Jacobson ideal

Definition 1.1.1 (Jacobson Ideal) Let A be a commutative ring. We define the *Jacobson ideal* of A to be $J(A) := \bigcap_{\mathfrak{m} \in \text{Max } A} \mathfrak{m}$.

Lemma 1.1.2 (Characterization of Jacobson ideal) $x \in J(A) \iff \forall y \in A, 1 - xy \in A^\times$.

Proof. Suppose for contradiction $x \in J(A)$ and there exists y such that $1 - xy \notin A^\times$. Then $\langle 1 - xy \rangle$ is a proper ideal and contained in some $\mathfrak{m} \in \text{Max } A$; in particular $1 - xy \in \mathfrak{m}$. But $x \in J(A) \subseteq \mathfrak{m}$, so $xy \in \mathfrak{m}$ and $1 \in \mathfrak{m} + xy = \mathfrak{m}$, a contradiction.

Conversely, suppose $x \notin \mathfrak{m}$ for some $\mathfrak{m}$. Then $x + \mathfrak{m} \neq 0 \in A/\mathfrak{m}$, so there exists y such that $xy \equiv 1 \pmod{\mathfrak{m}}$. But that implies $1 - xy \in \mathfrak{m}$ and therefore $1 - xy \notin A^\times$. $\square$

Proposition 1.1.3 (Properties of the Jacobson ideal) Let A be a commutative ring.

- (1). $\text{Nil } A[x] = (\text{Nil } A)[x]$.
- (2). $A[x]^\times = \{\sum_i a_i x^i : a_0 \in A^\times, a_i \in \text{Nil } A \text{ for } i \geq 1\}$.
- (3). $J(A[x]) = \text{Nil } A[x]$.

Proof. (1) Notice that $\text{Nil } A[x] \trianglelefteq A[x]$ is an ideal and $\text{Nil } A \subseteq \text{Nil } A[x]$ is a subset. Hence $(\text{Nil } A)[x] = \langle \text{Nil } A \rangle \subseteq \text{Nil } A[x]$. Conversely, since $A[x]/\mathfrak{p}[x] \simeq (A/\mathfrak{p})[x]$ is an integral domain for every $\mathfrak{p} \in \text{Spec } A$, we have that $\mathfrak{p}[x] \in \text{Spec } A[x]$ for every $\mathfrak{p}$. Hence

$$\text{Nil } A[x] \subseteq \bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p}[x] = (\text{Nil } A)[x].$$

(2) Here we only show that $\subseteq$ holds. Suppose $f = \sum_i a_i x^i \in A[x]^\times$ and $fg = 1$. Then clearly $a_0 \in A^\times$. Notice that $fg \equiv 1 \pmod{\mathfrak{p}}$ for every $\mathfrak{p} \in \text{Spec } A$. Since $A/\mathfrak{p}$ is an integral domain, we have that $(A/\mathfrak{p})[x]^\times = (A/\mathfrak{p})^\times$. Hence $f \pmod{\mathfrak{p}} \in (A/\mathfrak{p})[x]^\times$ implies that $a_i \in \mathfrak{p}$ for all $i \geq 1$. Since $\mathfrak{p}$ is arbitrary, we deduce $a_i \in \bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p} = \text{Nil } A$ for $i \geq 1$, so $f \in \text{RHS}$.

(3) Notice that

$$J(A[x]) = \bigcap_{\mathfrak{m} \in \text{Max } A[x]} \mathfrak{m} \supseteq \bigcap_{\mathfrak{p} \in \text{Spec } A[x]} \mathfrak{p} = \text{Nil } A[x].$$

Conversely, if $f \in J(A[x])$, then by Lemma 1.1.2 we have that $1 - xf \in A[x]^\times$. By (2), that means every coefficient of f lies in $\text{Nil } A$ (that is why we consider xf instead of f alone!). Hence $f \in (\text{Nil } A)[x] = \text{Nil } A[x]$, so we are done. $\square$

Proposition 1.1.4 (Basics of divisibility for ideals) *Let $\{\mathfrak{a}_i\}_i \subseteq \text{Ideal } A$ be a family of ideals.*

- (1). $\gcd(\{\mathfrak{a}_i\}_i) = \sum_i \mathfrak{a}_i$ and $V(\sum_i \mathfrak{a}_i) = \bigcap_i V(\mathfrak{a}_i)$.
 - (2). $\text{lcm}(\{\mathfrak{a}_i\}_i) = \bigcap_i \mathfrak{a}_i$ and $V(\bigcap_{i=1}^n \mathfrak{a}_i) = \bigcup_{i=1}^n V(\mathfrak{a}_i)$; the second equality need not hold for an infinite intersection of ideals.
 - (3). $V(0) = \text{Spec } A$ and $V(A) = \emptyset$.
- Recall that Ideal A is ordered by inverse inclusion, so the “greatest” common divisor of a family of ideals is in fact the *smallest* ideal in A containing every member of the family. Similarly, the “least” common multiple is in fact the *largest* ideal contained in every member.

Proof. (1) If $\mathfrak{b} \mid \mathfrak{a}_i$ for every i , then $\mathfrak{b} \mid \sum_i \mathfrak{a}_i$. On the other hand, it is clear that $\sum_i \mathfrak{a}_i \mid \mathfrak{a}_j$ for every j , so $\sum_i \mathfrak{a}_i$ is the gcd *by definition*. Now $V(\sum_i \mathfrak{a}_i) \subseteq \bigcap_i V(\mathfrak{a}_i)$ since $V(\sum_i \mathfrak{a}_i) \subseteq V(\mathfrak{a}_j)$ for every j . Conversely, if $\mathfrak{p} \in \bigcap_i V(\mathfrak{a}_i)$, then $\mathfrak{p} \supseteq \mathfrak{a}_i$ for every i which implies that $\mathfrak{p} \supseteq \sum_i \mathfrak{a}_i$, so $\mathfrak{p} \in V(\sum_i \mathfrak{a}_i)$.

(2) That $\text{lcm}(\{\mathfrak{a}_i\}_i) = \bigcap_i \mathfrak{a}_i$ follows from the RHS being an ideal. Now if $\mathfrak{p} \in \bigcup_{i=1}^n V(\mathfrak{a}_i)$, then $\mathfrak{p} \supseteq \mathfrak{a}_i$ for some i and so $\mathfrak{p} \supseteq \bigcap_{i=1}^n \mathfrak{a}_i$. Conversely, suppose $\mathfrak{p} \notin \bigcup_{i=1}^n V(\mathfrak{a}_i)$. Then for every $1 \leq i \leq n$, there exists $a_i \notin \mathfrak{p}$. So then $\prod_{i=1}^n a_i \in \bigcap_{i=1}^n \mathfrak{a}_i \setminus \mathfrak{p}$, which implies $\mathfrak{p} \notin V(\bigcap_{i=1}^n \mathfrak{a}_i)$. To see that this need not hold for an infinite family, let $\mathcal{P} \subseteq \mathbb{Z}$ be the set of prime numbers. Clearly $\bigcap_{p \in \mathcal{P}} p\mathbb{Z} = 0$, but $V(\bigcap_{p \in \mathcal{P}} p\mathbb{Z}) = 0 \notin \bigcup_{p \in \mathcal{P}} V(p\mathbb{Z})$.

(3) This is clear, so we are done. $\square$

The Zariski topology

Definition 1.2.1 (Zariski topology) Let A be a commutative ring. We define the *Zariski topology* on $\text{Spec } A$ by declaring the closed sets to be precisely $\{V(\mathfrak{a})\}_{\mathfrak{a} \trianglelefteq A}$. This is indeed a well-defined topology by Proposition 1.1.4.

Lemma 1.2.2 (Ideals contained in union of prime ideals) *Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n \in \text{Spec } A$ and $\mathfrak{a} \trianglelefteq A$. Suppose $\mathfrak{a} \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$. Then $\mathfrak{a} \subseteq \mathfrak{p}_i$ for some i .*

Proof. We proceed by induction. The base case is clear. Now WLOG we may assume $\mathfrak{a} \not\subseteq \bigcup_{i \neq j} \mathfrak{p}_i$ for every $1 \leq j \leq n$; otherwise by the inductive hypothesis we are done. Suppose $\mathfrak{a} \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$ but $\mathfrak{a} \not\subseteq \mathfrak{p}_i$ for any i . Then for every j we can pick $a_j \in \mathfrak{a} \setminus \bigcup_{i \neq j} \mathfrak{p}_i$. Consider $a' := \prod_{i=1}^{n-1} \mathfrak{p}_i \in \bigcup_{i=1}^{n-1} \mathfrak{p}_i$ and $a := a' + a_n \in \mathfrak{a} \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$. Notice that either $a \in \mathfrak{p}_n$ or $a \in \mathfrak{p}_i$ for some $i < n$. But if $a = a' + a_n \in \mathfrak{p}_n$, then $a' \in \mathfrak{p}_n$ which is a contradiction, since $\mathfrak{a}_i \not\subseteq \mathfrak{p}_n$ for every $i < n$ and $\mathfrak{p}_n$ is a prime ideal. If $a = a' + a_n \in \mathfrak{p}_i$, then $a_n \in \mathfrak{p}_i$, which is again a contradiction, so we are done. $\square$

- In what sense does this lemma show that a union of ideals is far from being an ideal?

Proposition 1.2.3 (Basic properties of Zariski topology)

- (1). Any non-empty closed set in $\text{Spec } A$ intersects with $\text{Max } A$.
- (2). $\{\mathfrak{m} \in \text{Spec } A : V(\mathfrak{m}) \text{ is a point}\} = \text{Max } A$.
- (3). In an integral domain D , 0 is dense in $\text{Spec } D$.

Proof. (1) If $V(\mathfrak{a}) \neq \emptyset$, then $\mathfrak{a}$ is a proper ideal, so it is contained in some maximal ideal $\mathfrak{m} \in V(\mathfrak{a})$.

(2) Clearly $\supseteq$ holds. Now if $V(\mathfrak{m}) = \{\mathfrak{m}\}$ is a singleton, by (1) $\mathfrak{m} \in \text{Max } A$.

(3) This is clear, so we are done. $\square$

Contraction and extension of ideals

Definition 1.3.1 (Contraction and extension) Let $f : A \rightarrow B$ be a ring homomorphism. For every $\mathfrak{a} \in \text{Ideal } A$ and $\mathfrak{b} \in \text{Ideal } B$, we define $\mathfrak{a}^e := \langle f(\mathfrak{a}) \rangle$ to be the *extension* of $\mathfrak{a}$ and $\mathfrak{b}^c := f^{-1}(\mathfrak{b})$ to be the *contraction* of $\mathfrak{b}$. It is easy to see that $\mathfrak{a}^e \in \text{Ideal } B$ and $\mathfrak{b}^c \in \text{Ideal } A$.

Lemma 1.3.2 (Properties of contraction and extension)

(1). $\mathfrak{a}^{ec} \supseteq \mathfrak{a}$ and $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$.

(2). $\mathfrak{a}^{ece} = \mathfrak{a}^e$ and $\mathfrak{b}^{cec} = \mathfrak{b}^c$.

As a result, contraction and extension induces a bijection between contracted and extended ideals.

Proof. (1) If $a \in \mathfrak{a}$, then clearly $f(a) \in \langle f(\mathfrak{a}) \rangle$, so $a \in f^{-1}(f(\langle \mathfrak{a} \rangle)) = \mathfrak{a}^{ec}$. If $a \in \mathfrak{b}^c$, then $f(a) \in \mathfrak{b}$, so $\mathfrak{b}^{ce} = \langle f(\mathfrak{b}^c) \rangle \subseteq \mathfrak{b}$.

(2) Since $\mathfrak{a}^{ec} \supseteq \mathfrak{a}$ we have that $\mathfrak{a}^{ece} \supseteq \mathfrak{a}^e$; but $\mathfrak{a}^e \in \text{Ideal } B$, so $(\mathfrak{a}^e)^{ce} \subseteq \mathfrak{a}^e$ and we have equality. Since $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$, we have that $\mathfrak{b}^{cec} \subseteq \mathfrak{b}^c$; but $\mathfrak{b}^c \in \text{Ideal } A$, so $(\mathfrak{b}^c)^{ec} \supseteq \mathfrak{b}^c$ and we have equality. $\square$

Lemma 1.3.3 (Embedding with contraction) In the above setting, $A/\mathfrak{b}^c \hookrightarrow B/\mathfrak{b}$.

Proof. Clearly $A \rightarrow B/\mathfrak{b}$ is a well-defined ring homomorphism. Its kernel is $f^{-1}(\mathfrak{b}) = \mathfrak{b}^c$, so by the first isomorphism theorem we are done. $\square$

Proposition 1.3.4 (Contraction is continuous) Let $f : A \rightarrow B$ be a ring homomorphism. Then the pullback $f^* : \text{Spec } B \rightarrow \text{Spec } A$ where $f^*(\mathfrak{q}) := \mathfrak{q}^c$ is continuous with respect to the Zariski topology.

Proof. It is clear that f^* is well-defined: by Lemma 1.3.2, $A/\mathfrak{q}^c \hookrightarrow B/\mathfrak{q}$ is an integral domain. Let $V(\mathfrak{a}) \subseteq \text{Spec } A$ be a closed set. Then

$$(f^*)^{-1}(V(\mathfrak{a})) = (f^*)^{-1}(\{\mathfrak{p} \in \text{Spec } A : \mathfrak{p} \supseteq \mathfrak{a}\}) = \{\mathfrak{q} \in \text{Spec } B : \mathfrak{q}^c \supseteq \mathfrak{a}\}.$$

But $\mathfrak{q}^c \supseteq \mathfrak{a}$ implies that $\mathfrak{q} \supseteq \mathfrak{q}^{ce} \supseteq \mathfrak{a}^e$; on the other hand, if $\mathfrak{q} \supseteq \mathfrak{a}^e$, then $\mathfrak{q}^c \supseteq \mathfrak{a}^{ec} \supseteq \mathfrak{a}$. Hence $\mathfrak{q}^c \supseteq \mathfrak{a} \iff \mathfrak{q} \supseteq \mathfrak{a}^e$, so we deduce $(f^*)^{-1}(V(\mathfrak{a})) = V(\mathfrak{a}^e)$ and we are done. $\square$

Lemma 1.3.5 (Contraction and quotient ring) Let $\pi : A \rightarrow A/\mathfrak{a}$. Then $\pi^* : \text{Spec } A/\mathfrak{a} \rightarrow V(\mathfrak{a})$ is a homeomorphism.

Proof. By Proposition 1.3.4, it suffices to show that π^* is bijective and closed. Bijectivity follows from the correspondence theorem: for every $\bar{\mathfrak{b}} \in \text{Ideal } A/\mathfrak{a}$, there exists unique $\mathfrak{b} \in \text{Ideal } A$ such that $\bar{\mathfrak{b}} = \mathfrak{b}/\mathfrak{a}$. But $(A/\mathfrak{b})/(\mathfrak{b}/\mathfrak{a}) \simeq A/\mathfrak{b}$; in particular, the LHS is an integral domain iff the RHS is, so $\bar{\mathfrak{p}} \in \text{Spec } A/\mathfrak{a} \iff \mathfrak{p} \in \text{Spec } A$ and $\mathfrak{p} \supseteq \mathfrak{a}$, so π^* is bijective. For closedness, notice that

$$\pi^*(V(\bar{\mathfrak{b}})) = \{\bar{\mathfrak{p}}^c : \bar{\mathfrak{p}} \in \text{Spec } A/\mathfrak{a} \text{ and } \bar{\mathfrak{p}} \supseteq \bar{\mathfrak{b}}\}.$$

But $\bar{\mathfrak{p}} \in \text{Spec } A/\mathfrak{a} \iff \mathfrak{p} \in V(\mathfrak{a})$ and $\bar{\mathfrak{p}} \supseteq \bar{\mathfrak{b}} \iff \mathfrak{p} \supseteq \mathfrak{b} \supseteq \mathfrak{a}$. Finally, notice that $\bar{\mathfrak{p}}^c = \mathfrak{p}$ by definition of π , so we have that $\pi^*(V(\bar{\mathfrak{b}})) = V(\mathfrak{b})$ and we are done. $\square$

Lemma 1.3.6 (Contraction and localization, I) *Let $S \subseteq A$ be a multiplicatively closed subset and $f : A \rightarrow S^{-1}A$ be given by $a \mapsto \frac{a}{1}$. Then for every $\mathfrak{a} \trianglelefteq A$, $\mathfrak{a}^e = S^{-1}\mathfrak{a}$, and for every $\tilde{\mathfrak{a}} \trianglelefteq S^{-1}A$, $\tilde{\mathfrak{a}}^{ce} = \tilde{\mathfrak{a}}$. As a result, $(-)^e : \text{Ideal } A \rightarrow \text{Ideal } S^{-1}A$ is surjective, and $(-)^c : \text{Ideal } S^{-1}A \rightarrow \text{Ideal } A$ is injective.*

Proof. Clearly $S^{-1}\mathfrak{a} \subseteq S^{-1}A$ is an ideal. Since $f(\mathfrak{a}) \subseteq S^{-1}\mathfrak{a}$, we have that $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{a}$; the other inclusion is immediate. To see that $\tilde{\mathfrak{a}}^{ce} = \tilde{\mathfrak{a}}$, it suffices to show $\tilde{\mathfrak{a}}^{ce} \supseteq \tilde{\mathfrak{a}}$. Let $\frac{a}{s} \in \tilde{\mathfrak{a}}$. Then $\frac{a}{1} = \frac{a}{s} \cdot \frac{1}{s} \in \tilde{\mathfrak{a}}$, since $s \in (S^{-1}A)^*$. So $\frac{a}{1} = f(a) \in \tilde{\mathfrak{a}}^c$. But then $\frac{a}{s} = \frac{a}{1} \cdot \frac{1}{s} \in \tilde{\mathfrak{a}}^{ce}$, so we are done. $\square$

Proposition 1.3.7 (Contraction and localization, II) *In the setting above, $f : A \rightarrow S^{-1}A$ induces a bijection between $\text{Spec } S^{-1}A$ and $\{\mathfrak{p} \in \text{Spec } A : \mathfrak{p} \cap S = \emptyset\}$. Moreover, $\mathfrak{p}^{ec} = \mathfrak{p}$.*

- That $\mathfrak{p} \cap S = \emptyset$ is a natural and very useful condition here!

Proof. By Lemma 1.3.6, it is clear that the map $\text{Spec } S^{-1}A \rightarrow \text{Spec } A$ is injective. If $\tilde{\mathfrak{p}} \in \text{Spec } S^{-1}A$, then we must have $(\tilde{\mathfrak{p}})^c \cap S = \emptyset$; otherwise, $\tilde{\mathfrak{p}} = S^{-1}(\tilde{\mathfrak{p}})^c = S^{-1}A$ is not proper. It remains to show f^* is surjective. Let $\mathfrak{p} \in \text{Spec } A$ and $\mathfrak{p} \cap S = \emptyset$. We claim that $S^{-1}\mathfrak{p} \in \text{Spec } S^{-1}A$. Consider the composite homomorphism $\theta : A \rightarrow A/\mathfrak{p} \rightarrow Q(A/\mathfrak{p})$. Since $\mathfrak{p} \cap S = \emptyset$, we have that $f(S) \subseteq (A/\mathfrak{p}) \setminus \{0\}$, which implies that $f(S) \subseteq Q(A/\mathfrak{p})^*$. So by the universal property of localization, there exists a homomorphism $g : S^{-1}A \rightarrow Q(A/\mathfrak{p})$ where $g(\frac{a}{s}) = 0 \iff a \in \mathfrak{p}$ (again since $\mathfrak{p} \cap S = \emptyset$). Hence $S^{-1}A/S^{-1}\mathfrak{p}$ is a subring of $Q(A/\mathfrak{p})$, which means it is an integral domain, so the claim is proven. Finally, noticing that $\mathfrak{p}^e = S^{-1}\mathfrak{p}$, we show that $\mathfrak{p}^{ec} = \mathfrak{p}$, which finishes the proof of surjectivity. To this end, it suffices to show that $\mathfrak{p}^{ec} \subseteq \mathfrak{p}$. If $a \in \mathfrak{p}^{ec}$, then $f(a) = \frac{a}{1} \in \mathfrak{p}^e = S^{-1}\mathfrak{p}$ implies that $as \in \mathfrak{p}$ for some s . But $\mathfrak{p} \cap S = \emptyset$, so we must have that $a \in \mathfrak{p}$ and we are done. $\square$

Theorem 1.3.8 (Fiber of contraction) *Let $f : A \rightarrow B$ be a ring homomorphism, $\mathfrak{p} \in \text{Spec } A$, and $S_{\mathfrak{p}} := A \setminus \mathfrak{p}$. Consider the composite homomorphism*

$$\theta : B \rightarrow f(S_{\mathfrak{p}})^{-1}B \rightarrow f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e.$$

Then

- (1). $\theta^* : \text{Spec}(f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e) \rightarrow (f^*)^{-1}(\mathfrak{p})$ is a bijection.
- (2). $B \otimes_A Q(A/\mathfrak{p}) \simeq f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e$, so there is bijection between $\text{Spec}(B \otimes_A Q(A/\mathfrak{p}))$ and $(f^*)^{-1}(\mathfrak{p})$.
- (3). $\mathfrak{p} \in \text{im } f^* \iff \mathfrak{p}^{ec} = \mathfrak{p}$.

Proof. (1) Write $\theta^* = i^* \circ \pi^*$. By Lemma 1.3.4,

$$\pi^* : \text{Spec}(f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e) \rightarrow V(f(S_{\mathfrak{p}})^{-1}\mathfrak{p})$$

is a bijection, where $V(f(S_{\mathfrak{p}})^{-1}\mathfrak{p}) \subseteq \text{Spec}(f(S_{\mathfrak{p}})^{-1}B)$. And by Proposition 1.3.7,

$$i^* : \text{Spec}(f(S_{\mathfrak{p}})^{-1}B) \rightarrow \{\mathfrak{q} \in \text{Spec } B : \mathfrak{q} \cap f(S_{\mathfrak{p}}) = \emptyset\}$$

is also a bijection. Hence for every $\tilde{\mathfrak{q}} \in \text{Spec}(f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e)$, there exists unique $\mathfrak{q} \in \text{Spec } B$ such that $\mathfrak{q} \cap f(S_{\mathfrak{p}}) = \emptyset$ and $f(S_{\mathfrak{p}})^{-1}\mathfrak{q} \supseteq f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e$. But $\mathfrak{q} \cap f(S_{\mathfrak{p}}) = \emptyset$ implies that $\mathfrak{q}^c \cap S_{\mathfrak{p}} = \emptyset$, so $\mathfrak{q}^c \subseteq \mathfrak{p}$; on the other hand, $f(S_{\mathfrak{p}})^{-1}\mathfrak{q} \supseteq f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e$ implies that $\mathfrak{q} \supseteq \mathfrak{p}^e$, so then $\mathfrak{q}^c \supseteq \mathfrak{p}^{ec} = \mathfrak{p}$ by Proposition 1.3.7. Therefore if $\mathfrak{q} \cap f(S_{\mathfrak{p}}) = \emptyset$ and $f(S_{\mathfrak{p}})^{-1}\mathfrak{q} \supseteq f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e$, then $\mathfrak{q}^c = \mathfrak{p}$; clearly the converse holds as well. As a result, we deduce

$$\theta^* : \text{Spec}(f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e) \rightarrow \{\mathfrak{q} \in \text{Spec } B : \mathfrak{q}^c = \mathfrak{p}\} = (f^*)^{-1}(\mathfrak{p})$$

is a bijection.

(2) First notice that $f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e \simeq f(S_{\mathfrak{p}})^{-1}(B/\mathfrak{p}^e)$. Viewing B as an A -module where $a \cdot b := f(a)b$, we have

$$B/\mathfrak{p}^e = B/f(\mathfrak{p})B = B/(\mathfrak{p} \cdot B) \simeq (A/\mathfrak{p}) \otimes_A B.$$

Hence

$$f(S_{\mathfrak{p}})^{-1}(B/\mathfrak{p}^e) \simeq f(S_{\mathfrak{p}})^{-1}(A/\mathfrak{p} \otimes_A B) \simeq f(S_{\mathfrak{p}})^{-1}B \otimes_B (A/\mathfrak{p} \otimes_A B) \simeq f(S_{\mathfrak{p}})^{-1}B \otimes_A A/\mathfrak{p}$$

since tensor product is associative and commutative. Now we use the fact (proven in remarks) that if $f : A \rightarrow B$ is a homomorphism and $S \subseteq A$ is multiplicatively closed, then $S^{-1}B \simeq f(S)^{-1}B$ as A -modules. As a result, we deduce

$$f(S_{\mathfrak{p}})^{-1}B \otimes_A A/\mathfrak{p} \simeq S_{\mathfrak{p}}^{-1}B \otimes_A A/\mathfrak{p} \simeq B \otimes_A S_{\mathfrak{p}}^{-1}(A/\mathfrak{p}) = B \otimes_A Q(A/\mathfrak{p}).$$

(3)

$$\begin{aligned} \mathfrak{p} \in \text{im } f^* &\iff (f^*)^{-1}(\mathfrak{p}) \neq \emptyset, \\ &\iff f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e \neq f(S_{\mathfrak{p}})^{-1}B \text{ by (1),} \\ &\iff \mathfrak{p}^e \cap f(S_{\mathfrak{p}}) = \emptyset, \\ &\iff \mathfrak{p}^{ec} \subseteq \mathfrak{p}. \end{aligned}$$

Since the other inclusion always holds, we are done. $\square$

- Here we prove the following: if $f : A \rightarrow B$ is a ring homomorphism and $S \subseteq A$ is multiplicatively closed, then $S^{-1}B \simeq f(S)^{-1}B$ as A -modules.

Proof. Define $g : S^{-1}B \rightarrow f(S)^{-1}B$ by $g(\frac{b}{s}) := \frac{b}{f(s)}$. It is clear that g is a homomorphism. We show that g admits an inverse, so then it is an isomorphism. We claim that $h : f(S)^{-1}B \rightarrow S^{-1}B$ by $h(\frac{b}{f(s)}) = \frac{b}{s}$ is a well-defined map. To this end, it suffices to show that if $f(s_1) = f(s_2)$, then $\frac{b}{s_1} = \frac{b}{s_2}$. But

$$\frac{b}{s_1} = \frac{b}{s_2} \iff b \cdot (s_2 - s_1) = b(f(s_2) - f(s_1)) = 0$$

since $f(S) \subseteq (S^{-1}B)^*$, so then claim is proven. It is clear that h is a homomorphism and $h \circ g = \text{id}_{S^{-1}B}$, so we are done. $\square$

Nakayama's Lemma

Proposition 1.4.1 (Endomorphisms trapped in an ideal) *Let M be a finitely-generated A -module and $\varphi \in \text{End}_A(M)$. If $\varphi(M) \subseteq \mathfrak{a}M$ for some $\mathfrak{a} \trianglelefteq A$, then there exist $a_i \in \mathfrak{a}$ such that*

$$\varphi^n + a_{n-1}\varphi^{n-1} + \cdots + a_0 = 0.$$

- Does this hold trivially in the case that $\mathfrak{a} = A$?

Proof. For every $a \in A$, define $\bar{a} : M \rightarrow M$ by $\bar{a}(m) := a \cdot m$. Let $\bar{A} := \{\bar{a} : a \in A\}$; then clearly $\bar{A} \subseteq \text{End}_A(M)$ is a commutative subring. Let $R := \bar{A}[\varphi] \subseteq \text{End}_A(M)$. Since M is finitely-generated, write $M = Am_1 + \cdots + Am_k$. Then for every $\psi \in \text{End}_A(M)$, $\psi = 0 \iff \psi(m_i) = 0$ for every i . Since $\psi(M) \subseteq \mathfrak{a}M$, for every i we can write $\varphi(m_i) = a_{i1}m_1 + \cdots + a_{ik}m_k$ for some $a_{ij} \in \mathfrak{a}$; in other words, we have

$$T\mathbf{m} := \begin{pmatrix} \bar{a}_{11} & \cdots & \bar{a}_{1k} \\ \vdots & \ddots & \vdots \\ \bar{a}_{k1} & \cdots & \bar{a}_{kk} \end{pmatrix} \begin{pmatrix} m_1 \\ \vdots \\ m_k \end{pmatrix} = \begin{pmatrix} \varphi(m_1) \\ \vdots \\ \varphi(m_k) \end{pmatrix}.$$

Hence $(\varphi I - T)\mathbf{m} = 0$. Multiplying both sides by the conjugate of $\varphi I - T$ in $\mathbf{M}_k(R)$, we deduce $\det(\varphi I - T)\mathbf{m} = 0$, so then $\det(\varphi I - T) = 0$. Now notice that $xI - T \equiv xI \pmod{\mathfrak{a}}$, and therefore $\det(xI - T) \equiv x^k \pmod{\mathfrak{a}}$. As a result, substituting x for φ we have

$$\det(\varphi I - T) = \varphi^k + \bar{a}_{k-1}\varphi^{k-1} + \cdots + \bar{a}_0.$$

Since $\det(\varphi I - T) = 0$, we are done. $\square$

Corollary 1.4.2 (Modules trapped in an ideal) *Let M be a finitely-generated A -module and $\mathfrak{a} \trianglelefteq A$. If $M = \mathfrak{a}M$, then there exists $a \in A$ such that $aM = 0$ and $a \equiv 1 \pmod{\mathfrak{a}}$.*

Proof. By Proposition 1.4.3, we have that $\text{id}_M + a_{n-1}\text{id}_M + \cdots + a_0 = 0$ for some $a_i \in \mathfrak{a}$. Define $a := 1 + \sum_{i=1}^{n-1} a_i \in A$; then clearly $aM = 0$ and $a \equiv 1 \pmod{\mathfrak{a}}$, so we are done. $\square$

Corollary 1.4.3 (Nakayama's Lemma) *Let M be a finitely-generated A -module such that $M = J(A)M$. Then $M = 0$.*

Proof. By Corollary 1.4.2, there exists $a \in J(A)$ such that $aM = 0$ and $a \equiv 1 \pmod{J(A)}$. Since $a \in 1 + J(A) \subseteq A^\times$ by Lemma 1.1.2, we deduce $M = 0$. $\square$

- Another useful form of Nakayama's Lemma: if M be a finitely generated A -module and $N \leq M$ is a submodule such that $M = N + J(A)M$, then $M = N$. Indeed, notice that

$$\begin{aligned} M = N + J(A)M &\iff \forall m \in M, \exists x \in J(A), m' \in M \text{ s.t. } m \equiv xm' \pmod{N}, \\ &\iff m + N = xm' + N \text{ in } M/N, \\ &\iff M/N = J(A)(M/N), \\ &\iff M = N. \end{aligned}$$

Primary Decomposition

Primary Ideals

Definition 2.1.1 (Primary ideals) We say a proper ideal $\mathfrak{q} \triangleleft A$ is *primary* if $xy \in \mathfrak{q}$ implies that $x \in \mathfrak{q}$ or $y^n \in \mathfrak{q}$ for some $n \in \mathbb{N}$.

Lemma 2.1.2 (Characterization of primary ideals) A proper ideal $\mathfrak{q} \triangleleft A$ is primary $\iff$ every zero-divisor of $A/\mathfrak{q}$ is nilpotent.

Proof. ($\Leftarrow$) Suppose $xy \in \mathfrak{q}$ and $x \notin \mathfrak{q}$. Since $\overline{xy} = 0 \in A/\mathfrak{q}$, we have that $\overline{y} \in D(A/\mathfrak{q})$, so then $\overline{y}^n = \overline{y^n} = 0$ for some n , which implies that $y^n \in \mathfrak{q}$.

($\Rightarrow$) Suppose $\overline{x} \in D(A/\mathfrak{q})$. Then there exists $y \notin \mathfrak{q}$ such that $xy \in \mathfrak{q}$; but $\mathfrak{q}$ is primary, so $x^n \in \mathfrak{q}$ for some n . Hence $\overline{x^n} = \overline{x^n} = 0 \in A/\mathfrak{q}$ and we are done. $\square$

Lemma 2.1.3 (Radical and intersection of prime divisors) Let $\mathfrak{a} \triangleleft A$ be a proper ideal. Then $\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p}$.

Proof. That $\sqrt{\mathfrak{a}} \subseteq \bigcap_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p}$ is clear. For the converse, suppose $x \notin \sqrt{\mathfrak{a}}$. Define

$$\Omega := \{\mathfrak{b} \in \text{Ideal } A : \mathfrak{b} \supseteq \mathfrak{a} \text{ and } \forall n \in \mathbb{N}, x^n \notin \mathfrak{b}\}.$$

Then Ω is non-empty since $\sqrt{\mathfrak{a}} \in \Omega$. It is easy to see that the union of every chain in Ω is again an element of Ω , so by Zorn's Lemma Ω contains a maximal element $\mathfrak{p}$, which we claim is prime. Indeed, if $a, b \notin \mathfrak{p}$, then by maximality, $\mathfrak{p} + \langle a \rangle$ and $\mathfrak{p} + \langle b \rangle$ are not in Ω , so then $x^n \in \mathfrak{p} + \langle a \rangle$ and $x^m \in \mathfrak{p} + \langle b \rangle$ for some $n, m \in \mathbb{N}$. Hence $x^{nm} \in \mathfrak{p} + \langle ab \rangle$ implies that $\mathfrak{p} + \langle ab \rangle \notin \Omega$, so $ab \notin \mathfrak{p}$ and the claim is proven. Finally, noticing that $\mathfrak{p} \supseteq \mathfrak{a}$, we deduce $\mathfrak{p} \in V(\mathfrak{a})$ and therefore $x \notin \bigcap_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p}$, so we are done. $\square$

Corollary 2.1.4 (Radical of primary ideals) Let $\mathfrak{q} \triangleleft A$ be a primary ideal. Then $\mathfrak{p} := \sqrt{\mathfrak{q}} \in \text{Spec } A$; in particular, $\mathfrak{p}$ is the smallest prime divisor of $\mathfrak{q}$.

- The converse of this does not hold; but if $\sqrt{\mathfrak{q}} = \mathfrak{m}$ is not only prime but maximal, then we can deduce $\mathfrak{q}$ is $\mathfrak{m}$ -primary. See Lemma 2.1.6.

Proof. Suppose there exist $a, b \notin \mathfrak{p}$ such that $ab \in \mathfrak{p}$. Then $(ab)^n = a^n b^n \in \mathfrak{q}$ for some $n \in \mathbb{N}$. Since $a \notin \mathfrak{p}$, we must have that $a^n \notin \mathfrak{q}$; but since $\mathfrak{q}$ is primary, this implies that $(b^n)^m \in \mathfrak{q} \subseteq \mathfrak{p}$, which is a contradiction. Now since $\mathfrak{p} = \sqrt{\mathfrak{q}} = \bigcap_{\mathfrak{p}' \in V(\mathfrak{q})} \mathfrak{p}'$ is prime and clearly $\mathfrak{p} \supseteq \mathfrak{q}$, it is the smallest prime divisor of $\mathfrak{q}$ and we are done. $\square$

Definition 2.1.5 ($\mathfrak{p}$ -primary ideals) A primary ideal $\mathfrak{q}$ is called $\mathfrak{p}$ -primary if $\sqrt{\mathfrak{q}} = \mathfrak{p}$.

Lemma 2.1.6 (m-primary ideals) *Let $\sqrt{\mathfrak{q}} = \mathfrak{m} \in \text{Max } A$. Then $\mathfrak{q}$ is $\mathfrak{m}$ -primary.*

Proof. For any $\mathfrak{p} \in \text{Spec } A$, if $\mathfrak{p} \supseteq \mathfrak{q}$, then $\mathfrak{p} \supseteq \sqrt{\mathfrak{q}} = \mathfrak{m}$. Since $\mathfrak{m} \in \text{Max } A$, we deduce $V(\mathfrak{q}) = \{\mathfrak{m}\}$. Now suppose $x \notin \mathfrak{q}$ and $xy \in \mathfrak{q}$. Consider $(\mathfrak{q} : x) := \{a \in A : ax \in \mathfrak{q}\}$ which is clearly an ideal; in fact it is proper. Hence $\emptyset \neq V(\mathfrak{q} : x) \subseteq V(\mathfrak{q}) = \{\mathfrak{m}\}$ implies that $V(\mathfrak{q} : x) = \{\mathfrak{m}\}$. But then $xy \in \mathfrak{q}$ implies that $y \in (\mathfrak{q} : x) \subseteq \mathfrak{m} = \sqrt{\mathfrak{q}}$, so $\mathfrak{q}$ is primary and we are done. $\square$

- Essentially what we need is the inclusions $\mathfrak{q} \subseteq (\mathfrak{q} : x) \subseteq \mathfrak{m}$, which holds because $\mathfrak{m}$ is maximal!

Lemma 2.1.7 (Primary ideals in a PID) *Let A be a PID. Then $\mathfrak{q} \triangleleft A$ is primary $\iff \mathfrak{q} = \langle p^n \rangle$ for some prime element p and $n \in \mathbb{N}$.*

Proof. Recall that $\text{Spec } A = \{0\} \cup \text{Max } A$, so WLOG we may assume $\mathfrak{q} = \langle q \rangle \neq 0$. Notice that $\mathfrak{q} \subseteq \sqrt{\mathfrak{q}} = \langle p \rangle$ implies that $p \mid q$. But since $\sqrt{\mathfrak{q}}$ is the smallest prime factor of $\mathfrak{q}$ and also a maximal ideal itself, we deduce that $V(\mathfrak{q}) = \{\sqrt{\mathfrak{p}}\}$, which implies that p is the only prime factor of q . Hence $\mathfrak{q} = \langle q \rangle = \langle p^n \rangle$ for some $n \in \mathbb{N}$ and we are done. $\square$

- One may say that primary ideals are meant to “mimic” the powers of prime ideals.

Lemma 2.1.8 (Characterization of $(\mathfrak{q} : x)$) *Let $\mathfrak{q}$ be $\mathfrak{p}$ -primary. Then*

$$(\mathfrak{q} : x) = \begin{cases} A & \text{if } x \in \mathfrak{q}, \\ \mathfrak{q} & \text{if } x \notin \mathfrak{p}, \\ \mathfrak{p}\text{-primary} & \text{if } x \notin \mathfrak{q}. \end{cases}$$

Proof. The first case is obvious.

For the second case, it suffices to show that $(\mathfrak{q} : x) \subseteq \mathfrak{q}$. Suppose $y \in (\mathfrak{q} : x)$ where $x \notin \mathfrak{p}$. Then $x^n \notin \mathfrak{p}$ for any $n \in \mathbb{N}$, which implies that $x \notin \sqrt{\mathfrak{q}}$, hence $y \in \mathfrak{q}$ since $\mathfrak{q}$ is primary.

For the third case, we first show that $\sqrt{(\mathfrak{q} : x)} = \mathfrak{p}$. To this end, it suffices to show $\sqrt{(\mathfrak{q} : x)} \subseteq \mathfrak{p}$. Suppose $y^n \in (\mathfrak{q} : x)$ where $x \notin \mathfrak{q}$. Then $y^n x \in \mathfrak{q}$ implies that $(y^n)^m \in \mathfrak{q}$ for some m , so $y \in \sqrt{\mathfrak{q}} = \mathfrak{p}$. Now suppose $yz \in (\mathfrak{q} : x)$ and $y^n \notin (\mathfrak{q} : x)$ for any n ; that is, $y^n \notin \mathfrak{p}$ for any n . But then $(xz)y \in \mathfrak{q}$ and $y^n \notin \mathfrak{q} \subseteq \mathfrak{p}$ for any n implies that $xz \in \mathfrak{q}$ since $\mathfrak{q}$ is primary, so $z \in (\mathfrak{q} : x)$ and we are done. $\square$

Primary decomposition

Definition 2.2.1 (Decomposable ideals) We say a proper ideal $\mathfrak{a} \triangleleft A$ is *decomposable* if $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ for some primary ideals $\mathfrak{q}_i$. We say such a decomposition is reduced if

- (1). For every i , $\mathfrak{q}_i \not\supseteq \bigcap_{j \neq i} \mathfrak{q}_j$, and
- (2). $\sqrt{\mathfrak{q}_i} \neq \sqrt{\mathfrak{q}_j}$ for all $i \neq j$.

- A decomposition need not be unique!

Lemma 2.2.2 (Intersection of $\mathfrak{p}$ -primary ideals) *If $\mathfrak{q}$ and $\mathfrak{q}'$ are two $\mathfrak{p}$ -primary ideals, then so is $\mathfrak{p} \cap \mathfrak{p}'$. As a result, every primary decomposition admits a reduced form.*

Proof. It is clear that $\sqrt{\mathfrak{q} \cap \mathfrak{q}'} \subseteq \mathfrak{p}$. For the converse, suppose $x \in \mathfrak{p} = \sqrt{\mathfrak{q}} = \sqrt{\mathfrak{q}'}$. Then $x^n \in \mathfrak{q}$ and $x^m \in \mathfrak{q}'$ for some $n, m \in \mathbb{N}$. Hence $x^{nm} \in \mathfrak{q} \cap \mathfrak{q}'$ implies that $x \in \sqrt{\mathfrak{q} \cap \mathfrak{q}'}$. To see that $\mathfrak{q} \cap \mathfrak{q}'$ is primary, suppose $xy \in \mathfrak{q} \cap \mathfrak{q}'$ and $x \notin \sqrt{\mathfrak{q} \cap \mathfrak{q}'} = \mathfrak{p}$. But since $xy \in \mathfrak{q} \cap \mathfrak{q}' \subseteq \mathfrak{q}$, this implies that $y \in \mathfrak{q}$; and similarly $y \in \mathfrak{q}'$. Hence we deduce $y \in \mathfrak{q} \cap \mathfrak{q}'$ and therefore $\mathfrak{p} \cap \mathfrak{p}'$ is primary.

Now if $\bigcap_{i=1}^n \mathfrak{q}_i$ is a primary decomposition, we partition the $\mathfrak{p}_i$'s by declaring that $\mathfrak{q}_i \sim \mathfrak{q}_j \iff \sqrt{\mathfrak{q}_i} = \sqrt{\mathfrak{q}_j}$ and then define the intersections within each equivalent class to be the new $\mathfrak{q}_i$'s. This gives condition (2). For (1), we simply drop some of the $\mathfrak{q}_i$'s if necessary, so we are done. $\square$

Theorem 2.2.3 (Prime ideals associated with a decomposable ideal) *Let $\bigcap_{i=1}^n \mathfrak{q}_i$ be a reduced decomposition of $\mathfrak{a}$ with $\mathfrak{p}_i := \sqrt{\mathfrak{q}_i}$. Then*

$$\text{Ass}(\mathfrak{a}) := \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\} = \text{Spec } A \cap \{\sqrt{(\mathfrak{a} : x)} : x \in A\}.$$

In particular, the $\mathfrak{p}_i$'s are independent of the choice of decomposition of $\mathfrak{a}$. We say that $\text{Ass}(\mathfrak{a})$ is the set of prime ideals associated with $\mathfrak{a}$.

Proof. We start with the following results: if $\{\mathfrak{b}_i\}_{i=1}^n \subseteq \text{Ideal } A$ and $x \in A$, then

$$\left(\bigcap_i \mathfrak{b}_i : x\right) = \bigcap_i (\mathfrak{b}_i : x) \quad \text{and} \quad \sqrt{\bigcap_i \mathfrak{b}_i} = \bigcap_i \sqrt{\mathfrak{b}_i}.$$

The first part is clear. For the second part, notice that

$$y \in \sqrt{\bigcap_i \mathfrak{b}_i} \iff (\exists m \in \mathbb{N}) : y^m \in \bigcap_i \mathfrak{b}_i \iff (\forall i)(\exists m_i) : y^{m_i} \in \mathfrak{b}_i \iff y \in \bigcap_i \sqrt{\mathfrak{b}_i}.$$

Hence for $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ and every $x \in A$, we have

$$\sqrt{(\mathfrak{a} : x)} = \sqrt{\left(\bigcap_i \mathfrak{q}_i : x\right)} = \sqrt{\bigcap_i (\mathfrak{q}_i : x)} = \bigcap_i \sqrt{\mathfrak{q}_i}.$$

Now by Lemma 2.1.8, if $x \in \mathfrak{q}_i$ then $\mathfrak{p}_i = A$, so furthermore we have $\sqrt{(\mathfrak{a} : x)} = \bigcap_{x \notin \mathfrak{q}_i} \mathfrak{p}_i$ provided that $x \notin \bigcap_i \mathfrak{p}_i$. Since $\bigcap_i \mathfrak{q}_i$ is reduced, for every i there exists $x_i \in (\bigcap_{j \neq i} \mathfrak{q}_j) \setminus \mathfrak{q}_i$. But then $\sqrt{(\mathfrak{a} : x_i)} = \mathfrak{p}_i$, which implies that the LHS is a subset of the RHS. Conversely, suppose $\mathfrak{p} := \sqrt{(\mathfrak{a} : x)} = \bigcap_{x \notin \mathfrak{q}_i} \mathfrak{p}_i$ is a prime ideal. Then $\bigcap_{x \notin \mathfrak{q}_i} \mathfrak{p}_i \subseteq \mathfrak{p}$ implies that $\mathfrak{p} \supseteq \mathfrak{p}_{i_0}$ for some $\mathfrak{p}_{i_0}$; but also $\mathfrak{p} \subseteq \bigcap_{x \notin \mathfrak{q}_i} \mathfrak{p}_i$ implies that $\mathfrak{p} \subseteq \mathfrak{p}_i$ for every i . Hence we deduce $\mathfrak{p} = \mathfrak{p}_{i_0}$ for some $\mathfrak{p}_{i_0} \in \text{Ass}(\mathfrak{a})$. So the RHS is a subset of the LHS and we are done. $\square$

Proposition 2.2.4 (Associated primes and prime divisors) *Let $\mathfrak{a} \triangleleft A$ be decomposable. Then*

- (1). $\text{Ass}(\mathfrak{a}) \subseteq V(\mathfrak{a})$.
- (2). For every $\mathfrak{p} \in V(\mathfrak{a})$, there exists $\mathfrak{p}' \in \text{Ass}(\mathfrak{a})$ such that $\mathfrak{p}' \subseteq \mathfrak{p}$.
- (3). The set of minimal elements in $V(\mathfrak{a})$ with respect to inclusion is the same as that of $\text{Ass}(\mathfrak{a})$.

Proof. (1) By Theorem 2.2.3 every $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ is of the form $\sqrt{(\mathfrak{a} : x)} \supseteq \mathfrak{a}$ for some $x \in A$, so $\mathfrak{p} \in V(\mathfrak{a})$.

(2) If $\mathfrak{p} \in V(\mathfrak{a})$, then $\mathfrak{p} \supseteq \mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$. Hence

$$\mathfrak{p} = \sqrt{\mathfrak{p}} = \sqrt{\bigcap_i \mathfrak{q}_i} = \bigcap_i \sqrt{\mathfrak{q}_i} \implies \mathfrak{p} \supseteq \mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$$

for some $\mathfrak{p}_i \in \text{Ass}(\mathfrak{a})$.

(3) Suppose $\mathfrak{p} \in V(\mathfrak{a})$ is a minimal element. Then by (2) there exists $\mathfrak{p}' \in \text{Ass}(\mathfrak{a})$ such that $\mathfrak{p}' \subseteq \mathfrak{p}$. Since $\text{Ass}(\mathfrak{a}) \subseteq V(\mathfrak{a})$, by minimality of $\mathfrak{p}$ we must have that $\mathfrak{p} = \mathfrak{p}'$, which implies that $\mathfrak{p}$ is also a minimal element in $\text{Ass}(\mathfrak{a})$. Conversely, if $\mathfrak{p}' \in \text{Ass}(\mathfrak{a})$ is a minimal element, we claim that $\mathfrak{p}'$ must also be minimal in $V(\mathfrak{a})$. Suppose not. Then there exists $\mathfrak{p} \in V(\mathfrak{a})$ such that $\mathfrak{p} \subsetneq \mathfrak{p}'$. But by (2), we can find $\bar{\mathfrak{p}} \in \text{Ass}(\mathfrak{a})$ such that $\bar{\mathfrak{p}} \subseteq \mathfrak{p} \subsetneq \mathfrak{p}'$, which contradicts the minimality of $\mathfrak{p}'$. Hence $\mathfrak{p}'$ is also minimal in $V(\mathfrak{a})$, so we are done. $\square$

Proposition 2.2.5 (Union of associated primes) *Let $\mathfrak{a} \triangleleft A$ be decomposable. Then*

$$\bigcup_{\mathfrak{p} \in \text{Ass}(\mathfrak{a})} \mathfrak{p} = \{x \in A : (\mathfrak{a} : x) \neq \mathfrak{a}\}.$$

Proof. Recall that

$$(\mathfrak{a} : x) = \left(\bigcap_{i=1}^n \mathfrak{q}_i : x \right) = \bigcap_i (\mathfrak{q}_i : x).$$

If $(\mathfrak{a} : x) \neq \mathfrak{a}$, then at least one $(\mathfrak{q}_i : x) \neq \mathfrak{q}_i$, so by Lemma 2.1.8 we must have $x \in \mathfrak{p}_i \in \text{Ass}(\mathfrak{a})$ for some i . Hence the LHS is a subset of the RHS.

Conversely, suppose $x \in \mathfrak{p} \in \text{Ass}(\mathfrak{a})$. Then by Theorem 2.2.3, there exists $y \in A$ such that $x \in \sqrt{(\mathfrak{a} : y)} = \mathfrak{p}$. Let $i \in \mathbb{N}$ be minimal such that $x^i y \in \mathfrak{a}$. Now suppose to the contrary $(\mathfrak{a} : x) = \mathfrak{a}$. In this case we claim that $y \in \mathfrak{a}$. Indeed, if $i = 0$ there is nothing to show; if not, then $x(x^{i-1}y) \in \mathfrak{a}$ implies that $x^{i-1}y \in (\mathfrak{a} : x) = \mathfrak{a}$, which contradicts the minimality of i and proves the claim. But then $y \in \mathfrak{a}$ implies that $\mathfrak{p} = (\mathfrak{a} : y) = A$ is not proper, which is a contradiction, so we are done. $\square$

Corollary 2.2.6 (Zero-divisors and nilpotent elements) *Let $0 \triangleleft A$ be decomposable. Then*

$$D(A) = \bigcup_{\mathfrak{p} \in \text{Ass}(0)} \mathfrak{p} \text{ and } \text{Nil } A = \bigcap_{\mathfrak{p} \in \text{Ass}(0)} \mathfrak{p}.$$

Proof. Notice that $x \in D(A) \iff (0 : x) \neq 0$. Hence by Proposition 2.2.5 we get the first part. For the second part, it is clear that

$$\text{Nil } A = \bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p} \subseteq \bigcap_{\mathfrak{p}' \in \text{Ass}(0)} \mathfrak{p}'.$$

On the other hand, by Proposition 2.2.4(2), for every $\mathfrak{p} \in \text{Spec } A$ there exists $\mathfrak{p}' \in \text{Ass}(0)$ such that $\mathfrak{p}' \subseteq \mathfrak{p}$, which implies that $\bigcap_{\mathfrak{p}' \in \text{Ass}(0)} \mathfrak{p}' \subseteq \mathfrak{p}$. Hence $\bigcap_{\mathfrak{p}' \in \text{Ass}(0)} \mathfrak{p}' \subseteq \text{Nil } A$ and we are done. $\square$

Localization and primary decomposition

Lemma 2.3.1 (Characterization of $\mathfrak{a}^{\text{ec}}$) Let $S \subseteq A$ be a multiplicatively closed subset and $f : A \rightarrow S^{-1}A$ be given by $f(a) := \frac{a}{1}$. Let $\mathfrak{a} \trianglelefteq A$. Then $S(\mathfrak{a}) := \mathfrak{a}^{\text{ec}} = \bigcup_{s \in S} (\mathfrak{a} : s)$.

Proof. If $a \in S(\mathfrak{a})$, then

$$\frac{a}{1} = f(a) \in \mathfrak{a}^e = S^{-1}\mathfrak{a} \implies (\exists s) : as \in \mathfrak{a} \implies a \in (\mathfrak{a} : s) \subseteq \bigcup_{s \in S} (\mathfrak{a} : s).$$

Conversely, if $a \in \bigcup_{s \in S} (\mathfrak{a} : s)$, then

$$(\exists s) : a \in (\mathfrak{a} : s) \implies as \in \mathfrak{a} \implies \frac{a}{1} = \frac{as}{s} \in S^{-1}\mathfrak{a} \implies a \in f^{-1}(S^{-1}\mathfrak{a}) = S(\mathfrak{a}).$$

□

Proposition 2.3.2 (Primary ideals in a localized ring)

- (1). If $\mathfrak{q}$ is $\mathfrak{p}$ -primary and $\mathfrak{p} \cap S \neq \emptyset$, then $S^{-1}\mathfrak{q} = S^{-1}A$.
- (2). If $\mathfrak{q}$ is $\mathfrak{p}$ -primary and $\mathfrak{p} \cap S = \emptyset$, then $S(\mathfrak{q}) = \mathfrak{q}$ and $S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$ -primary.
- (3). If $\mathfrak{p} \in \text{Spec } A$ and $\mathfrak{p} \cap S = \emptyset$, then extension and contraction induce bijection between $\{\mathfrak{q} \triangleleft A : \mathfrak{q} \text{ is } \mathfrak{p}\text{-primary}\}$ and $\{\tilde{\mathfrak{q}} \triangleleft S^{-1}A : \tilde{\mathfrak{q}} \text{ is } S^{-1}\mathfrak{p}\text{-primary}\}$.

Proof. (1) If $a \in \mathfrak{p} \cap S$, then $a^n \in \mathfrak{q} \cap S$ for some $n \in \mathbb{N}$, so $S^{-1}\mathfrak{q} = S^{-1}A$.

(2) By Lemma 2.3.1, we have that $S(\mathfrak{q}) = \bigcup_{s \in S} (\mathfrak{q} : s)$. Since $\mathfrak{p} \cap S = \emptyset$, by Lemma 2.1.8 we deduce $(\mathfrak{q} : s) = \mathfrak{q}$ for every $s \in S$, so $S(\mathfrak{q}) = \mathfrak{q}$. For the second part, since $S^{-1}\mathfrak{p}$ is prime, $S^{-1}\mathfrak{q} \subseteq S^{-1}\mathfrak{p}$ implies that $\sqrt{S^{-1}\mathfrak{q}} \subseteq S^{-1}\mathfrak{p}$. To see the converse, suppose $\frac{a}{s} \in S^{-1}\mathfrak{p}$. Then $a^n \in \mathfrak{q}$ for some $n \in \mathbb{N}$, so then $(\frac{a}{s})^n = \frac{a^n}{s^n} \in S^{-1}\mathfrak{q}$. Now we show that $S^{-1}\mathfrak{q}$ is primary. Suppose $\frac{a}{s} \cdot \frac{b}{s'} \in S^{-1}\mathfrak{q}$ and $\frac{b}{s'} \notin \sqrt{S^{-1}\mathfrak{q}} = S^{-1}\mathfrak{p}$. Notice that there exists $s'' \in S$ such that $s''ab \in \mathfrak{q}$. But $\frac{b}{s'} \notin S^{-1}\mathfrak{p}$ implies that $b \notin \mathfrak{p}$, so then $s''a \in \mathfrak{q}$ and therefore $\frac{a}{s} = \frac{as''}{ss''} \in S^{-1}\mathfrak{q}$.

(3) By (2), the LHS is a subset of the RHS. Since every ideal of $S^{-1}A$ is an extended ideal, it suffices to show that $\mathfrak{q} := \tilde{\mathfrak{q}}^c$ is $\mathfrak{p}$ -primary for every $S^{-1}\mathfrak{p}$ -primary ideal $\tilde{\mathfrak{q}}$. To this end, we first show $\sqrt{\mathfrak{q}} = \mathfrak{p}$. If $a \in \sqrt{\mathfrak{q}}$ then $a^n \in \mathfrak{q}$ for some $n \in \mathbb{N}$ and so $\frac{a^n}{1} = f(a^n) \in \tilde{\mathfrak{q}}$. But then $\frac{a}{1} \in \sqrt{\tilde{\mathfrak{q}}} = S^{-1}\mathfrak{p}$, so $as \in \mathfrak{p}$ for some $s \in S$. Now since $\mathfrak{p} \cap S = \emptyset$, we must have $a \in \mathfrak{p}$. Conversely, if $a \in \mathfrak{p}$, then $\frac{a}{1} \in S^{-1}\mathfrak{p}$ implies that $\frac{a^n}{1} \in \tilde{\mathfrak{q}}$ and therefore $a^n \in \mathfrak{q}$, so $a \in \sqrt{\mathfrak{q}}$. Finally to see that $\mathfrak{q}$ is primary, suppose $xy \in \mathfrak{q}$ and $y \notin \mathfrak{p} = \sqrt{\mathfrak{q}}$. Then $\frac{x}{1} \cdot \frac{y}{1} \in \tilde{\mathfrak{q}}$ and $\frac{y}{1} \notin S^{-1}\mathfrak{p}$ implies that $\frac{x}{1} \in \tilde{\mathfrak{q}}$, so $x \in \mathfrak{q}$ and we are done. □

Lemma 2.3.3 (Extension of primary ideals) Let $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ be a decomposable ideal. Suppose $\mathfrak{p}_i \cap S = \emptyset \iff 1 \leq i \leq m$. Then

- (1). $S^{-1}\mathfrak{a} = \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i$.
- (2). $S(\mathfrak{a}) = \bigcap_{i=1}^m S(\mathfrak{q}_i)$.

Proof. (1) Since $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$, for every i we have $S^{-1}\mathfrak{a} \subseteq S^{-1}\mathfrak{q}_i$. Hence $S^{-1}\mathfrak{a} \subseteq \bigcap_{i=1}^n S^{-1}\mathfrak{q}_i$. Now by Proposition 2.3.2(1), if $\mathfrak{p}_i \cap S \neq \emptyset$ then $S^{-1}\mathfrak{q}_i = S^{-1}A$; so we deduce $S^{-1}\mathfrak{a} \subseteq \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i$. To see the converse, suppose $\frac{a}{s} \in \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i$. Then $\frac{a}{1} \in \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i$, which implies that $a \in \bigcap_{i=1}^m S(\mathfrak{q}_i) = \bigcap_{i=1}^m \mathfrak{q}_i$ by Proposition 2.3.2(2). Now for $m < i < n$, pick $s_i \in \mathfrak{p}_i \cap S$ and define $s' := \prod_{i>m} s_i$. Notice that $as' \in \prod_{i=1}^n \mathfrak{q}_i = \mathfrak{a}$. As a result, $\frac{a}{s} = \frac{as'}{ss'} \in S^{-1}\mathfrak{a}$ and so $S^{-1}\mathfrak{a} = \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i$.

(2) This follows from taking contraction of both sides in (1). □

The second uniqueness theorem

Definition 2.4.1 (Isolated family of associated primes) Let $\mathfrak{a} \triangleleft A$ be decomposable. We say a non-empty subset $\Sigma \subseteq \text{Ass}(\mathfrak{a})$ is *isolated* if $(\mathfrak{p} \in \Sigma, \mathfrak{p}' \in \text{Ass}(\mathfrak{a}), \mathfrak{p}' \subseteq \mathfrak{p}) \implies \mathfrak{p}' \in \Sigma$.

- An example of isolated family is $\Sigma = \{\mathfrak{p}\}$ where $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ is a minimal element.

Theorem 2.4.2 (Second uniqueness) Let $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ be a decomposable ideal and $\Sigma \subseteq \text{Ass}(\mathfrak{a})$ be an isolated family. Then $\bigcap_{\sqrt{\mathfrak{q}_i} \in \Sigma} \mathfrak{q}_i$ only depends on $\mathfrak{a}$ and Σ and is independent of the choice of reduced primary decomposition. In particular, if $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ is a minimal element, then there exists an $\mathfrak{p}$ -primary ideal $\mathfrak{q}$ that appears in all reduced primary decompositions of $\mathfrak{a}$.

- For the “in particular” part, notice that $\bigcap_{\sqrt{\mathfrak{q}_i} \in \{\mathfrak{p}\}} \mathfrak{q}_i = \mathfrak{q}_i$ is fixed (since there is only **one** $\mathfrak{p}$ -primary ideal in every reduced primary decomposition!).
- Theorem 2.2.3 is sometimes referred to as the first uniqueness theorem.

Proof. It suffices to identify a multiplicatively closed subset $S_\Sigma \subseteq A$ such that $\sqrt{\mathfrak{q}_i} \in \Sigma \iff \sqrt{\mathfrak{q}_i} \cap S_\Sigma = \emptyset$; indeed, by Lemma 2.3.3(2), we have that

$$S_\Sigma(\mathfrak{a}) = \bigcap_{\sqrt{\mathfrak{q}_i} \cap S_\Sigma = \emptyset} \mathfrak{q}_i = \bigcap_{\sqrt{\mathfrak{q}_i} \in \Sigma} \mathfrak{q}_i,$$

which clearly on depends on $\mathfrak{a}$ and Σ . Consider $S_\Sigma := A \setminus \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$. We claim that S_Σ has the required properties. First we show it is multiplicatively closed. Indeed, suppose $s_1, s_2 \in S_\Sigma$ and $s_1 s_2 \notin S_\Sigma$. Then $s_1 s_2 \in \mathfrak{p}$ for some $\mathfrak{p} \in \Sigma$ and hence $s_1 \in \mathfrak{p}$ or $s_2 \in \mathfrak{p}$, contradicting that $s_1, s_2 \in S_\Sigma$. Second, we show $\mathfrak{p} \in \Sigma \iff \mathfrak{p} \cap S_\Sigma = \emptyset$. If $\mathfrak{p} \in \Sigma$ then it is clear that $\mathfrak{p} \cap S_\Sigma = \emptyset$. Conversely, if $\mathfrak{p}' \cap S_\Sigma = \emptyset$, then $\mathfrak{p}' \subseteq \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$. Notice that Σ is a finite family, so by Lemma 1.2.2 we deduce $\mathfrak{p}' \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \Sigma$. But Σ is isolated, so then $\mathfrak{p}' \in \Sigma$ and the claim is proven, so we are done. $\square$

Integral domain of Krull dimension one

Definition 2.5.1 (Krull dimension) Let A be a ring. We define the Krull dimension of A to be

$$\dim A = \sup\{n \in \mathbb{N} : \exists \mathfrak{p}_i \in \text{Spec } A : \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n\}.$$

Lemma 2.5.2 (Integral domain with special dimensions) Let D be an integral domain. Then

- (1). $\dim D = 0 \iff D$ is a field.
- (2). $\dim D = 1 \iff D$ is not a field and $\text{Spec } D = \{0\} \cup \text{Max } D$.

Proof. (1) It suffices to show if $\dim D = 0$ then D is a field. To this end, we claim any $x \in D \setminus \{0\}$ is a unit. If not, there exists a chain $0 \subsetneq \langle x \rangle \subseteq \mathfrak{m}$ where $\mathfrak{m}$ is a maximal ideal; hence $\dim D \geq 1$, a contradiction.

(2) Suppose $\dim D = 1$. Then D is not a field by (1). Suppose there exists $0 \neq \mathfrak{p} \in \text{Spec } D \setminus \text{Max } D$. Then $0 \subsetneq \mathfrak{p} \subsetneq \mathfrak{m}$ is a chain of length 2 for some maximal ideal $\mathfrak{m}$. Hence $\dim D \geq 2$, a contradiction. The other direction clearly holds, so we are done. $\square$

Proposition 2.5.3 (Primary decomposition and $\dim D = 1$) *Let D be an integral domain with $\dim D = 1$. Then any decomposable ideal in D admits a unique reduced primary decomposition.*

Proof. Notice that 0 is the reduced primary decomposition of 0 . Hence $\text{Ass}(0) = \{0\}$, which implies that 0 is minimal in $\text{Ass}(0)$ and therefore 0 is the unique decomposition.

Now suppose $\mathfrak{a} \triangleleft A$ is a non-zero decomposable ideal. Then $\text{Ass}(\mathfrak{a}) \subseteq \text{Max } D$; in particular, every $\mathfrak{p} \in \text{Ass}(D)$ is minimal. So by the second uniqueness theorem, for every $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ there exists a $\mathfrak{p}$ -primary ideal $\mathfrak{q}_{\mathfrak{p}}$ that appears in all the reduced primary decomposition of $\mathfrak{a}$. Hence the decomposition is unique and we are done. $\square$

Primary decomposition in Noetherian rings

Definition 2.6.1 (Irreducible ideals) *We say a proper ideal $\mathfrak{a} \triangleleft A$ is irreducible if $(\mathfrak{a} = \mathfrak{b} \cap \mathfrak{c}) \implies \mathfrak{a} = \mathfrak{b}$ or $\mathfrak{a} = \mathfrak{c}$.*

Lemma 2.6.2 (Irreducible decomposition in Noetherian rings) *Let A be Noetherian. Then every proper ideal can be written as a finite intersection of irreducible ideals.*

Proof. Suppose not. Then

$$\Sigma := \{\mathfrak{a} \triangleleft A : \mathfrak{a} \text{ cannot be written as a finite intersection of irreducible ideals}\}$$

is non-empty. Since A is Noetherian, Σ contains a maximal element $\mathfrak{a}_0$. Notice that $\mathfrak{a}_0$ is not irreducible, so we can write $\mathfrak{a}_0 = \mathfrak{b} \cap \mathfrak{c}$ where $\mathfrak{a}_0 \subsetneq \mathfrak{b}$ and $\mathfrak{a}_0 \subsetneq \mathfrak{c}$. Hence $\mathfrak{b}, \mathfrak{c} \notin \Sigma$ by the maximality of $\mathfrak{a}_0$. But that means they can be written as finite intersections of irreducible ideals, so then

$$\mathfrak{a} = \mathfrak{b} \cap \mathfrak{c} = \left(\bigcap_{i=1}^n \mathfrak{p}_i \right) \cap \left(\bigcap_{i=1}^m \mathfrak{p}'_i \right)$$

can also be written as a finite intersection, which is a contradiction. $\square$

Proposition 2.6.3 (Noetherian rings and primary decomposition) *Let A be Noetherian. Then every irreducible ideal is primary. As a result, every proper ideal in A admits a primary decomposition.*

Proof. Let $\mathfrak{q} \triangleleft A$ be irreducible. By Lemma 2.1.2, it suffices to show that every $x \in D(A/\mathfrak{q})$ is nilpotent. Notice that the ascending chain of ideals

$$(0 : x) \subseteq (0 : x^2) \subseteq \dots$$

terminates with $(0 : x^n)$ for some $n \in \mathbb{N}$ since A is Noetherian. Since $x \in D(A/\mathfrak{q})$, there exists $y \neq 0$ such that $xy = 0 \in A/\mathfrak{q}$. We claim that $\langle y \rangle \cap \langle x^n \rangle = 0$. Indeed, if $z \in \langle y \rangle \cap \langle x^n \rangle$, then $zx = 0$ since $z \in \langle y \rangle$ and $xy = 0$; moreover, $z = ax^n$ for some $a \in A/\mathfrak{q}$. But then $zx = ax^{n+1} = 0$ implies that $a \in (0 : x^{n+1}) = (0 : x^n)$, so $z = ax^n = 0$. Finally, since $\mathfrak{q} \triangleleft A$ is irreducible, we must have that $0 \triangleleft A/\mathfrak{q}$ is also irreducible. Hence $\langle y \rangle \cap \langle x^n \rangle = 0$ implies that $\langle y \rangle = 0$ or $\langle x^n \rangle = 0$. But $y \neq 0$, which implies that $\langle x^n \rangle = 0$ and therefore $x^n = 0$, so we are done. $\square$

Lemma 2.6.5 (Every prime ideal contains a minimal prime) *Let A be a ring and $\mathfrak{p} \in \operatorname{Spec} A$. Then $\mathfrak{p}$ contains a minimal prime ideal.*

Proof. Let $\mathfrak{p} \in \operatorname{Spec} A$ and define $S := \{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{q} \subseteq \mathfrak{p}\}$. Then $S \ni \mathfrak{p}$ is non-empty. We claim that if $\mathcal{C} \subseteq S$ is a chain, then $\mathfrak{q} := \bigcup_{\mathfrak{q}' \in \mathcal{C}} \mathfrak{q}'$ is a prime ideal. Indeed, if $xy \in \mathfrak{q}$, then $xy \in \mathfrak{q}'$ for some $\mathfrak{q}'$, so then $x \in \mathfrak{q}' \subseteq \mathfrak{q}$ or $y \in \mathfrak{q}' \subseteq \mathfrak{q}$. As a result, every chain in S admits an upper bound, so by Zorn's Lemma we are done (with $\mathfrak{q}_1 \preceq \mathfrak{q}_2 \iff \mathfrak{q}_1 \supseteq \mathfrak{q}_2$). $\square$

- If A is an integral domain, then $0 \in \operatorname{Spec} A$ is the unique minimal element.

Corollary 2.6.6 (Minimal prime ideals in a Noetherian ring) *Let A be Noetherian. Then $\operatorname{Spec} A$ has finitely many minimal elements $\mathfrak{p}_1, \dots, \mathfrak{p}_n$. Moreover, $\operatorname{Spec} A = \overline{\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}}$.*

Proof. Since A is Noetherian, $0 \triangleleft A$ is decomposable. Hence by Proposition 2.2.4(3), the minimal elements in $V(0) = \operatorname{Spec} A$ are the same as those in $\operatorname{Ass}(0)$, so they must be finite, say $\mathfrak{p}_1, \dots, \mathfrak{p}_n$. Now suppose $V(\mathfrak{a}) = \overline{\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}}$ and let $\mathfrak{p} \in \operatorname{Spec} A$. Then by Lemma 2.6.5, $\mathfrak{p} \supseteq \mathfrak{p}_i \supseteq \mathfrak{a}$ for some i and therefore $\mathfrak{p} \in V(\mathfrak{a})$. Hence $V(\mathfrak{a}) = \operatorname{Spec} A$. $\square$

Integral Extension of Rings

Integral closure

Definition 3.1.1 (Integral extension)

- (1). If $A \subseteq B$ is a subring, we say B/A is a *ring extension*.
- (2). We say $b \in B$ is *integral over* A if there exists non-zero, monic polynomial $f \in A[x]$ such that $f(b) = 0$.
- (3). We say B/A is an *integral extension* if every $b \in B$ is integral over A .

Proposition 3.1.2 (Characterization of integral elements) *Let B/A be a ring extension and $b \in B$. Then TFAE:*

- (1). b is integral over A ;
- (2). $A[b]$ is a finitely generated A -module;
- (3). There exists a subring $C \subseteq B$ such that $A[b] \subseteq C$ and C is a finitely generated A -module;
- (4). There exists a faithful $A[b]$ -module M that is finitely generated as an A -module.

- We say an A -module M is faithful if $\text{Ann}_A(M) = 0$; alternatively, consider the ring homomorphism

$$\varphi : A \rightarrow \text{End}_A(M), \quad a \mapsto (l_a : m \mapsto a \cdot m).$$

Then M is faithful $\iff \ker \varphi = 0$.

- As an aside, if B/A is a ring extension then any $A[b_1, \dots, b_n]$ is necessarily a faithful A -module.

Proof. (1) $\implies$ (2) Let $f \in A[x]$ be a non-zero, monic polynomial such that $f(b) = 0$. Since f is monic, by the division algorithm for every $g \in A[x]$, there exists $q, r \in A[x]$ with $\deg r < \deg f$ such that $g = qf + r$. Hence

$$g(b) = r(b) \in \sum_{i=0}^{\deg f - 1} Ab^i \implies A[b] = \sum_{i=0}^{\deg f - 1} Ab^i \text{ is a finitely generated } A\text{-module.}$$

(2) $\implies$ (3) Clear.

(3) $\implies$ (4) Let $M := C$. It remains to show that C is a faithful $A[b]$ -module. But notice that for every $r \in A[b]$, $r \cdot 1 = r \in C$. Hence $\text{Ann}_{A[b]}(C) = 0$.

(4) $\implies$ (1) Let $l_b : M \rightarrow M$ be given by $l_b(m) := b \cdot m$. Then since $A[b]$ is commutative, $l_b \in \text{End}_A(M)$. Now since M is a finitely generated A -module, l_b can be represented as a matrix in $\mathbf{M}_n(A)$ where n is the number of generators. So by the Cayley-Hamilton theorem, there exists a monic, non-zero polynomial $\chi_b(t) \in A[t]$ such that

$$\chi_b(b) = \sum_{i=0}^m \bar{a}_i b^i = 0 \in \text{End}_A(M), \quad \bar{a}_i := (l_{a_i} : m \mapsto a_i \cdot m).$$

But since M is a faithful $A[b]$ -module, this implies $\sum_i a_i b^i \in \text{Ann}_A(M) = \{0\}$, so b is integral over A and we are done. $\square$

Corollary 3.1.3 (Extension by integral elements) *Let B/A be a ring extension with $b_1, \dots, b_n$ integral over A . Then $A[b_1, \dots, b_n]$ is a finitely generated A -module.*

Proof. We proceed by induction. The base case is clear from Proposition 3.1.2. Suppose $C := A[b_1, \dots, b_{n-1}]$ is a finitely generated A -module. Notice that b_n is integral over C , so by Proposition 3.1.2 again we deduce $C[b_n]$ is a finitely generated C -module. As a result we have surjective map

$$(A^n)^m \twoheadrightarrow C^m \twoheadrightarrow C[b_n] = A[b_1, \dots, b_n].$$

So $A[b_1, \dots, b_n]$ is a finitely generated A -module and we are done. $\square$

Corollary 3.1.4 (Integral closure is a subring) *Let B/A be a ring extension. Then*

$$\overline{A} := \{b \in B : b \text{ is integral over } A\} \subseteq B$$

is a subring. It is called the integral closure of A in B .

Proof. Suppose $b_1, b_2 \in \overline{A}$. Then by Corollary 3.1.3, $A[b_1, b_2]$ is a finitely generated A -module. Hence by Proposition 3.1.2(3), any element $b \in A[b_1, b_2]$ is integral over A since $A[b] \subseteq A[b_1, b_2]$. As a result, $b_1 \pm b_2$ and $b_1 b_2$ are integral over A , so we are done. $\square$

- We say A is integrally closed in B if $\overline{A} = A$.

Definition 3.1.5 (Integrally closed integral domain) *Let D be an integral domain. We say D is integrally closed if it is integrally closed in its field of fraction.*

- Notice that every UFD is integrally closed due to the rational root criterion. In fact, being integral closed is a local property, whereas being a UFD is not.

Lemma 3.1.6 (Tower of integral extensions) *Let C/B and B/A be integral extensions. Then C/A is an integral extension.*

Proof. Let $c \in C$. Since C/B is integral, there exists a monic, non-zero polynomial $f(x) = \sum_{i=0}^n b_i x^i \in B[x]$ such that $f(c) = 0$. Now since B/A is also integral, by Corollary 3.1.3 $A' := A[b_1, \dots, b_n]$ is a finitely generated A -module. Notice that $f \in A'[x]$, which implies that c is integral over A' . So by Proposition 3.1.2, $A'[c]$ is a finitely generated A' -module. Overall, we deduce that $A'[c]$ is a finitely generated A -module. Hence by Proposition 3.1.2 again c is integral over A , so we are done. $\square$

Corollary 3.1.7 (Integral closure is integrally closed) *Let B/A be a ring extension. Then $\overline{A}$ is integrally closed in B .*

Proof. Let C denote the integral closure of $\overline{A}$ in B . Then $C/\overline{A}$ and $\overline{A}/A$ are both integral, which implies that C/A is also integral by Lemma 3.1.6. Hence $C \subseteq \overline{A}$ and we are done. $\square$

Integral extension, factor ring, and localization

Lemma 3.2.1 (Integrality in localized or factored rings) *Let B/A be an integral extension. Then*

(1). *For every $\mathfrak{b} \trianglelefteq B$, $A/\mathfrak{b}^c \hookrightarrow B/\mathfrak{b}$ is an integral extension.*

(2). *For every multiplicatively closed subset $S \subseteq A$, $S^{-1}B/S^{-1}A$ is an integral extension.*

Proof. (1) Let $b \in B$. Then since B/A is integral, there exists a monic, non-zero polynomial $f \in A[x]$ such that $f(b) = 0$. Pass f to the factored ring, we still have $\bar{f}(\bar{b}) = 0 \in B/\mathfrak{b}$. Notice that monicity is preserved by factoring, so $\bar{f} \in (A/\mathfrak{b}^c)[x]$ is still monic and non-zero, so $\bar{b}$ is integral over $A/\mathfrak{b}^c$.

(2) Let $\frac{b}{s} \in S^{-1}B$ and $f \in A[x]$ be a monic, non-zero polynomial such that $f(b) = 0$, say

$$b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0.$$

Then we deduce

$$\left(\frac{b}{s}\right)^n + \frac{a_{n-1}}{s} \left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_0}{s^n} = 0 \in S^{-1}B,$$

which implies that $\frac{b}{s}$ is integral over $S^{-1}A$, so we are done. $\square$

Proposition 3.2.2 (Integral closure and localization) *Let B/A be a ring extension. Then for every multiplicatively closed subset $S \subseteq A$, we have $\overline{S^{-1}A} = S^{-1}\bar{A}$.*

Proof. Let $C := \bar{A}$ and $\bar{C} := \overline{S^{-1}A}$. By Lemma 3.2.1(2), $S^{-1}C/S^{-1}A$ is an integral extension; hence $S^{-1}C \subseteq \bar{C}$. Now suppose $\frac{b}{s}$ is integral over $S^{-1}A$. Then

$$\left(\frac{b}{s}\right)^n + \frac{a_{n-1}}{s_{n-1}} \left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_0}{s_0} = 0 \in S^{-1}B$$

for some $n \in \mathbb{N}$ and $\frac{a_i}{s_i} \in S^{-1}B$. Hence there exists $s'' \in S$ such that

$$s''(s'_n b^n + s'_{n-1} b^{n-1} + \cdots + s'_0) = 0$$

where $s'_i = (s_{n-1} \cdots \widehat{s_i} \cdots s_0) s^{n-i} a_i$. As a result,

$$(s'' s'_n b)^n + s'_{n-1} (s'' s'_n b)^{n-1} + \cdots + s'_0 (s'' s'_n b)^n = 0 \implies s'' s'_n b \text{ is integral over } A.$$

Hence $s'' s'_n b \in C$ and therefore $\frac{a}{s} = \frac{s'' s'_n a}{s'' s'_n s} \in S^{-1}C$, so we are done. $\square$

Theorem 3.2.3 (Integral closedness is a local property) *Let D be an integral domain. Then TFAE:*

(1). *D is integrally closed;*

(2). *For every $\mathfrak{p} \in \text{Spec } D$, $D_{\mathfrak{p}}$ is integrally closed;*

(3). *For every $\mathfrak{m} \in \text{Max } D$, $D_{\mathfrak{m}}$ is integrally closed.*

Proof. (1) $\implies$ (2) This follows from Proposition 3.2.2 where $\overline{D_{\mathfrak{p}}} = (\bar{D})_{\mathfrak{p}} = D_{\mathfrak{p}}$.

(2) $\implies$ (3) Clear.

(3) $\implies$ (1) Let $C := \bar{D}$. Then by Proposition 3.2.2, for every $\mathfrak{m} \in \text{Max } D$, $C_{\mathfrak{m}}$ is the integral closure of $D_{\mathfrak{m}}$, which implies that $C_{\mathfrak{m}} = D_{\mathfrak{m}}$ since $D_{\mathfrak{m}}$ is integrally closed. View C as a D -module and consider the D -module homomorphism $i : D \rightarrow C$. Then since $i_{\mathfrak{m}} : D_{\mathfrak{m}} \rightarrow C_{\mathfrak{m}}$ is surjective for every $\mathfrak{m} \in \text{Max } D$, we deduce that i is also surjective, so we are done. $\square$

Contraction and integral embedding

Lemma 3.3.1 (Integral extension with fields) *Let B/A be an integral extension and B be an integral domain. Then B is a field $\iff A$ is a field.*

Proof. ($\implies$) If $a \in A \setminus \{0\}$, then $a^{-1} \in B$ since B is a field. Since B/A is integral, there exist $n \in \mathbb{N}$ and $a_i \in A$ such that

$$a^{-n} + a_{n-1}a^{-n+1} + \cdots + a_0 = 0 \implies a^{-1} = -a^{n-1}(a_{n-1}a^{-n+1} + \cdots + a_0) \in A,$$

so A is a field.

($\impliedby$) Let $b \in B$. Then $A[b]$ is a finite-dimensional A -algebra since B/A is integral. We claim that $A[b]$ is a field. Indeed, pick any $0 \neq x \in A[b]$ and consider $l_x : A[b] \rightarrow A[b]$ given by $l_x(y) := xy$. Clearly l_x is an A -linear map. It is also injective since B is an integral domain. Since $A[b]$ is a finite-dimensional vector space, l_x is surjective, which implies that x is a unit and the claim is proven. As a result, we deduce $b^{-1} \in A[b] \subseteq B$, so B is a field and we are done. $\square$

Corollary 3.3.2 (Contraction and maximal ideals) *Let $f : A \hookrightarrow B$ be an integral embedding. Then for every $\mathfrak{q} \in \text{Spec } B$, $f^*(\mathfrak{q}) \in \text{Max } A \iff \mathfrak{q} \in \text{Max } B$.*

Proof. Notice that by Lemma 3.2.1, $A/\mathfrak{q}^c \hookrightarrow B/\mathfrak{q}$ is an integral extension where $B/\mathfrak{q}$ is an integral domain. Hence by Lemma 3.3.1,

$$f^*(\mathfrak{q}) = \mathfrak{q}^c \in \text{Max } A \iff A/\mathfrak{q}^c \text{ is a field} \iff B/\mathfrak{q} \text{ is a field} \iff \mathfrak{q} \in \text{Max } B.$$

$\square$

Proposition 3.3.3 (Lying-over in integral extensions) *Let $f : A \hookrightarrow B$ be an integral embedding. Then $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is surjective and $f^*(\text{Max } B) = \text{Max } A$.*

Proof. Let $\mathfrak{p} \in \text{Spec } A$. Then by Lemma 3.2.1, $f_{\mathfrak{p}} : A_{\mathfrak{p}} \rightarrow S_{\mathfrak{p}}^{-1}B$ is an integral embedding. Hence by Corollary 3.3.2, we have that $f_{\mathfrak{p}}^*(\text{Max } S_{\mathfrak{p}}^{-1}B) \subseteq \text{Max } A_{\mathfrak{p}} = \{\mathfrak{p}_{\mathfrak{p}}\}$. But then by Proposition 1.3.7, that means there exists $\mathfrak{q} \in \text{Spec } B$ with $\mathfrak{q} \cap S_{\mathfrak{p}} = \emptyset$ and $S_{\mathfrak{p}}(\mathfrak{q}) = \mathfrak{q}$. Now $\mathfrak{q} \cap S_{\mathfrak{p}} = \emptyset$ implies that $\mathfrak{q} \cap A \subseteq \mathfrak{p}$; on the other hand, if $x \in \mathfrak{p}$, then

$$\frac{x}{1} \in \mathfrak{p}_{\mathfrak{p}} = f_{\mathfrak{p}}^*(\mathfrak{q}_{\mathfrak{p}}) = \mathfrak{q}_{\mathfrak{p}} \cap A_{\mathfrak{p}} \implies x \in S_{\mathfrak{p}}(\mathfrak{q}) = \mathfrak{q}.$$

Hence $\mathfrak{q} \cap A = \mathfrak{p}$ and f^* is surjective. Finally, using Corollary 3.3.2 again we deduce $f^*(\text{Max } B) = \text{Max } A$, so we are done. $\square$

Corollary 3.3.4 (Contraction is a closed map) *Let $f : A \hookrightarrow B$ be an integral embedding. Then $f^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is a closed map. In fact, for every $\mathfrak{b} \subseteq B$, $f^*(V(\mathfrak{b})) = V(\mathfrak{b}^c)$.*

Proof. Let $\mathfrak{b} \subseteq B$. Then by Lemma 3.2.1, $\bar{f} : A/\mathfrak{b}^c \hookrightarrow B/\mathfrak{b}$ is an integral extension. Hence by Proposition 3.3.3, we have commuting diagram

$$\begin{array}{ccc} \operatorname{Spec} B/\mathfrak{b} & \xrightarrow{\bar{f}^*} & \operatorname{Spec} A/\mathfrak{b}^c \\ \pi_{\mathfrak{b}}^* \downarrow & & \downarrow \pi_{\mathfrak{b}^c}^* \\ V(\mathfrak{b}) & \longrightarrow & V(\mathfrak{b}^c) \end{array}$$

where $\pi_{\mathfrak{b}} : B \rightarrow B/\mathfrak{b}$ and $\pi_{\mathfrak{b}^c} : A \rightarrow A/\mathfrak{b}^c$ are natural quotient maps and induce bijections. Since $\bar{f}^*$ is surjective, we deduce that $f^*(V(\mathfrak{b})) = V(\mathfrak{b}^c)$, so we are done. $\square$

Theorem 3.3.5 (Going-up) *Let $f : A \hookrightarrow B$ be an integral embedding. Suppose $\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ is a chain in $\operatorname{Spec} A$ and $\mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_{n-1}$ is a chain in $\operatorname{Spec} B$ such that $f^*(\mathfrak{q}_i) = \mathfrak{p}_i$ for every $i \leq n-1$. Then there exists $\mathfrak{q}_n \in \operatorname{Spec} B$ with $f^*(\mathfrak{q}_n) = \mathfrak{p}_n$ and $\mathfrak{q}_n \supsetneq \mathfrak{q}_{n-1}$.*

Proof. We proceed by induction. The base case is clear since $f^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is surjective by Proposition 3.3.3. Now since $f^*(V(\mathfrak{q}_{n-1})) = V(\mathfrak{p}_{n-1})$ by Corollary 3.3.4 and $\mathfrak{p}_n \in V(\mathfrak{p}_{n-1})$, there exists $\mathfrak{q}_n \in V(\mathfrak{q}_{n-1})$ such that $f^*(\mathfrak{q}_n) = \mathfrak{p}_n$. Finally, notice that $\mathfrak{q}_n \supsetneq \mathfrak{q}_{n-1}$ because $\mathfrak{q}_n \cap A = \mathfrak{p}_n \supsetneq \mathfrak{p}_{n-1} = \mathfrak{q}_{n-1} \cap A$, so we are done. $\square$

***Lemma 3.3.6 (Incomparability in integral extensions)** *Let $f : A \hookrightarrow B$ be an integral embedding, $\mathfrak{p} \in \operatorname{Spec} A$, and $\mathfrak{q}_1, \mathfrak{q}_2 \in (f^*)^{-1}(\mathfrak{p})$. If $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$, then $\mathfrak{q}_1 = \mathfrak{q}_2$.*

Proof. Notice that $\bar{f} : A/\mathfrak{q}_1^c \hookrightarrow B/\mathfrak{q}_1$ is an integral embedding where $\mathfrak{q}_1^c = \mathfrak{p}$. Since $\mathfrak{q}_2 \in (f^*)^{-1}(\mathfrak{p})$, we have that $\bar{f}^*(\bar{\mathfrak{q}}_2) = 0$. We claim that $\bar{\mathfrak{q}}_2 = 0$. If not, pick non-zero $\bar{a} \in \bar{\mathfrak{q}}_2 \subseteq B/\mathfrak{q}_1$. Now since $\bar{a}$ is integral over $A/\mathfrak{p}$, there exist minimal $n \in \mathbb{N}$ and $\bar{a}_i \in A/\mathfrak{p}$ such that $\bar{a}^n + \bar{a}_{n-1}\bar{a}^{n-1} + \cdots + \bar{a}_0 = 0$. But then we have

$$\bar{a}_0 = -\bar{a}(\bar{a}^{n-1} + \bar{a}_{n-1}\bar{a}^{n-2} + \cdots + \bar{a}_1) \in \bar{\mathfrak{q}}_2 \cap A/\mathfrak{p} = \bar{f}^*(\bar{\mathfrak{q}}_2) = 0,$$

which implies that $\bar{a}^{n-1} + \cdots + \bar{a}_{n-1}\bar{a}^{n-2} + \bar{a}_1 = 0$ since $B/\mathfrak{q}_1$ is an integral domain. But that contradicts the minimality of n and the claim is proven, so we are done. $\square$

- This proof yields a simple but important intuition: if B/A is an integral extension of **domains**, then any **non-zero ideal** of B must intersect A **non-trivially**
- Algebraically, this result follows immediately from the bijection $(f^*)^{-1}(\mathfrak{p}) \leftrightarrow \operatorname{Max} S_{\mathfrak{p}}^{-1}B$ which we establish in the proof of Proposition 3.3.3. Hence $\mathfrak{q}_1, \mathfrak{q}_2$ correspond to maximal ideals in a ring, so if one contains another then they must be equal (see Appendix).

Geometrically, there is another reason why this result should hold. Every commutative ring A corresponds to a topological space $\operatorname{Spec} A$, where the prime ideals are irreducible subspaces (affine varieties). To say that B/A is an integral extension means the projection map $f^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ satisfies certain “finiteness” condition; i.e. the fiber of an irreducible subspace is somehow rigid. If $\mathfrak{p}_2 \supsetneq \mathfrak{p}_1$, then $V(\mathfrak{p}_2) \subsetneq V(\mathfrak{p}_1)$...

- This result also implies that $\dim(f^*)^{-1}(\mathfrak{p}) = 0$ for every $\mathfrak{p} \in \operatorname{Spec} A$. Indeed, recall that there is bijection between $(f^*)^{-1}(\mathfrak{p})$ and $C := \operatorname{Spec}(B \otimes_A Q(A/\mathfrak{p}))$ by Theorem 1.3.8, so $(f^*)^{-1}(\mathfrak{p})$ inherits a topology. Now an irreducible closed subset in C is given by $V(\mathfrak{q})$ for some $\mathfrak{q} \in \operatorname{Spec} C$; but if $\mathfrak{q}' \in V(\mathfrak{q})$, we must have $\mathfrak{q}' = \mathfrak{q}$ since they both contract to $\mathfrak{p}$. Hence any irreducible subset of C consists of a singleton, which implies that $\dim C = 0$.

Theorem 3.3.7 (Dimension of rings in an integral extension) *Let $f : A \hookrightarrow B$ be an integral embedding. Then $\dim A = \dim B$.*

Proof. Let $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ be a chain in $\operatorname{Spec} A$. Then by Going-up, there exists a chain $\mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$ in $\operatorname{Spec} B$ such that $f^*(\mathfrak{q}_i) = \mathfrak{p}_i$ for every i . Hence $\dim A \leq \dim B$. Conversely, if $\mathfrak{q}_0 \subsetneq \cdots \subsetneq \mathfrak{q}_n$ is a chain in $\operatorname{Spec} B$, by Lemma 3.3.6 we must have $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ where $\mathfrak{p}_i := \mathfrak{q}_i \cap A \in \operatorname{Spec} A$. So then $\dim A \geq \dim B$ and we are done. $\square$

Openness of contraction

Definition 3.4.1 (Integrality over an ideal) *Let B/A be a ring extension. We say $b \in B$ is integral over $\mathfrak{a} \trianglelefteq A$ if $b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0$ for some $n \in \mathbb{N}^{>0}$ and $a_i \in \mathfrak{a}$.*

Lemma 3.4.2 (Characterization of integrality over an ideal) *Let B/A be a ring extension, $\mathfrak{a} \trianglelefteq A$, and $C := \overline{A}$. Then $b \in B$ is integral over $\mathfrak{a} \iff b \in \sqrt{\mathfrak{a}^e} \subseteq C$. In particular, if b_1, b_2 are integral over $\mathfrak{a}$, then so are $b_1 \pm b_2$ and $b_1 b_2$; so the integral closure of A with respect to $\mathfrak{a}$ is subring of B .*

Proof. ($\implies$) If b is integral over $\mathfrak{a}$, then $b^n = -(a_{n-1}b^{n-1} + \cdots + a_0) \in \mathfrak{a}^e$ for some $a_i \in \mathfrak{a}$, so $b \in \sqrt{\mathfrak{a}^e}$.

($\impliedby$) If $b \in \sqrt{\mathfrak{a}^e}$, then $b^n = c_1 a_1 + \cdots + c_m a_m$ for some $a_i \in \mathfrak{a}$ and $c_i \in B$. Consider $\overline{C} := A[c_1, \dots, c_m]$ and notice that it is a finitely generated A -module since C/A is an integral extension. Define $\varphi := l_{b^n} \in \operatorname{End}_A(\overline{C})$ by $\varphi(x) := b^n x$. Notice that $\varphi(\overline{C}) \subseteq \mathfrak{a}\overline{C}$. Hence by Proposition 1.4.1, there exist $a_i \in \mathfrak{a}$ and $k \in \mathbb{N}$ such that

$$\varphi^k + a_{k-1}\varphi^{k-1} + \cdots + a_0 = 0 \in \operatorname{End}_A(\overline{C}).$$

But $\overline{C}$ is a faithful A -module, which implies that $b^{nk} + a_{n-1}b^{n(k-1)} + \cdots + a_0 = 0 \in A$, so b is integral over $\mathfrak{a}$ and we are done. $\square$

Lemma 3.4.3 (Minimal polynomial and integrality over an ideal) *Let B/A be a ring extension, B be an integral domain, and A be integrally closed. Let $F := Q(A)$, $\mathfrak{a} \trianglelefteq A$, and $b \in B$ be integral over $\mathfrak{a}$. Suppose $m_{b,F}(x) = \sum_{i=0}^m c_i x^i \in F[x]$ is the minimal polynomial of b over F . Then $c_i \in \sqrt{\mathfrak{a}}$ for every i .*

- Here B being an integral domain is used to make sense of $m_{b,F}$; indeed, we cannot talk about the minimal polynomial of b over F unless b sits inside some field extension of F . If B is an integral domain, then $b \in Q(B)/F$.

Proof. Since b is integral over $\mathfrak{a}$, there exists monic $f(x) = \sum_{i=0}^n a_i x^i \in A[x]$ such that $f(b) = 0$ and $a_i \in \mathfrak{a}$. Let $E := \operatorname{Split}_F(f)$. Then over E we may write $f(x) = \prod_{i=0}^n (x - \beta_i)$ for some $\beta_i \in E$ and $\beta_0 = b$. Notice that every β_i is integral over $\mathfrak{a}$. Since $m_{b,F} \mid f$ by the definition of minimal polynomial, we deduce that every root of $m_{b,F} = \sum_{i=0}^m c_i x^i$ is also integral over $\mathfrak{a}$. By Lemma 3.4.2, this implies that every c_i is integral over $\mathfrak{a}$ (since $m_{b,F}$ is monic, all its coefficients can be expressed as linear combinations of its roots), so $c_i \in \sqrt{\mathfrak{a}^e}$. Finally, since A is integrally closed, we also have $c_i \in A$ for every i . As a result, $c_i \in \sqrt{\mathfrak{a}^e} \cap A = \sqrt{\mathfrak{a}}$, so we are done. $\square$

Theorem 3.4.4 (Going-down) *Let B/A be an integral extension, B be an integral domain, A be integrally closed. Suppose $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \in \text{Spec } A$ and $\mathfrak{q}_1 \in \text{Spec } B$ and $\mathfrak{q}_1 \cap A = \mathfrak{p}_1$. Then there exists $\mathfrak{q}_0 \in \text{Spec } B$ such that $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ and $\mathfrak{q}_0 \cap A = \mathfrak{p}_0$.*

- The philosophy is that if we want to understand ideals **contained** in a prime ideal $\mathfrak{p}$, we look at the localization at $\mathfrak{p}$ (zoom-in); for ideals **containing** $\mathfrak{p}$, we look at the quotient ring (zoom-out).
- For a second proof which uses some field theory and does not rely on 3.4.2 - 3.4.3, see Appendix.

Proof. Let $S_{\mathfrak{p}_1} := A \setminus \mathfrak{p}_1$ and $S_{\mathfrak{q}_1} := B \setminus \mathfrak{q}_1$. Notice that $S_{\mathfrak{p}_1} = S_{\mathfrak{q}_1} \cap A$ and we have $A_{\mathfrak{p}_1} \subseteq S_{\mathfrak{p}_1}^{-1}B \subseteq B_{\mathfrak{q}_1}$ where the first inclusion is an integral extension. Consider $f^* : \text{Spec } B_{\mathfrak{q}_1} \rightarrow \text{Spec } A_{\mathfrak{p}_1}$. We claim that it suffices to find $\tilde{\mathfrak{q}}_0 \in \text{Spec } B_{\mathfrak{q}_1}$ such that $f^*(\tilde{\mathfrak{q}}_0) = S_{\mathfrak{p}_1}^{-1}\mathfrak{p}_0$. Indeed, if this is the case then we must have $\tilde{\mathfrak{q}}_0 = S_{\mathfrak{q}_1}^{-1}\mathfrak{q}_0$ for some $\mathfrak{q}_0 \in \text{Spec } B$ with $\mathfrak{q}_0 \cap S_{\mathfrak{q}_1} = \emptyset$ (so then $\mathfrak{q}_0 \subseteq \mathfrak{q}_1$), and $S_{\mathfrak{q}_1}^{-1}\mathfrak{q}_0 \cap A_{\mathfrak{p}_1} = S_{\mathfrak{p}_1}^{-1}\mathfrak{p}_0$ (so then $\mathfrak{q}_0 \cap A = \mathfrak{p}_0$ and in particular $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$, Exercise); hence we will be done. Now we show such $\tilde{\mathfrak{q}}_0$ exists, i.e. $S_{\mathfrak{p}_1}^{-1}\mathfrak{q}_0 \in \text{im } f^*$, which is equivalent to $(S_{\mathfrak{p}_1}^{-1}\mathfrak{p}_0)^{\text{ec}} = S_{\mathfrak{p}_1}^{-1}\mathfrak{p}_0$ by Theorem 1.3.8(3). Suppose not. Then there exists $\frac{a}{s} \in (S_{\mathfrak{p}_1}^{-1}\mathfrak{p}_0)^{\text{ec}} \setminus (S_{\mathfrak{p}_1}^{-1}\mathfrak{p}_0)$ with $a \in A \setminus \mathfrak{p}_0$ and $s \in S_{\mathfrak{p}_1}$. In particular, recall that $f : A_{\mathfrak{p}_1} \rightarrow B_{\mathfrak{q}_1}$ is an embedding, so there exist $p_i \in \mathfrak{p}_0$, $b_i \in B$ and $s_i \in S_{\mathfrak{q}_1}$ such that

$$\frac{a}{s} = \sum_i p_i \frac{b_i}{s_i}.$$

As a result, since $S_{\mathfrak{p}_1} \subseteq S_{\mathfrak{q}_1}$, we can clear the denominators and get

$$as' = \sum_i p_i b_i s'_i$$

for some $s', s'_i \in S_{\mathfrak{q}_1}$. But that implies $as' \in \mathfrak{p}_0^e \subseteq \sqrt{\mathfrak{q}_0^e}$, so then as' is integral over $\mathfrak{p}_0$ by Lemma 3.4.2. Hence by Lemma 3.4.3, if

$$m_{as',F}(x) = x^m + a_{m-1}x^{m-1} + \cdots + a_0$$

is the minimal polynomial of as' over $F := Q(A)$, then $a_{m-i} \in \sqrt{\mathfrak{p}_0} = \mathfrak{p}_0$ for every i . We claim that

$$m_{s',F}(x) = \frac{1}{a^m} m_{as',F}(ax).$$

Indeed, it is clear that the RHS is zero when $x = s'$; moreover, notice that the RHS is a monic and irreducible polynomial since $a \neq 0$ is a unit in F and $m_{as',F}$ is a minimal polynomial. Now since $s' \in B$ is integral over A , we also have

$$m_{s',F}(x) = x^m + a'_{m-1}x^{m-1} + \cdots + a'_0$$

for some $a'_{m-i} \in A$, where $a^i a'_{m-i} = a_{m-i} \in \mathfrak{p}_0$. But $a \notin \mathfrak{p}_0$, so then $a'_{m-i} \in \mathfrak{p}_0$ for every i , which implies that s' is in fact integral over $\mathfrak{p}_0$. Finally, by Lemma 3.4.2 again we deduce

$$s' \in \sqrt{\mathfrak{p}_0^e} \subseteq \sqrt{\mathfrak{p}_1^e} \subseteq \sqrt{\mathfrak{q}_1} = \mathfrak{q}_1,$$

which is a contradiction since $s' \in S_{\mathfrak{q}_1}$. □

Theorem 3.4.5 (Openness of contraction) *Let B/A be an integral extension, B be an integral domain, and A be integrally closed. Then $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is open.*

Proof. Notice that $\{\mathcal{O}_b\}_{b \in B}$ where $\mathcal{O}_b := \{\mathfrak{q} \in \text{Spec } B : \mathfrak{q} \not\supseteq b\}$ forms a basis for open sets in $\text{Spec } B$; indeed, for any $\mathfrak{b} \subseteq B$, we have

$$\{\mathfrak{q} \in \text{Spec } B : \mathfrak{q} \not\supseteq \mathfrak{b}\} = \bigcup_{b \in \mathfrak{b}} \{\mathfrak{q} \in \text{Spec } B : \mathfrak{q} \not\supseteq \langle b \rangle\} = \bigcup_{b \in \mathfrak{b}} \mathcal{O}_b.$$

Hence it suffices to show that $f^*(\mathcal{O}_b)$ is open for every $b \in B$. Now since B/A is integral, by Lemma 3.4.3 we have that

$$m_{b,F}(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in A[x].$$

We claim that $f^*(\mathcal{O}_b) = \bigcup_{i=0}^{n-1} \mathcal{O}_{a_i}$.

($\subseteq$) If $\mathfrak{p} \in f^*(\mathcal{O}_b)$, then $\mathfrak{p} = \mathfrak{q}^c$ for some $\mathfrak{q} \not\supseteq b$. Suppose to the contrary $\mathfrak{p} \notin \bigcup_{i=0}^{n-1} \mathcal{O}_{a_i}$, then b is integral over $\mathfrak{b}$, which implies that $b \in \sqrt{\mathfrak{p}^e}$ by Lemma 3.4.2. But then $\sqrt{\mathfrak{p}^e} \subseteq \sqrt{\mathfrak{q}} = \mathfrak{q}$ implies that $b \in \mathfrak{q}$, a contradiction.

($\supseteq$) Suppose $\mathfrak{p} \in \bigcup_{i=0}^{n-1} \mathcal{O}_{a_i}$. Then b is not integral over $\mathfrak{p}$ and therefore $b \notin \sqrt{\mathfrak{p}^e}$. Recall that $\sqrt{\mathfrak{p}^e} = \bigcap_{\mathfrak{q} \in V(\mathfrak{p}^e)} \mathfrak{q}$, so then there exists $\tilde{\mathfrak{q}} \supseteq \mathfrak{p}^e$ with $b \notin \tilde{\mathfrak{q}}$. In particular, we have that $\mathfrak{p} \subseteq \tilde{\mathfrak{q}}^c$. If equality holds, then we are done. If not, then by Going-down there exists $\mathfrak{q} \subsetneq \tilde{\mathfrak{q}}$ with $\mathfrak{q}^c = \mathfrak{p}$ and $b \notin \mathfrak{q} \subseteq \tilde{\mathfrak{q}}$, so we are done as well. $\square$

- As an immediate corollary, consider the integral closure $\mathcal{O}_k$ of $\mathbb{Z}$ in a number field k . Then $f^* : \text{Spec } \mathcal{O}_k \rightarrow \text{Spec } \mathbb{Z}$ is a closed, open, and surjective map. Moreover, we have that $\dim \mathcal{O}_k = \dim \mathbb{Z} = 1$. In fact, they are both Noetherian, as we will show in the next section in much more generality.

Noetherianity and integral extension

Theorem 3.5.1 (Integral extension and quotient fields) *Let B/A be an integral extension, B be an integral domain, and A be integrally closed. Let $F := Q(A)$ and $E := Q(B)$. Suppose E/F is a finite separable extension. Then $B \subseteq \sum_{i=1}^n Ae_i$ for some $e_i \in E$.*

- This is the central theorem of this section; its proof however relies on several results below (3.5.3 through 3.5.9). For the sake of motivation we first present it here, assuming these results.

Proof. Let $\{\beta_1, \dots, \beta_n\}$ be an F -basis of E . Then by Lemma 3.5.9, there exists $a_i \in A$ such that $a_i\beta_i \in C$ where $C \supseteq B$ is the integral closure of A in E . In particular, taking $a := \prod_{i=1}^n a_i$, we have that $b_i := a\beta_i \in C$ for every i . Notice that $a \in A \setminus \{0\} \subseteq F$ by Lemma 3.5.9, hence $\{b_1, \dots, b_n\}$ is still an F -basis of E . By Lemma 3.5.7 and Lemma 3.5.8, we can find a dual basis $\{e_1, \dots, e_n\}$ with respect to the non-degenerate bilinear form $h(e, e') := \text{Tr}_{E/F}(ee')$. Now suppose $b \in B$. Then we can write $b = \sum_{j=1}^m c_j e_j$ for some $c_j \in F$. Notice that

$$h(b, b_i) = \sum_{j=1}^m c_j h(e_j, b_i) = c_i.$$

On the other hand, since $bb_i \in C$ and $h(b, b_i) = \text{Tr}_{E/F}(bb_i) \in F$ by Corollary 3.5.3, we deduce that $c_i = h(b, b_i) \in A$ because A is integrally closed, so then $b \in \sum_{i=1}^m Ae_i$ and we are done. $\square$

Corollary 3.5.2 (Noetherianity and integral extension) *Let B/A be an integral extension, B be an integral domain, and A be integrally closed. If A is Noetherian, then B is Noetherian.*

Proof. By Theorem 3.5.1, we have that $B \subseteq C := \sum_{i=1}^n Ae_i$ for some $e_i \in Q(B)$, where C is a Noetherian A -module since it is finitely generated. Hence B as the submodule of a Noetherian A -module is also a Noetherian A -module. In particular, it is a Noetherian B -module, so we are done. $\square$

Lemma 3.5.2 (Finite separable extension and basis) *Let E/F be a finite separable extension and $\mathcal{B} := \{e_1, \dots, e_n\}$ be an F -basis of E . Let $e \in E$ and define $l_e : E \rightarrow E$ via $l_e(e') := ee'$. Let $\text{Embed}_F(E, \bar{F}) = \{\sigma_1, \dots, \sigma_n\}$. Then $[l_e]_{\mathcal{B}} \sim \text{diag}(\sigma_1(e), \dots, \sigma_n(e))$ over $\bar{F}$.*

Proof. Since E/F is finite separable, by the primitive element theorem there exists $\alpha \in E$ such that $E = F[\alpha]$. Notice that $\deg m_{\alpha, F} = [E : F] = n$, so we can write $m_{\alpha, F}(x) = \prod_{i=1}^n (x - \alpha_i)$ for some $\alpha_i \in \bar{F}$. Also notice that for every α_i , there exists unique $\sigma_j \in \text{Embed}_F(E, \bar{F})$ such that $\sigma_j(\alpha) = \alpha_i$; so up to reordering we may assume $i = j$. Now we have

$$\begin{aligned} E \otimes_F \bar{F} &= F[\alpha] \otimes_F \bar{F} \\ &\simeq F[x] / \langle \prod_{i=1}^n (x - \alpha_i) \rangle \otimes_F \bar{F} \simeq \bar{F}[x] / \langle \prod_{i=1}^n (x - \alpha_i) \rangle \simeq \bigoplus_{i=1}^n \bar{F}[x] / \langle x - \alpha_i \rangle \simeq \bar{F}^n. \end{aligned}$$

Following the morphisms, we have

$$\theta : \alpha \otimes 1 \mapsto (x + \mathbf{a}) \otimes 1 \mapsto x + \mathbf{a}^e \mapsto (x + \langle x - \alpha_1 \rangle, \dots, x + \langle x - \alpha_n \rangle) \mapsto (\sigma_1(\alpha), \dots, \sigma_n(\alpha)).$$

Since this is an $\bar{F}$ -isomorphism and $\sigma_i \in \text{Embed}_F(E, \bar{F})$, we have $\theta(e \otimes 1) = (\sigma_1(e), \dots, \sigma_n(e))$ for every $e \in E$. As a result, the following diagram commutes:

$$\begin{array}{ccccc} E & \xrightarrow{i} & E \otimes_F \bar{F} & \xrightarrow{\theta} & \bar{F}^n \\ l_e \downarrow & & \downarrow l_e \otimes \text{id}_{\bar{F}} & & \downarrow d_e \\ E & \xrightarrow{i} & E \otimes_F \bar{F} & \xrightarrow{\theta} & \bar{F}^n \end{array}$$

where $d_e : (x_1, \dots, x_n) \mapsto (\sigma_1(e)x_1, \dots, \sigma_n(e)x_n)$. Hence $d_e \circ \theta \circ i = \theta \circ i \circ l_e$, where i can be viewed as the identity matrix. Now since $\mathcal{B} = \{e_1, \dots, e_n\}$ is an F -basis of E , we deduce that $\{e_1 \otimes 1, \dots, e_n \otimes 1\}$ is an $\bar{F}$ -basis of $E \otimes_F \bar{F}$; indeed, the $e_i \otimes 1$'s are linearly independent, and their span has dimension n . So then $\hat{\mathcal{B}} := \{\theta(e_1 \otimes 1), \dots, \theta(e_n \otimes 1)\}$ is an $\bar{F}$ -basis of $\bar{F}^n$. Since $l_e = \theta^{-1} \circ d_e \circ \theta$ as matrices, we have that $[l_e]_{\mathcal{B}} \sim [d_e]_{\hat{\mathcal{B}}}$. Finally, notice that $[d_e]_{\hat{\mathcal{B}}} \sim [d_e]_{\mathcal{B}'}$ where $\mathcal{B}' := \text{diag}(\sigma_1(e), \dots, \sigma_n(e))$ is the standard basis of $\bar{F}^n$ under d_e ; so then $[l_e]_{\mathcal{B}} \sim \text{diag}(\sigma_1(e), \dots, \sigma_n(e))$ and we are done. $\square$

Corollary 3.5.3 (Trace and norm) *Let E/F be a finite separable extension. For every $e \in E$, define*

$$\text{Tr}_{E/F}(e) := \sum_{\sigma \in \text{Embed}_F(E, \bar{F})} \sigma(e) \quad \text{and} \quad N_{E/F}(e) := \prod_{\sigma \in \text{Embed}_F(E, \bar{F})} \sigma(e).$$

Then $\text{Tr}_{E/F}(e) = \text{Tr}(l_e)$ and $N_{E/F}(e) = \det l_e$; in particular, $\text{Tr}_{E/F}(E) \subseteq F$ and $N_{E/F}(E) \subseteq F$.

Proof. This is clear from Lemma 3.5.2 since similar matrices have the same determinant and trace. $\square$

- In fact, this result holds for any simple extension $E = F[\alpha]$; see the proof of Lemma 3.5.2.

Definition 3.5.4 (Non-degenerate bilinear form) Let $f : V \times V \rightarrow F$ be a bilinear form. We say f is *non-degenerate* if the map $V \rightarrow V^*$ via $v \mapsto f(v, -)$ is injective. Equivalently, if $f(v, w) = 0$ for all $w \in V$, then $v = 0$.

Lemma 3.5.5 (Characterization of non-degeneracy) *Let V be a finite-dimensional F -vector space and $f : V \times V \rightarrow F$ be a bilinear form. Then f is non-degenerate $\iff \det[f]_{\mathcal{B}} \neq 0$.*

- Here $[f]_{\mathcal{B}} = [f(e_i, e_j)]_{ij}$ where $\mathcal{B} := \{e_1, \dots, e_n\}$ is an F -basis for V . Given any $v, w \in V$, we can write $f(v, w) = \langle v|_{\mathcal{B}}[f]_{\mathcal{B}}|w\rangle_{\mathcal{B}}$.

Proof. ($\implies$) If $\det[f]_{\mathcal{B}} = 0$, there exists $w \neq 0$ such that $[f]_{\mathcal{B}}|w\rangle_{\mathcal{B}} = 0$. But then $f(v, w) = \langle v|_{\mathcal{B}}[f]_{\mathcal{B}}|w\rangle_{\mathcal{B}} = 0$ for all v , which implies that f is degenerate.

($\impliedby$) If f is degenerate, then there exists $w \neq 0$ such that $f(v, w) = 0$, i.e., $\langle v|_{\mathcal{B}}[f]_{\mathcal{B}}|w\rangle_{\mathcal{B}} = 0$ for every v . Pick $v = e_i$ to be any element in the standard basis, then $\langle e_i|_{\mathcal{B}}[f]_{\mathcal{B}}|w\rangle_{\mathcal{B}} = 0$ implies that the i -th component of $[f]_{\mathcal{B}}|w\rangle_{\mathcal{B}}$ is zero; hence $[f]_{\mathcal{B}}|w\rangle_{\mathcal{B}} = 0$. So then $\det[f]_{\mathcal{B}} = 0$ since $w \neq 0$. $\square$

Lemma 3.5.6 (Non-degeneracy and reflexivity) *Let V be a finite-dimensional F -vector space and $f : V \times V \rightarrow F$ be a non-degenerate bilinear form. Then $T_f : V \rightarrow V^*$ via $v \mapsto f(v, -)$ is an isomorphism, where $V^* = \text{Hom}_F(V, F)$.*

Proof. Since F is finite-dimensional, we have that

$$V^* = \text{Hom}_F(V, F) \simeq \text{Hom}_F(F^n, F) \simeq \bigoplus_{i=1}^n \text{Hom}_F(F, F) \simeq F^n \simeq V.$$

Now if f is non-degenerate, then T_f is an injective map between vector spaces of the same finite-dimension, so it is also surjective and we are done. $\square$

Lemma 3.5.7 (Dual basis) *In the above setting, let $\{v_1, \dots, v_n\}$ be an F -basis of V . Then there exists a dual basis $\{w_1, \dots, w_n\}$ with respect to f such that*

$$f(v_i, w_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{else.} \end{cases}$$

Proof. Let $\{v_1^*, \dots, v_n^*\}$ be the dual basis of V^* ; i.e.

$$v_i^*(v_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{else.} \end{cases}$$

Since T_f is an isomorphism, there exist distinct $w_i \in V$ such that $T_f(w_i) = v_i^*$ for every i , so we are done. $\square$

Lemma 3.5.8 (Trace and non-degeneracy) *Let E/F be a finite separable extension. Then $f(e, e') := \text{Tr}_{E/F}(ee')$ defines a symmetric non-degenerate bilinear form on E .*

Proof. That f maps into F follows from Corollary 3.5.3, so it is indeed a bilinear form. Let $\mathcal{B} := \{e_1, \dots, e_n\}$ be an F -basis of E . To see that f is non-degenerate, by Lemma 3.5.5 it suffices to show that

$$\det[f]_{\mathcal{B}} = \det[\text{Tr}_{E/F}(e_i e_j)]_{ij} \neq 0.$$

Notice that $\text{Tr}_{E/F}(e_i e_j) = \sum_{k=1}^n \sigma_k(e_i) \sigma_k(e_j)$ where $\sigma_k \in \text{Embed}_F(E, \overline{F})$. Define $X = [\sigma_k(e_i)]_{ik}$. Then we have that

$$\det[\text{Tr}_{E/F}(e_i e_j)]_{ij} = \det X X^t = \det X^2.$$

Hence it suffices to show that the rows of X are linearly independent, but this follows from the proof of Lemma 3.5.2 as we show that $\{\theta(e_1 \otimes 1), \dots, \theta(e_n \otimes 1)\}$ is an $\overline{F}$ -basis of $\overline{F}^n$. Since this matrix is similar to X by Lemma 3.5.2, we are done. $\square$

Lemma 3.5.9 (Field of fraction and integral closure) *Let A be an integral domain, $F := Q(A)$, E/F be an algebraic extension, B be the integral closure of A in E . Then $E = (A \setminus \{0\})^{-1}B$.*

Proof. Let $\beta \in E$ and $m_{\beta, F} \in F[x]$ be the minimal polynomial, say $\beta^n + c_{n-1}\beta^{n-1} + \cdots + c_0 = 0$. Taking a common denominator $a \in A$ of the c_i 's, we have that $a\beta^n + a_{n-1}\beta^{n-1} + \cdots + a_0 = 0$ where $ac_i = a_i \in A$. Multiplying both sides by a^{n-1} , we have

$$(a\beta)^n + a_{n-1}(a\beta)^{n-1} + \cdots + a_1a^{n-2}(a\beta) + a^{n-1}a_0 = 0,$$

which implies that $a\beta$ is integral over A and therefore $a\beta \in B$. Finally, we deduce $\beta = \frac{a\beta}{a} \in (A \setminus \{0\})^{-1}B$, so we are done. $\square$

Corollary 3.5.10 ($\mathcal{O}_k$ as an additive group) *Let k be a number field. Then $\mathcal{O}_k$ as an additive group is isomorphic to $\mathbb{Z}^{[k:\mathbb{Q}]}$.*

Proof. Notice that $k = Q(\mathcal{O}_k)$ by Lemma 3.5.9; hence by Theorem 3.5.1 we have $\mathcal{O}_k \subseteq \sum_{i=1}^n \mathbb{Z}e_i$ for some $e_i \in k$. In particular, $\mathcal{O}_k$ is the subgroup of a finitely generated torsion-free abelian group. Hence by the characterization of finitely generated module over a PID, we have $\mathcal{O}_k \simeq \mathbb{Z}^d$ for some $d \in \mathbb{N}$. Since $k = (\mathbb{Z} \setminus \{0\})^{-1}\mathcal{O}_k$ by Lemma 3.5.9 again, we have $d = [k:\mathbb{Q}]$, so we are done. $\square$

- Notice that any submodule of a finitely generated module over an PID is necessarily finitely generated; indeed, we can write $M \simeq D^r \oplus \bigoplus_{i=1}^n D/\langle d_i \rangle$, and every component of M is Noetherian.

Valuation Rings

Valuation ring and Nullstellensatz

Definition 4.1.1 (Valuation ring) Let A be an integral domain and $F := Q(A)$. We say A is a valuation ring if for every $\alpha \in F$, $\alpha \in A$ or $\alpha^{-1} \in A$.

- For instance $\{\frac{m}{n} : m, n \in \mathbb{Z}, 2 \nmid n\} \subseteq \mathbb{Q}$ is a valuation ring. More generally, if D is a UFD and $p \in D$ is a prime element, then $D_{(p)} := \{\frac{a}{b} : a, b \in D, p \nmid b\} \subseteq Q(D)$ is also a valuation ring.
- In fact, an integral domain A is a valuation ring $\iff$ there is a valuation v on $Q(A)$ such that $A = \{\alpha \in Q(A) : v(\alpha) \geq 0\}$; the proof is not difficult, just tedious.

Proposition 4.1.2 (Properties of valuation ring) Let A be a valuation ring and $F := Q(A)$. Then

- (1). A is local.
- (2). If $A \subseteq A' \subseteq F$, then A' is a valuation ring.
- (3). A is integrally closed.

Proof. (1) Define $\mathfrak{m} := A \setminus A^\times$. It suffices to show that $\mathfrak{m}$ is an ideal. If $a \in A$ and $m \in \mathfrak{m}$, then clearly $am \in \mathfrak{m}$; otherwise, $(am)b = (ab)m = 1$ implies that $m \in A^\times$, a contradiction. If $m, n \in \mathfrak{m} \setminus \{0\}$, then $\frac{m}{n} \in A$ or $\frac{n}{m} \in A$ since A is a valuation ring. As a result, $m(1 + \frac{m}{n}) \in \mathfrak{m}$ or $n(1 + \frac{m}{n}) \in \mathfrak{m}$; either case we get $m + n \in \mathfrak{m}$, so $\mathfrak{m}$ is an ideal.

(2) This is clear.

(3) Let $\alpha \in F$ be integral over A . Suppose to the contrary $\alpha \notin A$; then since A is a valuation ring, $\alpha^{-1} \in A$. Notice that $\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0$ for some $a_i \in A$, which implies that

$$\alpha = -\alpha^{-(n-1)}(a_{n-1}\alpha^{n-1} + \dots + a_0) = -(a_{n-1} + \alpha^{-1}a_{n-2} + \dots + \alpha^{-(n-1)}a_0) \in A,$$

a contradiction, so we are done. $\square$

Theorem 4.1.3 (Integral domain and valuation ring) Let Ω be an algebraically closed field, A_0 be an integral domain, $\theta_0 \in \text{Hom}_{\text{Ring}}(A_0, \Omega)$. Suppose A_0 is a subring of F and define

$$\Sigma := \{(B, \theta) : A \subseteq B \subseteq F, \theta \in \text{Hom}_{\text{Ring}}(B, \Omega), \theta|_{A_0} = \theta_0\}$$

with ordering $(B, \theta) \preceq (B', \theta') \iff B \subseteq B'$ and $\theta'|_B = \theta$. Then Σ contains a maximal element (B, θ) where B is a valuation ring with unique maximal ideal $\ker \theta$; moreover, $Q(B) = F$.

- Here is a reason why understanding $\text{Hom}_{\text{Ring}}(A, \Omega)$ can be important. If $A \simeq F[x_1, \dots, x_n]/\mathfrak{a}$ is a finitely generated F -algebra, there is a bijection

$$\text{Hom}_{\text{Ring}}(A, \Omega) \longleftrightarrow \{\varphi \in \text{Hom}_{\text{Alg}_F}(F[x_1, \dots, x_n], \Omega) : \ker \varphi \supseteq \mathfrak{a}\}.$$

On the other hand, there is a bijection

$$\text{Hom}_{\text{Alg}_F}(F[x_1, \dots, x_n], \Omega) \longleftrightarrow \Omega^n$$

via $p \mapsto (e_p : f \mapsto f(p))$; indeed, every φ is uniquely determined by where it sends the x_i 's. As a result, we deduce that

$$\text{Hom}_{\text{Ring}}(A, \Omega) \longleftrightarrow \{p \in \Omega^n : \forall f \in \mathfrak{a}, f(p) = 0\} = V(\mathfrak{a}) \subseteq \mathbb{A}_\Omega^n;$$

In particular, when $F = \Omega$ and $\mathfrak{a} \triangleleft F[x_1, \dots, x_n]$ is a prime ideal, $\text{Hom}_{\text{Ring}}(A, \Omega)$ carries the structure of an affine variety.

Proof. Step 1. We show that Σ admits a maximal element. Clearly $\Sigma \ni (A_0, \theta_0)$ is non-empty. Now if $\mathcal{C} := \{(B_i, \theta_i)\}_i \subseteq \Sigma$ is a chain, define $B := \bigcup_i B_i \subseteq F$ and $\theta : B \rightarrow \Omega$ via $\theta(b) := \theta_i(b)$ if $b \in B_i$. First we show that B is a ring. If $a, b \in B$, then $a \in B_i$ and B_j for some i, j ; since $\mathcal{C}$ is a chain, WLOG we may assume $B_i \subseteq B_j$. Hence $a, b \in B_j$ implies that $a \pm b, ab \in B_j \subseteq B$; so B is a ring. Next we show that θ is a well-defined map. To this end, suppose $b \in B_i \cap B_j$. Since $\mathcal{C}$ is a chain, we may assume WLOG $(B_i, \theta_i) \preceq (B_j, \theta_j)$, which implies that $\theta_i(b) = \theta_j|_{B_i}(b) = \theta_j(b)$, so θ is indeed well-defined. Finally, it is clear that $\theta \in \text{Hom}_{\text{Ring}}(B, \Omega)$ and $\theta|_{A_0} = \theta_0$. So every chain in Σ has an upper bound; hence by Zorn's Lemma, Σ admits a maximal element.

Step 2. We show that a maximal element $(B, \theta) \in \Sigma$ is a local ring with unique maximal $\mathfrak{m} := \ker \theta$. Notice that $B/\mathfrak{m} \hookrightarrow \Omega$ is an integral domain, which implies that $\mathfrak{m} \in \text{Spec } B$; moreover, every non-zero element in $B/\mathfrak{m}$ is sent to a unit. Hence by the universal property of localization, we get a ring homomorphism $\widehat{\theta} : B_{\mathfrak{m}} \rightarrow \Omega$ which extends θ . It is clear that $\widehat{\theta}|_{A_0} = \theta_0$, so $(B_{\mathfrak{m}}, \widehat{\theta}) \in \Sigma$. But (B, θ) is maximal, so $B = B_{\mathfrak{m}}$ and hence $B \setminus \mathfrak{m}$ consists of units. As a result, B is a local ring with unique maximal ideal $\mathfrak{m}$.

Step 3. We show that for every $\alpha \in F$, $\alpha \in B$ or $\alpha^{-1} \in B$; this will imply that B is a valuation ring and moreover $F = Q(B)$. To this end, we first prove a claim:

$$\text{For every } \alpha \in F, \text{ either } \mathfrak{m}[\alpha] \neq B[\alpha] \text{ or } \mathfrak{m}[\alpha^{-1}] \neq B[\alpha^{-1}].$$

Suppose this is false for some α , i.e. $1 \in \mathfrak{m}[\alpha] \cap \mathfrak{m}[\alpha^{-1}]$. Then we can choose smallest $n, m \in \mathbb{N}$ such that $1 = \sum_{i=1}^n c_i \alpha^i = \sum_{j=1}^m c'_j \alpha^{-j}$ for some $c_i, c'_j \in \mathfrak{m}$. WLOG we may assume $n \geq m$. Notice that

$$\begin{aligned} 1 = \sum_j c'_j \alpha^{-j} &\implies B^\times \ni 1 - c'_0 = \sum_{j>0} c'_j \alpha^{-j} \\ &\implies 1 = (1 - c'_0)^{-1} \sum_{j>0} c'_j \alpha^{-j} =: \sum_{j>0} c''_j \alpha^{-j} \\ &\implies \alpha = c''_0 + \sum_{j>0} c''_j \alpha^{-j+1}. \end{aligned}$$

As a result,

$$1 = \sum_i c_i \alpha^i = \left(\sum_{i<n} c_i \alpha^i \right) + c_n \alpha^n = \left(\sum_{i<n} c_i \alpha^i \right) + c_n \alpha^{n-1} (c'_0 + \sum_{j>0} c''_j \alpha^{-j+1}) =: \sum_{i<n} c'''_i \alpha^i,$$

which is a contradiction and the claim is proven.

Now let $\alpha \in F$. WLOG we may assume $\mathfrak{m}[\alpha] \neq B[\alpha]$, so then there exists $\mathfrak{m}' \in \text{Max } B[\alpha]$ such that $\mathfrak{m}[\alpha] \subseteq \mathfrak{m}'$. Since $\mathfrak{m}' \cap B \supseteq \mathfrak{m} \in \text{Max } B$, we deduce that $\mathfrak{m} = \mathfrak{m}' \cap B$ and there is an embedding $B/\mathfrak{m} \hookrightarrow B[\alpha]/\mathfrak{m}' = k(\mathfrak{m})[\bar{\alpha}]$, where $k(\mathfrak{m})$ is the copy of the $B/\mathfrak{m}$ in $B[\alpha]/\mathfrak{m}'$ and $\bar{\alpha} := \alpha + \mathfrak{m}'$. Now since $k(\mathfrak{m})[\bar{\alpha}]$ is a field, $\bar{\alpha}$ must be an algebraic element, which implies that $k(\mathfrak{m})[\bar{\alpha}]/k(\mathfrak{m})$ is a finite extension. In particular it is algebraic and the following diagram commutes:

$$\begin{array}{ccc} B & \xrightarrow{\quad} & B[\alpha] \\ \downarrow & & \downarrow \\ B/\mathfrak{m} & \xrightarrow{\quad} & B[\alpha]/\mathfrak{m}' \\ \downarrow \theta & \swarrow \text{dashed} & \\ \Omega & & \end{array}$$

As a result, we get an extension $\widehat{\theta} : B[\alpha] \rightarrow \Omega$ of θ , so then by the maximality of (B, θ) we deduce $B = B[\alpha]$; therefore $\alpha \in B$ and we are done. $\square$

- For any field extension E/F , if $\alpha \in E$ is transcendental over F , then $F[\alpha]$ is never a field. Indeed, since α is transcendental, for any two distinct polynomials $f, g \in F[x]$ we must have $f(\alpha) \neq g(\alpha) \in E$. Hence it is easy to define a ring isomorphism $F[x] \simeq F[\alpha]$.

Proposition 4.1.4 (Characterization of integral closure) *Let A be an integral domain and $F := Q(A)$. Then*

$$\overline{A} = \bigcap_{\substack{A \subseteq B \subseteq F, \\ B \text{ valuation ring}}} B.$$

Proof. By Theorem 4.1.3, there exists a valuation ring B such that $A \subseteq B \subseteq F$. Since any valuation ring is integrally closed, we have that the LHS is a subset of the RHS.

On the other hand, if $f \in F$ is not integral over A , then $f \notin A[f^{-1}]$. In particular $f^{-1} \notin A[f^{-1}]^\times$, so there exists $\mathfrak{m} \in \text{Max } A[f^{-1}]$ such that $f^{-1} \in \mathfrak{m}$. Consider the ring homomorphism $\theta : A[f^{-1}] \rightarrow A[f^{-1}]/\mathfrak{m} \hookrightarrow \Omega$ where Ω is the algebraic closure of $A[f^{-1}]/\mathfrak{m}$. Then by Theorem 4.1.3 there exists a valuation ring B such that $A[f^{-1}] \subseteq B \subseteq F$ with $\hat{\theta} : B \rightarrow \Omega$ extending θ . Since $f^{-1} \in \mathfrak{m} \subseteq \ker \hat{\theta}$, we deduce $f^{-1} \in B$ is not invertible, so then $f \notin B$ and we are done. $\square$

Theorem 4.1.5 (Extension of morphism) *Let $A \subseteq B$ be integral domains, B be a finitely generated A -algebra, and $b_0 \in B \setminus \{0\}$. Then for some $a_0 \in A$, if $\varphi \in \text{Hom}_{\text{Ring}}(A, \Omega)$ satisfies $\varphi(a_0) \neq 0$, then there exists $\theta \in \text{Hom}_{\text{Alg}_A}(B, \Omega)$ such that $\theta(b_0) \neq 0$ and $\theta|_A = \varphi$.*

Proof. We proceed by induction on the number of generator of B as an A -algebra. Hence it suffices to prove the case $B = A[\beta]$.

Case 1: β is transcendental over A . Write $b_0 = \sum_{i=0}^n c_i \beta^i$ and let $a_0 := c_n$. Then if $\varphi(a_0) \neq 0$, $f(x) := \sum_{i=0}^n \varphi(c_i) x^i \in \Omega[x]$ is a non-zero polynomial. Pick any $\omega_0 \in \Omega$ such that $f(\omega_0) \neq 0$ and define $\theta(\beta) := \omega_0$. Notice that $\theta \in \text{Hom}_{\text{Alg}_A}(B, \Omega)$ since $B = A[\beta] \simeq A[x]$ is a free A -algebra, so it suffices to specify $\theta(\beta)$. As a result, we have that $\theta|_A = \varphi$ and clearly

$$\theta(b_0) = \theta\left(\sum_i c_i \beta^i\right) = \sum_i \varphi(c_i) \omega_0^i \neq 0 \in \Omega.$$

Case 2: β is algebraic over A . Then $A[\beta]/A$ is an algebraic extension, meaning every element in $A[\beta]$ satisfies a non-zero (not necessarily monic) polynomial in $A[x]$. Hence $0 = \sum_{i=0}^n a'_i \beta^i = \sum_{j=0}^m a''_j b_0^{-j}$ for some $a'_i, a''_j \in A$. Define $a_0 := a'_n a''_m$. Then if $\varphi(a'_n a''_m) \neq 0$, by the universal property of localization there is a map $\hat{\varphi} : A' := A[1/a'_n a''_m] \rightarrow \Omega$ extending φ . Then by Theorem 4.1.3, there is a valuation ring C with $A' \subseteq C \subseteq Q(A')$ and a map $\tilde{\varphi} : C \rightarrow \Omega$ extending $\hat{\varphi}$. Notice that $\beta, b_0^{-1} \in C$ since they are integral over C and C is integrally closed; moreover, $b_0 \in A[\beta] \subseteq C$. As a result, we deduce $b_0 \in C^\times$ and therefore $b_0 \notin \ker \tilde{\varphi}$. Finally, this implies $\tilde{\varphi}|_B \in \text{Hom}_{\text{Alg}_A}(B, \Omega)$ satisfies $\tilde{\varphi}|_B(b_0) \neq 0$, so we are done. $\square$

- In the setting of *Case 2*, we have that $\sum_{j=0}^m a''_j b_0^{-j} = 0$ for some $a''_j \in A$ because b_0 being algebraic over A implies that $ab_0 \in A[b_0^{-1}]$ for some $a \in A$. Now since $ab_0 \in A[\beta]$ satisfies $f(ab_0) = 0$ for some $f \in A[x]$, we can find $\tilde{f} \in A[x]$ such that $\tilde{f}(b_0^{-1}) = f(ab_0) = 0$.

This technique is a special case of the following: if B/A is an integral extension of domains, then $Q(B)/A$ is algebraic; i.e. every $\beta \in Q(B)$ satisfies a non-trivial polynomial in $A[x]$.

- If $\varphi \in \text{Hom}_{\text{Ring}}(A, \Omega)$ is an embedding, then $\theta \in \text{Hom}_{\text{Alg}_A}(B, \Omega)$ always exists for any $b_0 \in B \setminus \{0\}$.

Theorem 4.1.6 (Nullstellensatz, I) *Let k be a field, B be a finitely generated k -algebra. If B is a field, then B/k is a finite extension.*

Proof. Let $\varphi : k \hookrightarrow \Omega := \bar{k}^{\text{alg}}$ be an embedding and $b_0 := 1 \in B$. Then for any $a_0 \in k^\times$, $\varphi(a_0) \neq 0$. Hence by Theorem 4.1.5 there exists $\hat{\varphi} : B \rightarrow \Omega$ with $\hat{\varphi}(b_0) \neq 0$. But B is a field, so $\hat{\varphi}$ is in fact an embedding and therefore B/k is an algebraic extension. Since $B = k[b_1, \dots, b_n]$ is a finitely generated k -algebra, by the tower formula we deduce B/k is a finite extension, so we are done. $\square$

Theorem 4.1.7 (Nullstellensatz, II) *Let k be algebraically closed and $A \simeq k[x_1, \dots, x_n]/\mathfrak{a}$ be a finitely generated k -algebra. Then $\text{Max } A = \{\mathfrak{m}_p/\mathfrak{a} : p \in V(\mathfrak{a})\}$ where $\mathfrak{m}_p = \langle x_1 - p_1, \dots, x_n - p_n \rangle$; in particular, there is a bijection between $\text{Max } A$ and $V(\mathfrak{a})$.*

- Here $V(\mathfrak{a}) := \{p \in \mathbb{A}_k^n : \forall f \in \mathfrak{a}, f(p) = 0\}$.

Proof. Let $p \in V(\mathfrak{a})$ and define $e_p : k[x_1, \dots, x_n] \rightarrow k$ to be evaluation at p . Then $\ker e_p \supseteq \mathfrak{m}_p$ and $\ker e_p \supseteq \mathfrak{a}$. Suppose $f \in \ker e_p$. Consider the Taylor expansion of f around p :

$$f(x) = f(p) + \sum \partial_i f(p)(x_i - p_i) + \text{terms with higher order},$$

which implies that $f \in \mathfrak{m}_p$ since $f(p) = 0$. Hence $\ker e_p = \mathfrak{m}_p$ and $\mathfrak{m}_p \in \text{Max } k[x_1, \dots, x_n]$ by the first isomorphism theorem, so then $\mathfrak{m}_p/\mathfrak{a} \in \text{Max } A$.

Conversely, let $\mathfrak{m} \in \text{Max } A$. Then there exists $\tilde{\mathfrak{m}} \in \text{Max } k[x_1, \dots, x_n]$ such that $\mathfrak{m} = \tilde{\mathfrak{m}}/\mathfrak{a}$. Define $B := A/\mathfrak{m}$. Then B is a field and finitely generated k -algebra. Hence by Theorem 4.1.6 B/k is a finite extension; but k is algebraically closed, so then $B = k$. Now since $B \simeq k[x_1, \dots, x_n]/\tilde{\mathfrak{m}}$, for every i there exists $p_i \in k$ such that $p_i + \tilde{\mathfrak{m}} = x_i + \tilde{\mathfrak{m}}$, i.e. $x_i - p_i \in \tilde{\mathfrak{m}}$. As a result, $\tilde{\mathfrak{m}} \supseteq \mathfrak{m}_p$, which implies $\tilde{\mathfrak{m}} = \mathfrak{m}_p$ because $\mathfrak{m}_p$ is maximal. Hence $\mathfrak{m} = \mathfrak{m}_p/\mathfrak{a}$ and we are done. $\square$

- In fact, this proof shows that for an algebraically closed field k , every maximal ideal in $k[x_1, \dots, x_n]$ is of the form $\langle x_1 - p_1, \dots, x_n - p_n \rangle$ for some $(p_1, \dots, p_n) \in \mathbb{A}_k^n$.

Theorem 4.1.8 (Nullstellensatz, III) *Let k be algebraically closed and $\mathfrak{a} \triangleleft k[x_1, \dots, x_n]$ be a proper ideal. Then $V(\mathfrak{a}) \neq \emptyset$.*

Proof. This is immediate from Theorem 4.1.7. $\square$

Theorem 4.1.9 (Nullstellensatz, IV) *Let k be algebraically closed and $\mathfrak{a} \trianglelefteq k[x_1, \dots, x_n]$. Then $\sqrt{\mathfrak{a}} = I(V(\mathfrak{a}))$.*

- We start by the following observation: If D is an integral domain and $d \in D \setminus \{0\}$, then $D[1/d] \simeq D[x]/\langle 1 - dx \rangle$. Indeed, consider the evaluation $e_{\frac{1}{d}} : D[x] \rightarrow D[1/d]$, which is clearly a surjective ring homomorphism. Notice that its kernel is precisely the ideal generated by the minimal polynomial of $\frac{1}{d}$ in $Q(D)[x]$, i.e. $\langle 1 - dx \rangle$, so by the first isomorphism theorem we are done.

Proof. WLOG we may assume $\mathfrak{a}$ is proper; otherwise $V(A) = \emptyset$ and so $I(V(A)) = A$. If $f \in \sqrt{\mathfrak{a}}$, then $f^m \in \mathfrak{a}$ for some $m \in \mathbb{N}$. Hence $f^m|_{V(\mathfrak{a})} = 0 \in k$; but k is a field, so $f|_{V(\mathfrak{a})} = 0$ and $f \in I(V(\mathfrak{a}))$.

Conversely, suppose to the contrary there exists $f \in I(V(\mathfrak{a})) \setminus \sqrt{\mathfrak{a}}$. Using the observation we mention in remarks, we have

$$S_f^{-1}(k[x_1, \dots, x_n]) \simeq \frac{k[x_1, \dots, x_{n+1}]}{\langle 1 - x_{n+1}f \rangle}.$$

Now since $S_f \cap \mathfrak{a} = \emptyset$, $S_f^{-1}\mathfrak{a}$ is a proper ideal; more precisely:

$$S_f^{-1}k[x_1, \dots, x_n]/S_f^{-1}\mathfrak{a} \simeq \frac{k[x_1, \dots, x_{n+1}]}{\langle 1 - x_{n+1}f \rangle + \mathfrak{a}[x_{n+1}]},$$

where $\mathfrak{a}[x_{n+1}]$ is the ideal generated by $\mathfrak{a}$ in $k[x_1, \dots, x_{n+1}]$. So by Theorem 4.1.8, there exists $(p, p_{n+1}) \in V(\langle 1 - x_{n+1}f \rangle + \mathfrak{a}[x_{n+1}])$, which implies that $1 - p_{n+1}f(p) = 0$ and $p \in V(\mathfrak{a})$. But then $f(p) = 0$ since $f \in I(V(\mathfrak{a}))$, which is a contradiction and so we are done. $\square$

Proposition 4.1.10 (Equivalency of four Nullstellensatz) *The four versions of Nullstellensatz above are all equivalent.*

Proof. (IV $\implies$ II) Let $\mathfrak{m} \in \text{Max } A$. It suffices to show that $\mathfrak{m}$ corresponds to a unique $p \in V(\mathfrak{a})$. Notice that $\mathfrak{m} = \tilde{\mathfrak{m}}/\mathfrak{a}$ for some $\tilde{\mathfrak{m}} \in \text{Max } k[x_1, \dots, x_n]$, and $\tilde{\mathfrak{m}} = \sqrt{\tilde{\mathfrak{m}}} = I(V(\tilde{\mathfrak{m}}))$ by IV. We claim that $V(\tilde{\mathfrak{m}})$ consists of a point $p \in V(\mathfrak{a})$. Indeed, if $|V(\tilde{\mathfrak{m}})| \geq 2$, pick $q \in V(\tilde{\mathfrak{m}})$ and notice that $I(q) \supseteq I(V(\tilde{\mathfrak{m}})) = \tilde{\mathfrak{m}}$ is a proper ideal, so $I(q) = \tilde{\mathfrak{m}}$. But then $\{q\} = V(I(q)) = V(\tilde{\mathfrak{m}})$, a contradiction. Hence $V(\tilde{\mathfrak{m}})$ is a point; now since $\tilde{\mathfrak{m}} \supseteq \mathfrak{a}$, we deduce $V(\tilde{\mathfrak{m}}) \subseteq V(\mathfrak{a})$. So there is a bijection between $\text{Max } A$ and $V(\mathfrak{a})$, where each $p \in V(\mathfrak{a})$ gives rise to $\mathfrak{m}_p/\mathfrak{a} \in \text{Max } A$.

(II $\implies$ I) First notice that $\bar{k}[x_1, \dots, x_n]/k[x_1, \dots, x_n]$ is an integral extension; indeed, $\bar{k}/k$ is algebraic and each x_i is trivially integral over $k[x_1, \dots, x_n]$. Suppose $B \simeq k[x_1, \dots, x_n]/\mathfrak{m}$ is a finitely generated k -algebra and a field. Define $B^e = \bar{k}[x_1, \dots, x_n]/\mathfrak{m}^e$. Then B^e is a finitely generated $\bar{k}$ -algebra; in fact, it is also a field by Lemma 3.3.1. Now by II, there is a bijection between $\text{Max } B^e$ and $\{\tilde{\mathfrak{m}}_p/\mathfrak{m}^e : p \in V(\mathfrak{m}^e)\}$. But $\mathfrak{m}^e$ is a maximal ideal, so $\tilde{\mathfrak{m}}_p/\mathfrak{m}^e = 0$ and therefore $\mathfrak{m}^e = \tilde{\mathfrak{m}}_p$ for some $p \in \bar{k}$. As a result, $B^e \simeq \bar{k}[x_1, \dots, x_n]/\tilde{\mathfrak{m}}_p \simeq \bar{k}$. Using the embedding $B \hookrightarrow B^e \simeq \bar{k}$, we deduce B/k is an algebraic extension. Finally, since B is finitely generated as a k -algebra, B/k is a finite extension and we are done. $\square$

Finitely generated k -algebra and normalization

Definition 4.2.1 (Jacobson ring) We say a ring A is a *Jacobson ring* if for every $\mathfrak{p} \in \text{Spec } A$,

$$\mathfrak{p} = \bigcap_{\mathfrak{p} \subseteq \mathfrak{m} \in \text{Max } A} \mathfrak{m}.$$

In particular, we have that $J(A) = \text{Nil } A$.

Theorem 4.2.2 (Finitely generated k -algebra and Jacobson ring) *Let k be algebraically closed and A be finitely generated k -algebra. Then A is a Jacobson ring.*

Proof. Suppose $A \simeq k[x_1, \dots, x_n]/\mathfrak{a}$. Notice that there is a bijection between $\text{Spec } A$ and $V(\mathfrak{a})$; also there is bijection between $\text{Max } A$ and $\text{Max } k[x_1, \dots, x_n] \cap V(\mathfrak{a})$. Hence it suffices to show that $A' := k[x_1, \dots, x_n]$ is a Jacobson ring. But by Nullstellensatz IV and II, for every $\mathfrak{p} \in \text{Spec } A'$ we have

$$\mathfrak{p} = \sqrt{\mathfrak{p}} = I(V(\mathfrak{p})) = \bigcap_{p \in V(\mathfrak{p})} \mathfrak{m}_p = \bigcap_{\mathfrak{p} \subseteq \mathfrak{m} \in \text{Max } A'} \mathfrak{m}.$$

$\square$

Corollary 4.2.3 (Jacobson ring and Jacobson ideal) *Let k be algebraically closed and A be a finitely generated k -algebra. Then $J(A)^n = 0$ for some $n \in \mathbb{N}^{>0}$.*

Proof. By Theorem 4.2.2, A is a Jacobson ring, so then $\text{Nil } A = J(A)$. But A is also Noetherian, so $(\text{Nil } A)^n = 0$ for some n and we are done. $\square$

- This result holds for any Noetherian Jacobson ring.

Lemma 4.2.4 (Some automorphisms of $k[x_1, \dots, x_n]$) Let $M \in \mathbb{N}^{>0}$ and define

$$\varphi_M^\pm : k[x_1, \dots, x_n] \rightarrow k[x_1, \dots, x_n], \quad \varphi_M^\pm(x_i) = \begin{cases} x_i \pm x_n^{M^i} & \text{if } i < n, \\ x_n & \text{if } i = n. \end{cases}$$

Then $\varphi_M^+ \circ \varphi_M^- = \varphi_M^- \circ \varphi_M^+ = \text{id}_{k[x_1, \dots, x_n]}$; so $\varphi_M^\pm \in \text{Aut}_k(k[x_1, \dots, x_n])$.

Proof. This is clear. □

Corollary 4.2.5 (Normalization of polynomial) Let $f \in k[x_1, \dots, x_n]$ be a non-zero polynomial. Then there exists $\varphi \in \text{Aut}_k(k[x_1, \dots, x_n])$ such that the leading coefficient of $\varphi(f) \in (k[x_1, \dots, x_{n-1}])[x_n]$ is in $k^\times$.

Proof. Choose $M > \deg f$. Then by Lemma 4.2.4,

$$\begin{aligned} \varphi(f) &= \sum a_{i_1, \dots, i_n} x_n^{i_n} (x_{n-1} + x_n^{M^{n-1}})^{i_{n-1}} \dots (x_1 + x_n^M)^{i_1}, \\ &= \sum a_{i_1, \dots, i_n} x_n^{i_n + i_{n-1}M^{n-1} + \dots + i_1M} + \text{terms with lower degree.} \end{aligned}$$

Since $0 \leq i_j < M$, $i_n + i_{n-1}M^{n-1} + \dots + i_1M$ uniquely determines $i_1, \dots, i_n$, so all these terms are distinct and we are done. □

Theorem 4.2.6 (Noether's normalization lemma) Let k be a field, A be a finitely generated k -algebra. Then there exist $\xi_1, \dots, \xi_n \in A$ such that

- (1). $\xi_1, \dots, \xi_n$ are algebraically independent over k .
- (2). $A/k[\xi_1, \dots, \xi_n]$ is an integral extension.

Proof. We proceed by induction on the number of generators of A . The base case $A = k[\alpha]$ is clear: if α is algebraic over A , then A/k is an integral extension; if α is transcendental, then $A/k[\alpha]$ is trivially integral. Now for the inductive case $A = k[\alpha_1, \dots, \alpha_n]$, we may assume WLOG that the α_i 's are algebraically dependent; otherwise there is nothing to show. So there exists non-zero $f \in k[x_1, \dots, x_n]$ such that $f(\alpha_1, \dots, \alpha_n) = 0$. By Corollary 4.2.5, pick $\varphi \in \text{Aut}_k(k[x_1, \dots, x_n])$ such that the leading coefficient of $\varphi(f) \in (k[x_1, \dots, x_{n-1}])[x_n]$ is in $k^\times$. Notice that φ induces an k -algebra isomorphism

$$\bar{\varphi} : k[\alpha_1, \dots, \alpha_n] \rightarrow k[\beta_1, \dots, \beta_n], \quad g(\alpha_1, \dots, \alpha_n) \mapsto \varphi(g)(\alpha_1, \dots, \alpha_n).$$

Now by the inductive hypothesis, there exist $t_1, \dots, t_{n-1} \in k[\beta_1, \dots, \beta_{n-1}]$ such that the t_i 's are algebraically independent over k and $k[\beta_1, \dots, \beta_{n-1}]/k[t_1, \dots, t_{n-1}]$ is integral. But since β_n is integral over $k[\beta_1, \dots, \beta_{n-1}]$, we deduce $k[\beta_1, \dots, \beta_n]/k[t_1, \dots, t_{n-1}]$ is an integral extension, so we are done. □

Corollary 4.2.7 (Dimensions of finitely generated k -algebras) *Let A be a finitely generated k -algebra. Then there exists an integral embedding $k[x_1, \dots, x_n] \hookrightarrow A$; hence $\dim A = \dim k[x_1, \dots, x_n] = n$.*

- This actually implies that the cardinality n in Theorem 4.2.6 is unique!

Proof. That $\dim A = \dim k[x_1, \dots, x_n]$ follows immediately from Theorem 4.2.6 and Theorem 3.3.7, so it suffices to show that $\dim k[x_1, \dots, x_n] = n$. It is clear that $\dim k[x_1, \dots, x_n] \geq n$ given the following chain of prime ideals:

$$0 \subsetneq \langle x_1 \rangle \subsetneq \langle x_1, x_2 \rangle \subsetneq \cdots \subsetneq \langle x_1, \dots, x_n \rangle.$$

For the converse, we proceed by induction on n . The base case is clear since $k[x]$ is a PID. Now let $0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m$ be a chain of maximal length in $k[x_1, \dots, x_n]$; indeed, since $k[x_1, \dots, x_n]$ is Noetherian, its dimension must be finite. Pick $0 \neq f \in \mathfrak{p}_1$ and apply Corollary 4.2.5 to obtain monic $\varphi(f)$ in $(k[x_1, \dots, x_{n-1}])[x_n]$. Passing to the quotient ring, we deduce that $\bar{x}_n \in k[x_1, \dots, x_n]/\langle f \rangle$ is integral over $k[\bar{x}_1, \dots, \bar{x}_{n-1}]$. As a result, by Theorem 3.3.7 and the inductive hypothesis, we have

$$m - 1 = \dim k[x_1, \dots, x_n]/\langle f \rangle = \dim k[\bar{x}_1, \dots, \bar{x}_{n-1}] \leq n - 1,$$

so $m \leq n$ and we are done. □

Noetherian Rings of Special Dimensions

Composition series and Artinian modules

Lemma 5.1.1 (Noetherian rings of dimension 0) *Let A be a Noetherian ring. Then $\dim A = 0 \iff$ there exist (not necessarily distinct!) $\mathfrak{m}_i \in \text{Max } A$ such that $\mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$; in this case $\text{Spec } A = \{\mathfrak{m}_1, \dots, \mathfrak{m}_n\}$.*

Proof. ($\implies$) Since A is Noetherian, $0 \triangleleft A$ is decomposable, say $0 = \bigcap_{i=1}^n \mathfrak{q}_i$. Since $\dim A = 0$, we have that $\text{Spec } A = \text{Max } A$; so there exist $\mathfrak{m}_i \in \text{Max } A$ such that $\sqrt{\mathfrak{q}_i} = \mathfrak{m}_i$. But every $\mathfrak{m}_i \in \text{Ideal } A$ is finitely generated, so there exists $l \in \mathbb{N}^{>0}$ such that $\mathfrak{m}_1^l \cdots \mathfrak{m}_n^l \subseteq \bigcap_i \mathfrak{q}_i = 0$.

($\impliedby$) It suffices to show that $\text{Spec } A = \text{Max } A$. Let $\mathfrak{p} \in \text{Spec } A$. Then $\mathfrak{p} \supseteq 0 = \mathfrak{m}_1 \cdots \mathfrak{m}_n$ implies that $\mathfrak{p} \supseteq \mathfrak{m}_i$ for some i ; by maximality we deduce $\mathfrak{p} = \mathfrak{m}_i \in \text{Max } A$, so we are done. $\square$

Corollary 5.1.2 (Composition series and Noetherian rings) *Let A be a Noetherian ring of dimension 0. Then there exist a chain of ideals $0 = \mathfrak{a}_n \subseteq \mathfrak{a}_{n-1} \subseteq \cdots \subseteq \mathfrak{a}_0 = A$ such that $\mathfrak{a}_i/\mathfrak{a}_{i+1}$ is a simple A -module.*

Proof. By Lemma 5.1.1, there exist $\mathfrak{m}_i \in \text{Max } A$ such that $\mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$. Define $\tilde{\mathfrak{a}}_i := \mathfrak{m}_1 \cdots \mathfrak{m}_i$. Notice that $\tilde{\mathfrak{a}}_i/\tilde{\mathfrak{a}}_{i+1} = \tilde{\mathfrak{a}}_i/\tilde{\mathfrak{a}}_i \mathfrak{m}_{i+1}$ is a vector space over $A/\mathfrak{m}_{i+1}$; and it is also finite-dimensional since $\tilde{\mathfrak{a}}_i$ is finitely generated. As a result, it has a “full flag”; that means there exists a sequence $\tilde{\mathfrak{a}}_i = \mathfrak{a}_{i,1} \supsetneq \mathfrak{a}_{i,2} \supsetneq \cdots \supsetneq \mathfrak{a}_{i,d_i} = \tilde{\mathfrak{a}}_{i+1}$ where each $\mathfrak{a}_{i,j}/\mathfrak{a}_{i,j+1}$ is a one-dimensional $A/\mathfrak{m}_{i+1}$ -vector space and hence a simple A -module, so we are done. $\square$

Definition 5.1.3 (Composition series of modules) Let M be an A -module. We say $0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ is a *composition series* of M if every $M_i \subseteq M$ is a submodule and M_i/M_{i+1} is a simple A -module. Note that the composition series may not exist.

Proposition 5.1.4 (Length of composition series) *Let M be an A -module with a composition series of length n . Then any composition series of M has length n and any series can be extended into a composition series.*

Proof. Let $l(M)$ denote the length of the shortest composition series of M .

Step 1. We show that if $N \leq M$ is a proper submodule, then $l(N) < l(M)$. Let $0 = M_0 \subsetneq \cdots \subsetneq M_n = M$ be a composition series of M and define $N_i := M_i \cap N$. Notice that for every i , we have $N_{i+1}/N_i \hookrightarrow M_{i+1}/M_i$. But M_{i+1}/M_i is a simple A -module, so either $N_{i+1}/N_i = 0$ or $N_{i+1}/N_i \simeq M_{i+1}/M_i$. Now consider the chain $0 = N_0 \subseteq \cdots \subseteq N_n = N$. We claim that $N_{i+1}/N_i = 0$ for at least one i ; indeed, otherwise we will have $N = M$ if we trace up the chain, contradicting that $N \leq M$ is a proper submodule. Hence $l(N) < l(M)$.

Step 2. We show that every composition series of M must have length n . Notice that if $0 = M'_0 \subsetneq \cdots \subsetneq M'_k = M$ be a composition series of M , then by *Step 1* we deduce $0 = l(M'_0) < \cdots < l(M'_k) = l(M)$, which implies that $k \leq l(M)$. Hence $k = l(M)$ by definition.

Step 3. Suppose $0 = K_0 \subsetneq \cdots \subsetneq K_m = M$ is chain. Then $0 = l(K_0) < \cdots < l(K_m) = n$ implies that $m \leq n$. If $\{K_i\}$ does not form a composition series, then it can be enlarged; but we cannot enlarge it more than $l(M)$ times, which implies we must get a composition series by the end. $\square$

Definition 5.1.5 (Artinian modules) We say an A -module M is *Artinian* if it satisfies the descending chain condition (DCC): if $M_1 \supseteq M_2 \supseteq \cdots$ is a descending chain of submodules, then $M_n = M_{n+1} = \cdots$ for some $n \in \mathbb{N}^{>0}$.

Lemma 5.1.6 (Basic properties of Artinian modules) *Let M be an A -module. Then*

- (1). M is Artinian $\iff$ any non-empty family of submodules of M admits a minimal element.
- (2). M is Artinian $\iff$ for any submodule $N \leq M$, both N and M/N are Artinian.
- (3). If $M_1, \dots, M_n$ are Artinian A -modules, then $\bigoplus_{i=1}^n M_i$ is Artinian.

Proof. These are similar to the Noetherian case. $\square$

Lemma 5.1.7 (Noetherian vector space) *Let V be an F -vector space. Then V is Noetherian $\iff \dim V < \infty \iff V$ is Artinian.*

Proof. V is finite-dimensional $\iff$ every chain of subspaces of V has finite length. $\square$

Proposition 5.1.8 (Modules with composition series) *Let M be an A -module. Then $l(M) < \infty \iff M$ is Artinian and Noetherian.*

Proof. ($\implies$) If the length of any chain of submodules in M is bounded by $l(M) < \infty$, then M is Noetherian and Artinian.

($\impliedby$) Let $M_0 = 0$. We construct a composition series of M inductively as follows: for every $i > 0$, define $\Sigma_i := \{N \leq M : M_{i-1} \leq N \leq M\}$. Notice that $\Sigma_i \ni M_{i-1}$ is non-empty; now since M is Artinian, we can pick a minimal element $M_i \in \Sigma_i$. We claim that every M_i/M_{i-1} is a simple A -module; otherwise, M_i will not be minimal in Σ_i . Finally, since M is Noetherian, the ascending chain $0 = M_0 \subseteq M_1 \subseteq \cdots$ has finite length, i.e. $M_n = M$ for some $n \in \mathbb{N}^{>0}$. So we have produced a composition series and we are done. $\square$

Artinian rings

Lemma 5.2.1 (Noetherian rings of dimension 0, again) *Suppose there exist $\mathfrak{m}_i \in \text{Max } A$ such that $\mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$. Then A is Noetherian $\iff A$ is Artinian.*

Proof. ($\implies$) By Corollary 5.1.2, A admits a composition series, so by Proposition 5.1.8 it is Artinian.

($\impliedby$) Define $\tilde{\mathfrak{a}}_i := \mathfrak{m}_1 \cdots \mathfrak{m}_i$. Then $\tilde{\mathfrak{a}}_i/\tilde{\mathfrak{a}}_{i+1} = \tilde{\mathfrak{a}}_i/\tilde{\mathfrak{a}}_i \mathfrak{m}_{i+1}$ is an Artinian $A/\mathfrak{m}_{i+1}$ -vector space since A is Artinian, which implies that it is also finite-dimensional. As a result, every $\tilde{\mathfrak{a}}_i/\tilde{\mathfrak{a}}_{i+1}$ admits a full flag, which implies that $l(A) < \infty$ and so by Proposition 5.1.8 again we are done. $\square$

- In fact, A is Artinian $\iff A$ is a Noetherian ring of dimension 0. This lemma proves one direction; the other one we will prove below.

Lemma 5.2.2 (Dimension of Artinian rings) *Let A be an Artinian ring. Then $\dim A = 0$.*

Proof. It suffices to show that $\operatorname{Spec} A = \operatorname{Max} A$. Let $\mathfrak{p} \in \operatorname{Spec} A$. We claim that $A/\mathfrak{p}$ is a field. Indeed, pick any $x \in (A/\mathfrak{p}) \setminus \{0\}$ and consider the chain $\langle x \rangle \supseteq \langle x^2 \rangle \supseteq \cdots$. But A is Artinian and so is $A/\mathfrak{p}$, which implies that $\langle x^n \rangle = \langle x^{n+1} \rangle$ for some $n \in \mathbb{N}^{>0}$; hence $x^n = x^{n+1}y$ for some $y \in (A/\mathfrak{p})^\times$. Since $A/\mathfrak{p}$ is an integral domain, by the cancellation property we deduce $1 = xy$, so we are done. $\square$

Lemma 5.2.3 (Spectrum of Artinian rings) *Let A be an Artinian ring. Then $\operatorname{Spec} A$ is finite and discrete.*

Proof. By Lemma 5.2.2 $\operatorname{Spec} A = \operatorname{Max} A$, so $V(\mathfrak{m}) = \{\mathfrak{m}\}$ for every $\mathfrak{m} \in \operatorname{Spec} A$ and hence $\operatorname{Spec} A$ is discrete. Now consider $\Sigma := \{\bigcap_{i=1}^n \mathfrak{m}_i : \mathfrak{m}_i \in \operatorname{Spec} A, n \in \mathbb{N}^{>0}\}$, which contains a minimal element since A is Artinian, say $\bigcap_{i=1}^n \mathfrak{m}'_i$. Hence for every $\mathfrak{m} \in \operatorname{Spec} A$ we have $\mathfrak{m} \cap \bigcap_i \mathfrak{m}'_i = \bigcap_i \mathfrak{m}'_i$. In particular, $\mathfrak{m} \supseteq \bigcap_i \mathfrak{m}'_i \supseteq \mathfrak{m}'_1 \cdots \mathfrak{m}'_n$. Since $\mathfrak{m}$ is prime, we deduce $\mathfrak{m} \supseteq \mathfrak{m}'_i$ for some i and therefore $\mathfrak{m} = \mathfrak{m}'_i$, so we are done. $\square$

Proposition 5.2.4 (Jacobson ideal of Artinian rings) *Let A be an Artinian ring. Then $J(A) = \operatorname{Nil} A$ and $J(A)^n = 0$ for some $n \in \mathbb{N}^{>0}$.*

Proof. That $J(A) = \operatorname{Nil} A$ follows immediately from Lemma 5.2.2. For the second part, suppose to the contrary $J(A)^n \neq 0$ for any $n \in \mathbb{N}^{>0}$. Consider the chain $J(A) \supseteq J(A)^2 \supseteq \cdots$. Then since A is Artinian, there exists n_0 such that $J_0 := J(A)^{n_0} = J(A)^{n_0+1} = \cdots$. Define $\Sigma = \{\mathfrak{a} \triangleleft A : \mathfrak{a}J_0 \neq 0\}$. It is clear that Σ is non-empty, so it admits a minimal element $\mathfrak{a}_0$. Since $\mathfrak{a}_0J_0 \neq 0$, there exists $x \in \mathfrak{a}_0$ such that $xJ_0 \neq 0$, so then $\langle x \rangle J_0 \neq 0$ and $\langle x \rangle \in \Sigma$, which implies that $\langle x \rangle = \mathfrak{a}_0$ by minimality. But notice that $xJ_0 = xJ(A)J_0 = J(A)xJ_0$. Hence by Nakayama's Lemma we deduce $xJ_0 = 0$, which is a contradiction and so we are done. $\square$

Theorem 5.2.5 (Artinian and Noetherian rings) *Let A be a ring. Then A is Artinian $\iff A$ is Noetherian and $\dim A = 0$.*

Proof. ($\Leftarrow$) is clear by Lemma 5.2.1.

($\Rightarrow$) By Lemma 5.2.3 and Proposition 5.2.4, we have that $|\operatorname{Max} A| = m < \infty$ and $J(A)^n = 0$ for some $n \in \mathbb{N}^{>0}$. But $J(A) = \bigcap_{i=1}^m \mathfrak{m}_i$ implies that $\mathfrak{m}_1^n \cdots \mathfrak{m}_m^n = 0$, so by Lemma 5.2.1, we are done. $\square$

Theorem 5.2.6 (Structure of Artinian rings) *Every Artinian ring is a unique direct product of local Artinian rings up to isomorphism.*

- We will need the following remark: if $\mathfrak{a}_1, \mathfrak{a}_2 \trianglelefteq A$ are coprime then $\mathfrak{a}_1 \cap \mathfrak{a}_2 = \mathfrak{a}_1\mathfrak{a}_2$. Indeed, for every $z \in \mathfrak{a}_1 \cap \mathfrak{a}_2$, writing $z = z \cdot 1 = z(a_1 + a_2)$ for some $a_i \in \mathfrak{a}_i$ implies $z \in \mathfrak{a}_1\mathfrak{a}_2$.

Proof. (Existence). Let A be an Artinian ring. Then by Theorem 5.2.5, for some $\mathfrak{m}_i \in \text{Max } A$ and $l \in \mathbb{N}^{>0}$ we have that $\mathfrak{m}_1^l \cdots \mathfrak{m}_n^l = 0$. Notice that the $\mathfrak{m}_i^l$'s are pairwise coprime since $\sqrt{\mathfrak{m}_i^l + \mathfrak{m}_j^l} \supseteq \mathfrak{m}_i + \mathfrak{m}_j = A$. Hence by our remark we have that $\bigcap_{i=0}^n \mathfrak{m}_i^l = \prod_{i=0}^n \mathfrak{m}_i^l = 0$, so by the Chinese remainder theorem we get $A \simeq \bigoplus_{i=0}^n A/\mathfrak{m}_i^l$. We claim that each $A/\mathfrak{m}_i^l$ is a local ring; indeed, notice that

$$\text{Max } A/\mathfrak{m}_i^l = \text{Spec } A/\mathfrak{m}_i^l \leftrightarrow V(\mathfrak{m}_i^l) = \{\mathfrak{m}_i\}$$

since $\mathfrak{m}_i$ is maximal, so $\text{Max } A/\mathfrak{m}_i^l = \{\mathfrak{m}_i/\mathfrak{m}_i^l\}$ and the claim is proven.

(Uniqueness). Let $A = \prod_{i=0}^n A_i$ where each A_i is a local Artinian ring with $\text{Max } A_i = \{\mathfrak{m}_i\}$. Define $\tilde{\mathfrak{m}}_i := (\prod_{j \neq i} A_j) \times \mathfrak{m}_i$ and $\mathfrak{q}_i := (\prod_{j \neq i} A_j) \times \{0\} \subseteq \tilde{\mathfrak{m}}_i$. Notice that $\tilde{\mathfrak{m}}_i \in \text{Max } A$. Since $J(A_i) = \mathfrak{m}_i$ and A_i is Artinian, we have that $\mathfrak{m}_i^{n_i} = 0$ for some $n_i \in \mathbb{N}^{>0}$ by Proposition 5.2.4. Hence $\tilde{\mathfrak{m}}_i^{n_i} = \mathfrak{q}_i$, which implies that $\mathfrak{q}_i$ is $\tilde{\mathfrak{m}}_i$ -primary by Lemma 2.1.6. As a result, $0 = \bigcap_{i=1}^n \mathfrak{q}_i$ is a reduced primary decomposition with $\text{Ass}(0) = \{\tilde{\mathfrak{m}}_1, \dots, \tilde{\mathfrak{m}}_n\}$. But each $\tilde{\mathfrak{m}}_i \in \text{Ass}(0)$ is a minimal element, hence every $\mathfrak{q}_i \subseteq A$ is unique by Theorem 2.4.2 and so we are done. $\square$

Proposition 5.2.7 (Properties of local Artinian rings) *Let A be a local Artinian ring with $\text{Max } A = \{\mathfrak{m}\}$. Then TFAE:*

- (1). A is a PID;
- (2). $\mathfrak{m}$ is principal;
- (3). $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) \leq 1$.

Proof. Clearly (1) $\implies$ (2) $\implies$ (3).

(3) $\implies$ (1) Notice that $J(A) = \mathfrak{m}$. If $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = 0$, then $\mathfrak{m} = \mathfrak{m}^2 = J(A)\mathfrak{m}$. Hence by Nakayama's Lemma we deduce $\mathfrak{m} = 0$, so A is a field. If $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = 1$, then there exists $x \in \mathfrak{m}$ such that $\langle x \rangle/\mathfrak{m}^2 = \mathfrak{m}/\mathfrak{m}^2$, i.e. $\langle x \rangle \equiv \mathfrak{m} \pmod{\mathfrak{m}^2}$. As a result, $\mathfrak{m}/\langle x \rangle = \mathfrak{m}^2/\langle x \rangle\mathfrak{m} = J(A)(\mathfrak{m}/\langle x \rangle)$, so by Nakayama's Lemma again we deduce $\mathfrak{m} = \langle x \rangle$. Now pick any non-zero proper ideal $\mathfrak{a} \triangleleft A$. Since $J(A)^n = \mathfrak{m}^n = 0$ for some $n \in \mathbb{N}^{>0}$ by Proposition 5.2.4, we can pick the largest k such that $\mathfrak{m}^k \supseteq \mathfrak{a}$ and $\mathfrak{m}^{k+1} \not\supseteq \mathfrak{a}$. Let $a \in \mathfrak{a} \setminus \mathfrak{m}^{k+1}$. Then there exists $y \in A/\mathfrak{m} = A^\times$ such that $x^k y = a$; in particular, $x^k = ay^{-1} \in \mathfrak{a}$, which implies that $\mathfrak{a} \supseteq \langle x^k \rangle = \mathfrak{m}^k$. So $\mathfrak{a} = \langle x^k \rangle$ and we are done. $\square$

- If a ring satisfies these properties, its proper ideals are extremely rigid: they are all of the form $\langle x^n \rangle$ for some $n \in \mathbb{N}^{>0}$ where $\mathfrak{m} = \langle x \rangle$. Indeed, if $\mathfrak{a} \triangleleft A$ is proper then $\mathfrak{a} = \langle y \rangle$ for some $y \in \mathfrak{m}$; in particular we can write $y = a_1 x$ for some $a_1 \in A$. If $a_1 \in A \setminus \mathfrak{m} = A^\times$, then $\mathfrak{a} = \langle x \rangle$; if not, then $a_1 = a_2 x$ for some $a_2 \in A$ and $y = a_2 x^2$, and we can proceed inductively. But since x is prime, there can only be finitely many x in the factorization of y , so by the end we must have $y = a_n x^n$ for some unit a_n , and therefore $\mathfrak{a} = \langle x^n \rangle$. In fact, this characterization holds for any local PID.

Dedekind domains

Lemma 5.3.1 (Properties of local Noetherian domains) *Let D be a local Noetherian domain of dimension 1 and $\text{Max } D = \{\mathfrak{m}\}$. Then*

- (1). $\text{Spec } D = \{0, \mathfrak{m}\}$, where $\mathfrak{m}$ is a closed point and 0 is a dense point.
- (2). If $0 \neq \mathfrak{a} \triangleleft D$, then $\mathfrak{a}$ is $\mathfrak{m}$ -primary and $\mathfrak{a} \supseteq \mathfrak{m}^k$ for some $k \in \mathbb{N}^{>0}$.
- (3). $\mathfrak{m} \supsetneq \mathfrak{m}^2 \supsetneq \cdots$ and $\bigcap_{i=1}^\infty \mathfrak{m}^i = 0$.

Proof. (1) This is clear.

(2) Notice that $\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \in V(\sqrt{\mathfrak{a}})} \mathfrak{p}$; hence by (1) we deduce $\sqrt{\mathfrak{a}} = \mathfrak{m}$ and $\mathfrak{a}$ is $\mathfrak{m}$ -primary since $\mathfrak{m}$ is maximal. Now since D is Noetherian, $\mathfrak{m}$ is finitely generated, so there exists $k \in \mathbb{N}^{>0}$ such that $\mathfrak{m}^k \subseteq \mathfrak{a}$.

(3) Suppose to the contrary $\mathfrak{m}^l = \mathfrak{m}^{l+1}$ for some l . Since $J(D) = \mathfrak{m}$ and $\mathfrak{m}$ is finitely generated, by Nakayama's Lemma we deduce $\mathfrak{m}^l = 0$; but D is an integral domain, so then $\mathfrak{m} = 0$ which implies that D is a field, contradicting that $\dim D = 1$. Now we claim that $\bigcap_{i=1}^{\infty} \mathfrak{m}^i = 0$; if not, then by (2) there exists k such that $\mathfrak{m}^k \subseteq \bigcap_{i=1}^{\infty} \mathfrak{m}^i$. But that implies $\mathfrak{m}^k = \mathfrak{m}^{k+1}$ which is again a contradiction, so we are done. $\square$

Theorem 5.3.2 (Characterization of discrete valuation rings) *Let D be a local Noetherian domain of dimension 1 and $\text{Max } D = \{\mathfrak{m}\}$. Then TFAE:*

- (1). D is integrally closed;
- (2). $\mathfrak{m}$ is principal;
- (3). $\dim_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) = 1$ where $k(\mathfrak{m}) = D/\mathfrak{m}$.
- (4). For every $0 \neq \mathfrak{a} \triangleleft D$, there exists $k \in \mathbb{N}^{>0}$ such that $\mathfrak{a} = \mathfrak{m}^k$;
- (5). There exists $\pi \in D$ such that for every $0 \neq \mathfrak{a} \triangleleft D$, $\mathfrak{a} = \langle \pi^k \rangle$ for some $k \in \mathbb{N}^{>0}$;
- (6). There exists a valuation $v : Q(D) \rightarrow \mathbb{Z} \cup \{\infty\}$ such that $D = \{\alpha \in Q(D) : v(\alpha) \geq 0\}$. A valuation is a function satisfying:

- (i). $v(\alpha) = \infty \iff \alpha = 0$.
- (ii). $v(a_1 a_2) = v(a_1) + v(a_2)$.
- (iii). $v(a_1 + a_2) \geq \min\{v(a_1), v(a_2)\}$.

- An example of discrete valuation is the degree function $\deg : A(x) \rightarrow \mathbb{Z} \cup \{\infty\}$ where A is any commutative ring and $\deg(f/g) := \deg(f) - \deg(g)$.
- Because of the last property, such rings are called discrete valuation rings (DVR); clearly they satisfy the properties of general valuation rings.

Proof. (1) $\implies$ (2) Let $a \in \mathfrak{m} \setminus \{0\}$. By Lemma 5.3.1(2), there exists smallest $k \in \mathbb{N}^{>0}$ such that $\langle a \rangle \supseteq \mathfrak{m}^k$. Pick $b \in \mathfrak{m}^{k-1} \setminus \langle a \rangle$ and let $\alpha := \frac{b}{a}$. Notice that $\alpha \mathfrak{m} \subseteq D$ and $\alpha \notin D$. If $\alpha \mathfrak{m} = D$, then $\mathfrak{m} = D\alpha^{-1}$ where $\alpha^{-1} \in D$ (indeed, $\alpha \mathfrak{m} = D$ implies that $\alpha m = 1$ for some $m \in \mathfrak{m}$, so $m = \alpha^{-1} \in D$), so $\mathfrak{m}$ is principal. If not, then $\alpha \mathfrak{m} \triangleleft D$ is a proper ideal and so $\alpha \mathfrak{m} \subseteq \mathfrak{m}$. Since $\mathfrak{m}$ is finitely generated, by Proposition 1.4.1 there exist $d_i \in D$ such that $\alpha^n + d_{n-1}\alpha^{n-1} + \dots + d_0 = 0 \in \text{End}_D(\mathfrak{m})$. But $\mathfrak{m}$ is a faithful D -module since D is an integral domain, so $\alpha^n + d_{n-1}\alpha^{n-1} + \dots + d_0 = 0 \in D$, which implies that $\alpha \in D$ since D is integrally closed, which is a contradiction.

(2) $\implies$ (3) This is clear (Compared with Proposition 5.2.7(3), we get equality since D is not a field!).

(3) $\implies$ (4) If $0 \neq \mathfrak{a} \triangleleft D$, then $\mathfrak{a} \supseteq \mathfrak{m}^k$ for some k and $\overline{D} := D/\mathfrak{m}^k$ is a local Noetherian ring of dimension 0. So $\overline{D}$ is a local Artinian ring by Theorem 5.2.5 and

$$\dim_{k(\overline{\mathfrak{m}})}(\overline{\mathfrak{m}}/\overline{\mathfrak{m}}^2) = \dim_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) = 1$$

where $\overline{\mathfrak{m}} = \mathfrak{m}/\mathfrak{m}^k$. Hence by Proposition 5.2.7, $\overline{\mathfrak{m}} = \langle x \rangle$ is principal and $\mathfrak{a}/\mathfrak{m}^k = \langle x^i \rangle = \mathfrak{m}^i/\mathfrak{m}^k$ for some i . Since $\mathfrak{a}, \mathfrak{m} \supseteq \mathfrak{m}^k$, we deduce $\mathfrak{a} = \mathfrak{m}^i$.

(4) $\implies$ (5) Let $\pi \in \mathfrak{m} \setminus \mathfrak{m}^2$. Then $\langle \pi \rangle = \mathfrak{m}^i$ for some i ; in fact, we must have $i = 1$. Hence $\langle \pi^k \rangle = \mathfrak{m}^k$ for any $k \in \mathbb{N}^{>0}$.

(5) $\implies$ (6) Notice that for every $d \in D \setminus \{0\}$, there exists unique $v(d) \in \mathbb{N}^{>0}$ such that $\langle d \rangle = \langle \pi^{v(d)} \rangle$ by Lemma 5.3.1(3). This defines a function $v : D \rightarrow \mathbb{Z} \cup \{\infty\}$ with $v(0) := \infty$. Now we extend v to $Q(D)$ via $v(\frac{a}{b}) := v(a) - v(b)$. It is easy to check that v is a valuation and $D = \{\alpha \in Q(D) : v(\alpha) \geq 0\}$.

(6) $\implies$ (1) Clearly a DVR is a valuation ring and therefore integrally closed, so we are done. $\square$

Theorem/Definition 5.3.3 (Characterization of Dedekind domains) *Let A be a Noetherian domain of dimension 1. Then TFAE, in which case we say A is a Dedekind domain:*

- (1). A is integrally closed;
- (2). $A_{\mathfrak{m}}$ is a DVR for every $\mathfrak{m} \in \text{Max } A$;
- (3). $\mathfrak{q} \triangleleft A$ is a primary ideal $\iff \mathfrak{q} = \mathfrak{m}^n$ for some $\mathfrak{m} \in \text{Max } A$ and $n \in \mathbb{N}^{>0}$.

- This result implies that a Dedekind domain is locally a PID; but in general, it is not a PID. For how far a Dedekind domain is from being a PID, see 5.3.6 - 5.3.12.

Proof. (1) $\implies$ (2) If A is integrally closed, then $A_{\mathfrak{m}}$ is integrally closed by Theorem 3.2.3. It is clear that $A_{\mathfrak{m}}$ is a Noetherian domain, and $\dim A_{\mathfrak{m}} = 1$ since $\dim A = 1$ and $\mathfrak{m}$ is the unique maximal ideal, so $A_{\mathfrak{m}}$ is a DVR by Theorem 5.3.2.

(2) $\implies$ (3) If $\mathfrak{q} \triangleleft A$ is primary, then $\sqrt{\mathfrak{q}} = \mathfrak{m} \in \text{Max } A$ since $\dim A = 1$. Hence $\sqrt{\mathfrak{q}_{\mathfrak{m}}} = \mathfrak{m}A_{\mathfrak{m}}$ and so $\mathfrak{q}_{\mathfrak{m}}$ is $\mathfrak{m}A_{\mathfrak{m}}$ -primary since $\mathfrak{m}A_{\mathfrak{m}}$ is maximal. Now by (2) $A_{\mathfrak{m}}$ is a DVR, so $\mathfrak{q}_{\mathfrak{m}} = \mathfrak{m}^n A_{\mathfrak{m}}$ for some $n \in \mathbb{N}^{>0}$, which implies that $\mathfrak{q} = \mathfrak{m}^n$. Conversely, it is clear that $\mathfrak{m}^n$ is $\mathfrak{m}$ -primary.

(3) $\implies$ (1) Notice that if $\mathfrak{m} \in \text{Max } A$, every proper ideal $\tilde{\mathfrak{a}} \triangleleft A_{\mathfrak{m}}$ is $\mathfrak{m}A_{\mathfrak{m}}$ -primary since $A_{\mathfrak{m}}$ is a local ring. So then $\tilde{\mathfrak{a}} = \mathfrak{q}_{\mathfrak{m}}$ for some $\mathfrak{m}$ -primary $\mathfrak{q}$, where $\mathfrak{q} = \mathfrak{m}^n$ by (3). As a result, $\tilde{\mathfrak{a}} = \mathfrak{m}^n A_{\mathfrak{m}} = (\mathfrak{m}A_{\mathfrak{m}})^n$, so by Theorem 5.3.2(4) we deduce $A_{\mathfrak{m}}$ is integrally closed and therefore A is integrally closed, so we are done. $\square$

Corollary 5.3.4 (Ideals in Dedekind domains) *Let A be a Dedekind domain. Then any proper ideal $0 \neq \mathfrak{a} \triangleleft A$ is of the form $\mathfrak{a} = \prod_{\mathfrak{m} \in \text{Max } A} \mathfrak{m}^{v_{\mathfrak{m}}(\mathfrak{a})}$, where $v_{\mathfrak{m}}(\mathfrak{a}) = 0$ except for finitely many $\mathfrak{m}$.*

Proof. Since A is Noetherian and $\text{Spec } A = \text{Max } A$, every $0 \neq \mathfrak{a} \triangleleft A$ admits a unique primary decomposition by Theorem 2.4.2. Now by Theorem 5.3.3, every primary ideal in A is of the form $\mathfrak{m}^n$ for some $\mathfrak{m} \in \text{Max } A$; moreover, for any $n_i, n_j \in \mathbb{N}^{>0}$, $\mathfrak{m}_i^{n_i}$ and $\mathfrak{m}_j^{n_j}$ are coprime if $i \neq j$. Hence $\mathfrak{a} = \bigcap_{i=1}^k \mathfrak{m}_i^{n_i} = \prod_{i=1}^k \mathfrak{m}_i^{n_i}$, so we are done. $\square$

Corollary 5.3.5 ($\mathcal{O}_k$ is an Dedekind domain) *Let k be a number field. Then $\mathcal{O}_k$ is an Dedekind domain; so for every proper ideal $0 \neq \mathfrak{a} \triangleleft \mathcal{O}_k$, $\mathfrak{a} = \prod_{\mathfrak{m} \in \text{Max } \mathcal{O}_k} \mathfrak{m}^{v_{\mathfrak{m}}(\mathfrak{a})}$.*

Proof. Recall that $\mathcal{O}_k \simeq \mathbb{Z}^{[k:\mathbb{Q}]}$, so it is a Noetherian domain. Since $\mathcal{O}_k/\mathbb{Z}$ is an integral extension, $\dim \mathcal{O}_k = \dim \mathbb{Z} = 1$. Hence $\mathcal{O}_k$ is a Dedekind domain since it is also integrally closed, so we are done by Corollary 5.3.4. $\square$

Definition 5.3.6 (Frac A and Prin A) *Let A be an integral domain and $F := Q(A)$. We define*

$$\text{Frac } A := \{M \subseteq F : M \in \text{Mod}_A, \alpha M \subseteq A \text{ for some } \alpha \in F^\times\} \text{ and } \text{Prin } A := \{\alpha A : \alpha \in F^\times\}.$$

Notice that $\text{Prin } A \subseteq \text{Frac } A$.

Lemma 5.3.7 (Frac A and Prin A as algebraic objects) *For $M_1, M_2 \in \text{Frac } A$, define $M_1 \cdot M_2 := \sum_{m_i \in M_i} Am_1 m_2$. Then $(\text{Frac } A, \cdot)$ is a monoid and $(\text{Prin } A, \cdot)$ is an abelian group with A as identity.*

Proof. This is clear. $\square$

Lemma 5.3.8 (Invertible elements in $\text{Frac } A$) *Let $M \in \text{Frac}(A)$ and define $(A : M) := \{\alpha \in F : \alpha M \subseteq A\}$. Then $(A : M) \in \text{Frac } A$ and $M \in (\text{Frac } A)^\times \iff (A : M)M = A$.*

Proof. It is clear that $(A : M)$ is an A -submodule of F . Moreover, for every $\beta \in M \setminus \{0\}$, $\beta(A : M) \subseteq A$, so $(A : M) \in \text{Frac } A$. Now if $(A : M)M = A$, then $M \in (\text{Frac } A)^\times$ by definition; conversely, if $MM' = A$ for some M' , then $M' \subseteq (A : M)$, which implies that $A \subseteq (A : M)M \subseteq A$, so we are done. $\square$

Proposition 5.3.9 (Invertibility is a local property) *Let $M \in \text{Frac } A$. Then TFAE:*

- (1). $M \in (\text{Frac } A)^\times$;
- (2). M is finitely generated and for every $\mathfrak{p} \in \text{Spec } A$, $M_{\mathfrak{p}} \in (\text{Frac } A_{\mathfrak{p}})^\times$;
- (3). M is finitely generated and for every $\mathfrak{m} \in \text{Max } A$, $M_{\mathfrak{m}} \in (\text{Frac } A_{\mathfrak{m}})^\times$.

Proof. (1) $\implies$ (2) By Lemma 5.3.8, if M is invertible then $(A : M)M = A$. So there exist $m_i \in M$ and $m'_i \in (A : M)$ such that $1 = \sum_{i=1}^n m_i m'_i$. Hence for every $x \in M$, we have $x = x \cdot 1 = \sum_i (am'_i)m_i \in \sum_i Am_i$, which implies that $M = \langle m_1, \dots, m_n \rangle$ is finitely generated. Moreover, for every $\mathfrak{p} \in \text{Spec } A$ we have that $(A : M)_{\mathfrak{p}} M_{\mathfrak{p}} = ((A : M)M)_{\mathfrak{p}} = A_{\mathfrak{p}}$, so $M_{\mathfrak{p}} \in (\text{Frac } A_{\mathfrak{p}})^\times$.

(2) $\implies$ (3) This is clear.

(3) $\implies$ (1) Suppose $M = \langle m_1, \dots, m_n \rangle$ is finitely generated. To show that $(A : M)M = A$, it suffices to show $(A : M)_{\mathfrak{m}} M_{\mathfrak{m}} = A_{\mathfrak{m}}$ for every $\mathfrak{m} \in \text{Max } A$. We claim that $(A : M)_{\mathfrak{m}} = (A_{\mathfrak{m}} : M_{\mathfrak{m}})$; indeed,

$$\begin{aligned} (A_{\mathfrak{m}} : M_{\mathfrak{m}}) &= \{\alpha \in F : \alpha M_{\mathfrak{m}} \subseteq A_{\mathfrak{m}}\} = \{\alpha \in F : \alpha M \subseteq A_{\mathfrak{m}}\} = \{\alpha \in F : \forall i, \alpha m_i \in A_{\mathfrak{m}}\} \\ &= \{\alpha \in F : \forall i, \alpha m_i = a_i/s_i \text{ for some } a_i \in A, s_i \in A \setminus \mathfrak{m}\} \\ &= \{\alpha \in F : \exists s \in A \setminus \mathfrak{m}, \forall i, s\alpha m_i \in A\} = (A : M)_{\mathfrak{m}}. \end{aligned}$$

Hence $(A : M)_{\mathfrak{m}} M_{\mathfrak{m}} = (A_{\mathfrak{m}} : M_{\mathfrak{m}})M_{\mathfrak{m}} = A_{\mathfrak{m}}$, since $M_{\mathfrak{m}} \in (\text{Frac } A_{\mathfrak{m}})^\times$, so we are done. $\square$

Corollary 5.3.10 (Invertibility and Dedekind domains) *Let A be a Dedekind domain. Then every element in $\text{Frac } A$ is invertible; so $\text{Frac } A$ is a group.*

Proof. Let $M \in \text{Frac } A$ and $\alpha M \subseteq A$ for some $\alpha \in F^\times$. Since A is Noetherian, αM is finitely generated, which implies that M is finitely generated since α is a unit. So by Proposition 5.3.9, it suffices to show that $M_{\mathfrak{m}} \in (\text{Frac } A_{\mathfrak{m}})^\times$ for every $\mathfrak{m} \in \text{Max } A$. Now since $M_{\mathfrak{m}}$ is a finitely generated $A_{\mathfrak{m}}$ -module and $A_{\mathfrak{m}}$ is a DVR, $M_{\mathfrak{m}} = \sum_{i=1}^k A_{\mathfrak{m}} \alpha_i$ for some $\alpha_i = u_i \pi^{n_i}$ where $v(u_i) = 0$ and $v(\pi) = 1$. As a result, $M_{\mathfrak{m}} = A_{\mathfrak{m}} \pi^{\min\{n_1, \dots, n_k\}}$ and so $M_{\mathfrak{m}}^{-1} = A_{\mathfrak{m}} \pi^{-\min\{n_1, \dots, n_k\}}$; hence $M_{\mathfrak{m}}$ is invertible and we are done. $\square$

Definition/Corollary 5.3.11 (Class group of a Dedekind domain) We define the *class group* of a Dedekind domain A to be $\text{Cl}(A) = \text{Frac } A / \text{Prin } A$. Hence A is a PID $\iff \text{Cl}(A) = 0$.

Height of Ideals

Krull's height theorem and principal ideal theorem

Definition 6.1.1 (Height of an ideal) Let A be a commutative ring. We define the height of a prime ideal $\mathfrak{p}$ to be $\text{ht}(\mathfrak{p}) := \sup\{n \in \mathbb{N} : \exists \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}\}$. For a proper ideal $\mathfrak{a} \triangleleft A$, we define $\text{ht}(\mathfrak{a}) := \min\{\text{ht}(\mathfrak{p}) : \mathfrak{p} \in V(\mathfrak{a})\}$.

- Note that if $\mathfrak{a} \triangleleft A$ is decomposable, then every $\mathfrak{p} \in V(\mathfrak{a})$ contains some $\mathfrak{p}' \in \text{Ass}(\mathfrak{a})$, so we can write $\text{ht}(\mathfrak{a}) = \min\{\text{ht}(\mathfrak{p}) : \mathfrak{p} \in \text{Ass}(\mathfrak{a})\}$.

Proposition 6.1.2 (Associated primes in a Noetherian ring) Let A be Noetherian and $\mathfrak{a} \triangleleft A$ be a proper ideal. Then

$$\text{Ass}(\mathfrak{a}) = \text{Spec } A \cap \{(\mathfrak{a} : x) : x \in A\}.$$

- This is an improvement of Theorem 2.2.3 given the Noetherian condition on A .

Proof. Since A is Noetherian, we can find a primary decomposition $\mathfrak{a} = \bigcap_{i=1}^m \mathfrak{q}_i$. Recall from the proof of Theorem 2.2.3 that for every $x \in A$, $(\mathfrak{a} : x) = \bigcap_i (\mathfrak{q}_i : x) = \bigcap_{\mathfrak{q}_i \not\supseteq x} \mathfrak{q}_i$. Since the decomposition is reduced, for every i there exists $x_i \in (\bigcap_{j \neq i} \mathfrak{q}_j) \setminus \mathfrak{q}_i$, so then $(\mathfrak{a} : x_i) = (\mathfrak{q}_i : x_i)$. Now $\sqrt{(\mathfrak{a} : x_i)} = \mathfrak{p}_i \in \text{Ass}(\mathfrak{a})$ by Theorem 2.2.3. Notice that $\mathfrak{p}_i$ is finitely generated, so $\mathfrak{p}_i^n \subseteq (\mathfrak{a} : x_i)$ for some $n \in \mathbb{N}^{>0}$. If $n = 1$, then $(\mathfrak{a} : x_i) \in \text{Spec } A$ and we are done; otherwise, we can pick $y \in \mathfrak{p}_i^{n-1} \setminus (\mathfrak{a} : x_i)$. Since $(\mathfrak{p}_i y)x_i \subseteq \mathfrak{p}_i^n x_i \subseteq \mathfrak{a}$, we deduce $\mathfrak{p}_i \subseteq (\mathfrak{a} : yx_i)$; moreover, $yx_i \in (\bigcap_{j \neq i} \mathfrak{q}_j) \setminus \mathfrak{q}_i$ because $y \notin (\mathfrak{a} : x_i)$. Hence by Lemma 2.1.8, $(\mathfrak{a} : yx_i)$ is $\mathfrak{p}_i$ -primary. As a result, $\mathfrak{p}_i = (\mathfrak{a} : yx_i) \in \text{Spec } A$, so we are done. $\square$

Lemma 6.1.3 (Ideals with height zero) Let A be a Noetherian ring and $\mathfrak{a} \triangleleft A$ be a proper ideal. If $\text{ht}(\mathfrak{a}) = 0$, then $\mathfrak{a} \subseteq D(A)$; in particular, if $\text{ht}(\langle a \rangle) = 0$, then $a \in D(A)$.

Proof. If $\text{ht}(\mathfrak{a}) = 0$, then $\text{ht}(\mathfrak{p}) = 0$ for every $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$. So every $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ is minimal in $\text{Spec } A$, which implies that $\mathfrak{p} \in \text{Ass}(0)$ by Proposition 2.2.4(3). As a result, we deduce $\mathfrak{a} \subseteq \mathfrak{p} \subseteq \bigcup_{\mathfrak{p}' \in \text{Ass}(0)} \mathfrak{p}' = D(A)$ by Corollary 2.2.6, so we are done. $\square$

Theorem 6.1.4 (Krull's height theorem (HT)) Let A be Noetherian and $\mathfrak{a} := \langle a_1, \dots, a_n \rangle \triangleleft A$ be a proper ideal. Then $\text{ht}(\mathfrak{a}) \leq n$; moreover, if $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ is minimal, then $\text{ht}(\mathfrak{p}) \leq n$.

- In this proof we will assume Krull's principal ideal theorem (PIT): let A be Noetherian and $a \notin A^\times$; if $\mathfrak{p} \in \text{Ass}(\langle a \rangle)$ is minimal, then $\text{ht}(\mathfrak{p}) \leq 1$. PIT is a special case of HT, which we will prove later independently.

Proof. We proceed by induction on n . The base case is given by Krull's PIT. For the inductive case, we start by making some reductions. First notice that $\text{ht}(\mathfrak{p}) = \text{ht}(\mathfrak{p}A_{\mathfrak{p}})$ for every $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$, so we may assume WLOG that A is local with $\text{Max } A = \{\mathfrak{p}\}$. Secondly, we may assume $\text{ht}(\mathfrak{a}) \geq 1$; otherwise we are done by Lemma 6.1.3; as a result $\text{ht}(\mathfrak{p}) \geq 1$ for every $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$, so there exists $\mathfrak{p}' \in \text{Spec } A$ with $\mathfrak{p}' \subsetneq \mathfrak{p}$; in fact, since A is Noetherian we can assume there is no other prime ideal in-between. Hence we have that $\text{ht}(\mathfrak{p}) = \sup\{\text{ht}(\mathfrak{p}') : \mathfrak{p}' \subsetneq \mathfrak{p}\} + 1$. Now it suffices to show that for any such $\mathfrak{p}'$, $\mathfrak{p}' \in \text{Ass}(\mathfrak{b})$ is a minimal element for some $\mathfrak{b} \triangleleft A$ generated by $n - 1$ elements, so then by the inductive hypothesis we are done.

Let $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ be minimal. Since $\mathfrak{p}' \subsetneq \mathfrak{p}$ we must have $a_i \notin \mathfrak{p}'$ for some i ; by reordering if necessary we may assume $i = n$. Notice that $V(\langle a_n \rangle + \mathfrak{p}') = \{\mathfrak{p}\}$ since there is no other prime ideal between $\mathfrak{p}'$ and $\mathfrak{p}$, so

$\sqrt{\langle a_n \rangle + \mathfrak{p}'} = \mathfrak{p}$. In particular, for every i we have $a_i^{m_i} = a_n r_i + b_i$ for some $m_i \in \mathbb{N}^{>0}$, $r_i \in A$, and $b_i \in \mathfrak{p}'$. We claim that $V(\langle a_n, b_1, \dots, b_{n-1} \rangle) = \{\mathfrak{p}\}$; indeed, $\mathfrak{p} = \sqrt{\langle a_1, \dots, a_n \rangle} \subseteq \sqrt{\langle a_n, b_1, \dots, b_{n-1} \rangle} \subseteq \mathfrak{p}$. Finally, it remains to show that $\mathfrak{p}' \in V(\langle b_1, \dots, b_{n-1} \rangle)$ is a minimal element. Let $\bar{A} := A/\langle b_1, \dots, b_{n-1} \rangle$. Then by our claim, $\bar{\mathfrak{p}} \in V(\langle \bar{a}_n \rangle)$ is a minimal element, so $\text{ht}(\bar{\mathfrak{p}}) \leq 1$ by Krull's PIT. But $\bar{\mathfrak{p}}' \in \text{Spec } \bar{A}$ and $\bar{\mathfrak{p}}' \subsetneq \bar{\mathfrak{p}}$, which implies that $\bar{\mathfrak{p}}' \in \text{Spec } \bar{A}$ is a minimal element; so by the correspondence theorem, $\mathfrak{p}' \in V(\langle b_1, \dots, b_{n-1} \rangle)$ is minimal. As a result, by the inductive hypothesis we deduce $\text{ht}(\mathfrak{p}) \leq \text{ht}(\mathfrak{p}') + 1 \leq (n-1) + 1 = n$, so we are done. $\square$

Corollary 6.1.5 (Dimension of $k[x_1, \dots, x_n]$) *Let k be a field. Then $\dim k[x_1, \dots, x_n] = n$.*

Proof. Notice that $\bar{k}[x_1, \dots, x_n]/k[x_1, \dots, x_n]$ is an integral extension. Hence by Nullstellensatz II and Krull's HT, we have

$$\begin{aligned} \dim k[x_1, \dots, x_n] &= \dim \bar{k}[x_1, \dots, x_n] = \sup\{\text{ht}(\mathfrak{m}) : \mathfrak{m} \in \bar{k}[x_1, \dots, x_n]\} \\ &= \sup\{\text{ht}(\langle x_1 - p_1, \dots, x_n - p_n \rangle) : p \in \mathbb{A}_k^n\} \leq n. \end{aligned}$$

On the other hand, it is clear that $\dim k[x_1, \dots, x_n] \geq n$ considering the chain of prime ideals $\langle x_1 \rangle \subsetneq \langle x_1, x_2 \rangle \subsetneq \dots \subsetneq \langle x_1, \dots, x_n \rangle$, so we are done. $\square$

- This provides a much cleaner proof for Corollary 4.2.7!
- In fact, we can say more: for every $\mathfrak{m} \in \text{Max } k[x_1, \dots, x_n]$, $\text{ht}(\mathfrak{m}) = n$. This is a consequence of the following result: suppose B/A is integral, B is an integral domain, and A is integrally closed; then for every $\mathfrak{q} \in \text{Spec } B$, $\text{ht}(\mathfrak{q}) = \text{ht}(f^*(\mathfrak{q}))$. Since $k[x_1, \dots, x_n]$ is a UFD, in particular it is integrally closed, so we are done.

Theorem 6.1.6 (Krull's principal ideal theorem (PIT)) *Let A be Noetherian, $a \notin A^\times$, and $\mathfrak{p} \in \text{Ass}(\langle a \rangle)$ be a minimal element. Then $\text{ht}(\mathfrak{p}) \leq 1$.*

Proof. We start by making some reductions. Let $\mathfrak{p} \in V(\langle a \rangle)$ be a minimal element. First notice that $\text{ht}(\mathfrak{p}) = \text{ht}(\mathfrak{p}A_{\mathfrak{p}})$, so we may assume A is local with $\text{Max } A = \{\mathfrak{p}\}$. Secondly if $\text{ht}(\mathfrak{p}) \geq 2$, then we can find a chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}$ in $\text{Spec } A$; by passing to $A/\mathfrak{p}_0$, we still have a chain of length two, so we may also assume A is an integral domain.

Now we show $\text{ht}(\mathfrak{p}_1) = 0$, which will be a contradiction. Notice that

$$\text{ht}(\mathfrak{p}_1) = 0 \iff \dim A_{\mathfrak{p}_1} = 0 \iff A_{\mathfrak{p}_1} \text{ is Artinian} \iff \exists n \in \mathbb{N} \text{ s.t. } \mathfrak{p}_1^n A_{\mathfrak{p}_1} = \mathfrak{p}_1^{n+1} A_{\mathfrak{p}_1}.$$

Let $\mathfrak{p}_1^{(k)} := \mathfrak{p}_1^k A_{\mathfrak{p}_1} \cap A$. Since $\mathfrak{p}_1 A_{\mathfrak{p}_1}$ is maximal, every $\mathfrak{p}_1^k A_{\mathfrak{p}_1}$ is $\mathfrak{p}_1 A_{\mathfrak{p}_1}$ -primary; as a result, we deduce each $\mathfrak{p}_1^{(k)}$ is $\mathfrak{p}_1$ -primary. Since $\mathfrak{p} \in V(\langle a \rangle)$ is minimal, $\dim A/\langle a \rangle = 0$, which implies that $A/\langle a \rangle$ is Artinian, so then $\mathfrak{p}_1^{(n)} + \langle a \rangle = \mathfrak{p}_1^{(n+1)} + \langle a \rangle$ for some n . In particular, for every $x_n \in \mathfrak{p}_1^{(n)}$, $x_n = x_{n+1} + v_n a$ for some $x_{n+1} \in \mathfrak{p}_1^{(n+1)}$ and $v_n \in A$. We claim that $v_n \in \mathfrak{p}_1^{(n)}$. Indeed, notice that $v_n a = x_n - x_{n+1} \in \mathfrak{p}_1^{(n)}$; on the other hand, $\mathfrak{p} \in V(\langle a \rangle)$ being minimal and $\mathfrak{p}_1 \subsetneq \mathfrak{p}$ implies that $a \notin \mathfrak{p}_1$, and therefore $v_n \in \mathfrak{p}_1^{(n)}$. But then

$$\mathfrak{p}_1^{(n)} \subseteq \mathfrak{p}_1^{n+1} + a\mathfrak{p}_1^{(n)} \subseteq \mathfrak{p}_1^{(n+1)} + J(A)\mathfrak{p}_1^{n+1}$$

implies that $\mathfrak{p}_1^{(n)} = \mathfrak{p}_1^{(n+1)}$ by Nakayama's Lemma. Hence $\mathfrak{p}_1^n A_{\mathfrak{p}_1} = \mathfrak{p}_1^{n+1} A_{\mathfrak{p}_1}$, so we are done. $\square$

Theorem 6.1.7 (Converse to Krull's HT) *Let A be Noetherian, $\mathfrak{p} \in \text{Spec } A$, and $\text{ht}(\mathfrak{p}) = d$. Then there exists $\mathfrak{a} := \langle a_1, \dots, a_d \rangle$ such that $\mathfrak{p}$ is a minimal prime in $V(\mathfrak{a})$.*

Proof. Let $\mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_d = \mathfrak{p}$ be a chain of prime ideals. Since $\text{ht}(\mathfrak{p}) = d$, we deduce $\text{ht}(\mathfrak{p}_i) = i$ for every $i \leq d$. We claim that there exists $\langle a_1, \dots, a_d \rangle \triangleleft A$ such that (1) $\text{ht}(\langle a_1, \dots, a_k \rangle) = k$ and (2) $\mathfrak{p}_k$ is minimal in $V(\langle a_1, \dots, a_k \rangle)$; we proceed by induction on d .

Base case. Let $\text{ht}(\mathfrak{p}) = 1$. Then $\mathfrak{p} \not\subseteq \bigcup_{\text{ht}(\mathfrak{p}')=0} \mathfrak{p}'$ (note that this is a finite union since A is Noetherian); otherwise, we must have $\mathfrak{p} = \mathfrak{p}'$ for some $\text{ht}(\mathfrak{p}') = 0$, a contradiction. Pick $a_1 \in \mathfrak{p} \setminus \bigcup_{\text{ht}(\mathfrak{p}')=0} \mathfrak{p}'$ and notice that $\text{ht}(\langle a_1 \rangle) \leq 1$ by Krull's PIT. But $\langle a_1 \rangle \not\subseteq \mathfrak{p}'$ for any $\text{ht}(\mathfrak{p}') = 0$, so then $\text{ht}(\langle a_1 \rangle) = 1$. Now since $\mathfrak{p} \supseteq \langle a_1 \rangle$ and $\text{ht}(\mathfrak{p}) = 1$, we deduce $\mathfrak{p} \in V(\langle a_1 \rangle)$ is a minimal element.

Inductive case. Let $\mathfrak{p}'_1, \dots, \mathfrak{p}'_m$ be minimal elements in $V(\langle a_1, \dots, a_{d-1} \rangle)$ where $\text{ht}(\langle a_1, \dots, a_{d-1} \rangle) = d-1$. Then by Krull's HT, $\text{ht}(\mathfrak{p}'_i) \leq d-1$ for every i . Since $\text{ht}(\langle a_1, \dots, a_{d-1} \rangle) = d-1$ by the inductive hypothesis, we deduce $\text{ht}(\mathfrak{p}'_i) = d-1$. So similarly we must have $\mathfrak{p} \not\subseteq \bigcup_{i=1}^m \mathfrak{p}'_i$ since $\text{ht}(\mathfrak{p}) = d$. Pick $a_d \in \mathfrak{p} \setminus \bigcup_i \mathfrak{p}'_i$ and consider $\mathfrak{a} := \langle a_1, \dots, a_d \rangle$. Then by Krull's HT again, $\text{ht}(\mathfrak{a}) \leq d$; in fact, we claim that $\text{ht}(\mathfrak{a}) = d$. Indeed, otherwise $\mathfrak{a} \subseteq \mathfrak{p}'$ for some $\text{ht}(\mathfrak{p}') \leq d-1$; but since $\mathfrak{a} \not\subseteq \mathfrak{p}'_i$ for any i , we must have $\mathfrak{p}' \supsetneq \mathfrak{p}_i$ for some i because the $\mathfrak{p}'_i$'s are minimal over $\langle a_1, \dots, a_{d-1} \rangle$ and $\mathfrak{p} \supseteq \mathfrak{a}$; but $\text{ht}(\mathfrak{p}) \leq \text{ht}(\mathfrak{p}_i) = d-1$, which is a contradiction. Finally, since $\mathfrak{p} \in V(\mathfrak{a})$ and $\text{ht}(\mathfrak{p}) = d$, we deduce $\mathfrak{p}$ is minimal over $\mathfrak{a}$, so we are done. $\square$

Corollary 6.1.8 (Noetherian domain and UFD) *Let A be a Noetherian domain. Then A is a UFD $\iff$ every prime ideal of height 1 is principal.*

Proof. ($\implies$) Suppose $\text{ht}(\mathfrak{p}) = 1$. By Theorem 6.1.7, there exists $a \in A$ such that $\mathfrak{p}$ is minimal in $\text{Ass}(\langle a \rangle)$. Write $a = u \cdot p_1^{e_1} \cdots p_n^{e_n}$, so then $\langle a \rangle = \langle p_1^{e_1} \rangle \cdots \langle p_n^{e_n} \rangle$. Since $\mathfrak{p}$ is prime, we deduce $\mathfrak{p} \supseteq \langle p_i^{e_i} \rangle$ for some i and therefore $\mathfrak{p} \supseteq \langle p_i \rangle \in \text{Ass}(\mathfrak{a})$. But $\mathfrak{p}$ is minimal, so we must have that $\mathfrak{p} = \langle p_i \rangle$ is principal.

($\impliedby$) Let $a \in A$ be irreducible. It suffices to show that a is prime, or equivalently $\langle a \rangle$ is a prime ideal. Notice that every minimal $\mathfrak{p}_i \in \text{Ass}(\langle a \rangle)$ has height 1, so $\mathfrak{p}_i = \langle p_i \rangle$ is principal by assumption. Now $\langle a \rangle \subseteq \langle p_i \rangle$, but $\langle a \rangle$ is maximal among principal ideals, which implies that $\langle a \rangle = \langle p_i \rangle$ for all i . In particular, $\langle a \rangle$ is a prime ideal, so we are done. $\square$

Theorem 6.1.9 (Dimension of Noetherian local rings) *Let A be a Noetherian local ring with $\text{Max } A = \{\mathfrak{m}\}$. Then*

$$\dim A = \min_{\mathfrak{q}:\mathfrak{m}\text{-primary}} d(\mathfrak{q})$$

where $d(\mathfrak{q})$ denotes the number of minimal generators of $\mathfrak{q}$.

Proof. Notice that $\dim A = \text{ht}(\mathfrak{m})$. If $\mathfrak{q} \triangleleft A$ is $\mathfrak{m}$ -primary, we must have $V(\mathfrak{q}) = \{\mathfrak{m}\}$ since $\text{Max } A = \{\mathfrak{m}\}$. So by Krull's HT, we deduce $\text{ht}(\mathfrak{m}) \leq d(\mathfrak{q})$ and therefore

$$\dim A \leq \min_{\mathfrak{q}:\mathfrak{m}\text{-primary}} d(\mathfrak{q}).$$

Conversely, if $\text{ht}(\mathfrak{m}) = d$, then by Theorem 6.1.7 there exists $\mathfrak{q}_0 := \langle a_1, \dots, a_d \rangle \triangleleft A$ such that $\mathfrak{m}$ is minimal over $\mathfrak{q}_0$. Since $\mathfrak{m}$ is the unique maximal ideal, we deduce $\sqrt{\mathfrak{q}_0} = \mathfrak{m}$ and so $\mathfrak{q}_0$ is $\mathfrak{m}$ -primary. As a result,

$$\dim A = \text{ht}(\mathfrak{m}) = d \geq d(\mathfrak{q}_0) \geq \min_{\mathfrak{q}:\mathfrak{m}\text{-primary}} d(\mathfrak{q}),$$

so we are done. $\square$

- One immediate consequence of this result is that $\dim A \leq \dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$, since by Nakayama's Lemma $d(\mathfrak{m}) = \dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$.

- Here are some further implications. Consider a prime ideal $\mathfrak{p} \triangleleft k[x_1, \dots, x_n]$ and the integral domain $A := k[x_1, \dots, x_n]/\mathfrak{p}$. This is called the coordinate ring of an affine variety $V(\mathfrak{p})$. If we consider all regular functions around a point $p \in V(\mathfrak{p})$, then they form a ring which can be naturally identified with $A_{\mathfrak{m}_p}$, since around p these functions can be represented as quotients of polynomials whose denominators do not vanish at p . Now $A_{\mathfrak{m}_p}$ is a local Noetherian domain, and we can identify the “tangent plane” at p with the dual of $\tilde{\mathfrak{m}}_p/\tilde{\mathfrak{m}}_p^2$ where $\tilde{\mathfrak{m}}_p = \mathfrak{m}_p A_{\mathfrak{m}_p}$. If $\dim A = \dim_{A/\mathfrak{m}_p}(\tilde{\mathfrak{m}}_p/\tilde{\mathfrak{m}}_p^2)$, we say that $V(\mathfrak{p})$ does not have a singularity at p , or in more geometric terms, $V(\mathfrak{p})$ is *smooth* at p . Hence $V(\mathfrak{p})$ is smooth $\iff A_{\mathfrak{m}_p}$ is a regular local ring for every $p \in V(\mathfrak{p})$.

Cohen-Macaulay rings

Theorem 6.2.1 (Dimension of quotient Noetherian rings) *Let A be a local Noetherian ring with $\text{Max } A = \{\mathfrak{m}\}$ and $a \notin D(A) \cup A^\times$. Then $\dim A/\langle a \rangle = \dim A - 1$.*

Proof. Since A is local Noetherian, so is $A' := A/\langle a \rangle$. Hence $\dim A' = \text{ht}(\bar{\mathfrak{m}}) = d < \infty$ where $\bar{\mathfrak{m}} := \mathfrak{m}/\langle a \rangle$. Let $\mathfrak{p}_0/\langle a \rangle \subsetneq \dots \subsetneq \mathfrak{p}_d/\langle a \rangle = \bar{\mathfrak{m}}$ be a saturated chain of prime ideals. Then $\mathfrak{p}_0$ is minimal over $\langle a \rangle$, which implies that $\text{ht}(\mathfrak{p}_0) \leq 1$ by Krull’s PIT. But $a \notin D(A)$, so $\text{ht}(\langle a \rangle) \geq 1$ and therefore $\text{ht}(\mathfrak{p}_0) = 1$. As a result, we deduce $\text{ht}(\mathfrak{m}) \geq \text{ht}(\bar{\mathfrak{m}}) + 1 = d + 1$. On the other hand, since $\text{ht}(\bar{\mathfrak{m}}) = d$, by Theorem 6.1.7 there exists $\bar{\mathfrak{a}} := \langle \bar{a}_1, \dots, \bar{a}_d \rangle \triangleleft A'$ such that $\bar{\mathfrak{m}}$ is minimal over $\bar{\mathfrak{a}}$; but that implies that $\mathfrak{m}$ is minimal over $\langle a, a_1, \dots, a_d \rangle \triangleleft A$. Hence by Krull’s HT, $\text{ht}(\mathfrak{m}) \leq d + 1$, so then $\dim A = \text{ht}(\mathfrak{m}) = d + 1$ and we are done. $\square$

Definition 6.2.2 (A -regular sequence) *Let A be a local ring with $\text{Max } A = \{\mathfrak{m}\}$. We say $(x_1, \dots, x_r)$ is an A -regular sequence if $x_i \in \mathfrak{m}$ and $x_i \notin D(A/\langle x_1, \dots, x_{i-1} \rangle)$ for every i .*

Corollary 6.2.3 (A -regular sequences and dimension of quotient rings) *Let A be a local Noetherian ring and $(x_1, \dots, x_r)$ be an A -regular sequence. Then for every $s \leq r$, $\dim A/\langle x_1, \dots, x_s \rangle = \dim A - s$.*

Proof. This follows by invoking Theorem 6.2.1 inductively. $\square$

Definition 6.2.4 (Depth and Cohen-Macaulay rings) *Let A be a local ring. We define its *depth*, denoted $\text{depth } A$, to be the maximal length of an A -regular sequence.*

Note that in general $\dim A \geq \text{depth } A$, by Corollary 6.2.3. We say a local Noetherian ring A is *Cohen-Macaulay* if $\dim A = \text{depth } A$.

Theorem 6.2.5 (Characterization of Cohen-Macaulay ring) *Let A be a local Noetherian ring. Suppose any proper ideal $\mathfrak{a} \triangleleft A$ is unmixed; that is, if $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ then $\mathfrak{p}$ is minimal in $\text{Ass}(\mathfrak{a})$. Then A is a Cohen-Macaulay ring.*

- As it will turn out in the proof, the conditions we list here are stronger than needed: we only need to assume any $\mathfrak{a} \triangleleft A$ with $d(\mathfrak{a}) \leq \dim A - \dim A/\mathfrak{a}$ is unmixed. The converse also holds, but the proof is beyond our scope here.

Proof. By induction on r , we show that if $r \leq d := \dim A$, then there exists an A -regular sequence $(x_1, \dots, x_r)$.

Base case. If $d = 0$, we are done. If $d \geq 1$, we claim that $\mathfrak{m} \not\subseteq D(A) = \bigcup_{\mathfrak{p} \in \text{Ass}(0)} \mathfrak{p}$. Indeed, otherwise we must have $\mathfrak{m} = \mathfrak{p}$ for some $\mathfrak{p}$; but $0 \triangleleft A$ is unmixed, so $\mathfrak{m} \in \text{Ass}(0)$ is a minimal prime, which implies that $\mathfrak{m}$ is minimal in $\text{Spec } A$ and therefore $d = 0$, a contradiction.

Inductive case. Let $r \leq d$. If equality holds, we are done; otherwise $d - r > 0$. Then by the inductive hypothesis, there exists an A -regular sequence $(x_1, \dots, x_r)$. We claim that $\mathfrak{m} \not\subseteq \bigcup_{\mathfrak{p} \in \text{Ass}(\mathfrak{a})} \mathfrak{p}$ where $\mathfrak{a} := \langle x_1, \dots, x_r \rangle$. Indeed, if not then $\mathfrak{m} = \mathfrak{p}$ for some $\mathfrak{p}$, so then $\mathfrak{m}$ is minimal over $\langle x_1, \dots, x_r \rangle$ since $\mathfrak{a}$ is unmixed. As a result, $\dim A/\mathfrak{a} = 0$. But $\dim A/\mathfrak{a} = d - r$ by Corollary 6.2.3, contradicting that $d - r > 0$. Hence by the correspondence theorem,

$$\bar{\mathfrak{m}} := \mathfrak{m}/\mathfrak{a} \not\subseteq \bigcup_{\bar{\mathfrak{p}} \in \text{Ass}(\bar{0})} \bar{\mathfrak{p}} = D(A/\mathfrak{a}),$$

so we can pick $\bar{x}_{r+1} \in \bar{\mathfrak{m}} \setminus D(A/\mathfrak{a})$ and we are done. $\square$

Finitely generated k -domains

Theorem 6.3.1 (Noether's normalization lemma, stronger version) *Let k be a field and $\mathfrak{a} \triangleleft \tilde{A} := k[x_1, \dots, x_n]$ be a proper ideal. Then there exist $y_1, \dots, y_n \in \tilde{A}$ such that*

(1). $y_1, \dots, y_n$ are algebraically independent over k .

(2). $\tilde{A}/k[y_1, \dots, y_n]$ is an integral extension.

(3). $\mathfrak{a} \cap k[y_1, \dots, y_n] = \langle y_{d+1}, \dots, y_n \rangle$.

- If $A \simeq k[x_1, \dots, x_n]/\mathfrak{a}$ is a finitely generated algebra, choose $y_1, \dots, y_n$ algebraically independent over k for $\mathfrak{a}$. Moreover, since $\tilde{A}/k[y_1, \dots, y_n]$ is integral, we deduce

$$\frac{k[y_1, \dots, y_n]}{\mathfrak{a} \cap k[y_1, \dots, y_n]} = \frac{k[y_1, \dots, y_n]}{\langle y_{d+1}, \dots, y_n \rangle} \simeq k[y_1, \dots, y_d],$$

so we get back to Theorem 4.2.6.

Proof. Step 1. Suppose $\mathfrak{a} = \langle f \rangle$ is principal. Recall from the proof of Theorem 4.2.6 that there exists $\varphi \in \text{Aut}_k(\tilde{A})$ such that $\varphi(f)$ as a polynomial in x_n is monic, so WLOG we may assume f is monic in x_n . Define $y_i = x_i$ for $i \leq n-1$ and $y_n = f$. Then x_n is integral over $k[y_1, \dots, y_n]$ and the y_i 's are algebraically independent over k . Moreover, we have that $\langle f \rangle \cap k[y_1, \dots, y_n] = y_n k[x_1, \dots, y_n] = \langle y_n \rangle$.

Step 2. We proceed by induction on n . The base case is clear: if $n = 1$, then $\tilde{A} = k[x_1]$ is a PID, so we are done by Step 1. For the inductive case, pick $f \in \mathfrak{a}$ and choose $y_1, \dots, y_n$ for $\langle f \rangle$ as in Step 1. Define $\mathfrak{a}' := \mathfrak{a} \cap k[y_1, \dots, y_{n-1}] \trianglelefteq k[y_1, \dots, y_{n-1}]$ and notice that it is in fact a proper ideal; indeed, otherwise $\mathfrak{a} \supseteq \mathfrak{a}' + \langle f \rangle = \tilde{A}$ is not proper. So by the inductive hypothesis, there exist $z_1, \dots, z_{n-1} \in k[y_1, \dots, y_{n-1}]$ such that

(1). $z_1, \dots, z_{n-1}$ are algebraically independent over k .

(2). $k[y_1, \dots, y_{n-1}]/k[z_1, \dots, z_{n-1}]$ is an integral extension.

(3). $\mathfrak{a}' \cap k[z_1, \dots, z_{n-1}] = \langle z_d, \dots, z_{n-1} \rangle$.

Since $y_1, \dots, y_n$ are algebraically independent over k , we deduce $z_1, \dots, z_{n-1}, y_n$ are also algebraically independent. Now $k[x_1, \dots, x_n]/k[y_1, \dots, y_n]$ is an integral extension and so is $k[y_1, \dots, y_{n-1}]/k[z_1, \dots, z_{n-1}]$; hence $k[x_1, \dots, x_n]/k[z_1, \dots, z_{n-1}, y_n]$ is integral. Finally, it remains to check that $\mathfrak{a} \cap k[z_1, \dots, z_{n-1}, y_n] = \langle z_{d+1}, \dots, z_{n-1}, y_n \rangle$. It is clear that the RHS is a subset of the LHS. For the converse, suppose $g \in \mathfrak{a} \cap k[z_1, \dots, z_{n-1}, y_n]$. Then $g = g_0 + y_n p$ for some $g_0 \in k[z_1, \dots, z_{n-1}]$ and $p \in k[z_1, \dots, z_{n-1}, y_n]$. Since $y_n \in \mathfrak{a}$, we deduce $g_0 \in \mathfrak{a} \cap k[z_1, \dots, z_{n-1}] = \langle z_d, \dots, z_{n-1} \rangle$. But that implies that $g \in \langle z_{d+1}, \dots, z_{n-1}, y_n \rangle$, so we are done. $\square$

Theorem 6.3.2 (Dimension of a finitely generated k -domain) *Let k be a field and A be a finitely generated k -domain. Then for every $\mathfrak{p} \in \operatorname{Spec} A$, $\operatorname{ht}(\mathfrak{p}) + \dim A/\mathfrak{p} = \dim A$.*

- This result is motivated by the following observation. If A is a Noetherian ring, then $\dim A = \max_{\mathfrak{p} \in \operatorname{Ass}(0)} \{\operatorname{ht}(\mathfrak{p}) + \dim A/\mathfrak{p}\}$ as every maximal chain of prime ideals contains a minimal prime. If A is an integral domain, then $0 \triangleleft A$ is the unique minimal prime ideal. So this theorem is essentially showing that every maximal chain has the same length for special types of Noetherian domain.

Proof. By Noether's normalization lemma, there exists $x_1, \dots, x_d \in A$ such that (1) $x_1, \dots, x_d$ are algebraically independent over k and (2) $A/k[x_1, \dots, x_d]$ is an integral extension. Consider $f^* : \operatorname{Spec} A \rightarrow \operatorname{Spec} k[x_1, \dots, x_d]$. Then for every $\mathfrak{p} \in \operatorname{Spec} A$, $\operatorname{ht}(\mathfrak{p}) = \operatorname{ht}(f^*(\mathfrak{p}))$ and $\dim A/\mathfrak{p} = \dim k[x_1, \dots, x_d]/f^*(\mathfrak{p})$. Now by Theorem 6.3.1, there exist $y_1, \dots, y_d \in k[x_1, \dots, x_d]$ such that (1) $y_1, \dots, y_d$ are algebraically independent over k , (2) $k[x_1, \dots, x_d]/k[y_1, \dots, y_d]$ is integral, and (3) $g^*(f^*(\mathfrak{p})) = \langle y_{d'+1}, \dots, y_d \rangle$ where $g^* : \operatorname{Spec} k[x_1, \dots, x_d] \rightarrow \operatorname{Spec} k[y_1, \dots, y_d]$ is the contraction map. Now we have that

$$\operatorname{ht}(\mathfrak{p}) = \operatorname{ht}(f^*(\mathfrak{p})) = \operatorname{ht}(f^*(g^*(\mathfrak{p}))) = \operatorname{ht}(\langle y_{d'+1}, \dots, y_d \rangle),$$

$$\dim A/\mathfrak{p} = \dim k[x_1, \dots, x_d]/f^*(\mathfrak{p}) = \dim \frac{k[y_1, \dots, y_d]}{\langle y_{d'+1}, \dots, y_d \rangle}.$$

But

$$\frac{k[y_1, \dots, y_d]}{\langle y_{d'+1}, \dots, y_d \rangle} \simeq k[y_1, \dots, y_{d'}] \implies \dim A/\mathfrak{p} = d'.$$

And by Krull's HT, $\operatorname{ht}(\langle y_{d'+1}, \dots, y_d \rangle) \leq d - d'$, on the other hand, $\langle y_{d'+1} \rangle \subsetneq \langle y_{d'+1}, y_{d'+2} \rangle \subsetneq \dots \subsetneq \langle y_{d'+1}, \dots, y_d \rangle$ is a chain of length $d - d'$, which implies that $\operatorname{ht}(\langle y_{d'+1}, \dots, y_d \rangle) = d - d'$. As a result,

$$\operatorname{ht}(\mathfrak{p}) + \dim A/\mathfrak{p} = (d - d') + d' = d = \dim k[x_1, \dots, x_d] = \dim A,$$

so we are done. □

Appendix

This section is devoted to reflections on certain results which have a special appeal for me by the end of the course. They have a strong personal flavor and are not written or meant to be read in full rigor. Most of them are attempts to explain why I think certain statements or results should admit an intuitive approach, or what the fundamental reasons are behind their validity.

Integral extension

Every commutative ring A gives rise to a topological space $\text{Spec } A$, and the study of an integral extension B/A is the study of the pullback or the projection map $f^* : \text{Spec } B \rightarrow \text{Spec } A$. Here is a list of my favorite properties about f^* .

- (1). *Lying-over (Proposition 3.3.3): the fiber of any $\mathfrak{p} \in \text{Spec } A$ is non-empty.*

Consider the map $\text{Spec } S_{\mathfrak{p}}^{-1}B \rightarrow \text{Spec } A_{\mathfrak{p}}$ where $S_{\mathfrak{p}}^{-1}B/A_{\mathfrak{p}}$ is again integral. The key technical lemma is that $f^*(\text{Max } S_{\mathfrak{p}}^{-1}B) = \text{Max } A_{\mathfrak{p}} = \{\mathfrak{p}A_{\mathfrak{p}}\}$, and as a result we obtain a bijection $(f^*)^{-1}(\mathfrak{p}) \leftrightarrow \text{Max } S_{\mathfrak{p}}^{-1}B \neq \emptyset$.

- (2). *Incomparability (Lemma 3.3.6): if $\mathfrak{p} \in \text{Spec } A$ and $\mathfrak{q}_1 \subseteq \mathfrak{q}_2 \in (f^*)^{-1}(\mathfrak{p})$, then $\mathfrak{q}_1 = \mathfrak{q}_2$.*

This is a direct consequence of lying-over. Indeed, both $\mathfrak{q}_1$ and $\mathfrak{q}_2$ must correspond to maximal ideal in $\text{Max } S_{\mathfrak{p}}^{-1}B$; if one contains another then they must be equal.

- (3). *Going-up (Theorem 3.3.5): if $\mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \in \text{Spec } A$ and $\mathfrak{q}_1 \in (f^*)^{-1}(\mathfrak{p}_1)$, then there exists $\mathfrak{q}_1 \subsetneq \mathfrak{q}_2 \in (f^*)^{-1}(\mathfrak{p}_2)$.*

This is another direct consequence of lying-over applying to $\bar{\mathfrak{p}}_2 \in \text{Spec } A/\mathfrak{p}_1$ where $A/\mathfrak{p}_1 \hookrightarrow B/\mathfrak{q}_1$ is integral.

- (4). *Dimensionality (Theorem 3.3.7): $\dim A = \dim B$.*

Consequence of incomparability and Going-up.

You get (1) - (4) for free whenever you have an integral extension. The following properties require B be an integral domain and A be integrally closed.

- (5). *Going-down (Theorem 3.4.4): let B/A be an integral extension, B be an integral domain, and A be integrally closed; if $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \in \text{Spec } A$ and $\mathfrak{q}_1 \cap A = \mathfrak{p}_1$, then there exists $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ such that $\mathfrak{q}_0 \cap A = \mathfrak{p}_0$.*

Here is a second proof which may provide a clearer picture. By lying-over we can always find $\mathfrak{q}_0$ that contracts to $\mathfrak{p}_0$; the main difficulty is to ensure $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$, which is what the following lemma allows us to do:

Let A be an integral domain which is integrally closed, $F := Q(A)$, E/F be a normal extension, and B be the integral closure of F in E . Then for every $\mathfrak{p} \in \text{Spec } A$, $\text{Aut}_F(E) \curvearrowright (f^)^{-1}(\mathfrak{p})$ transitively.*

Proof. Let $F := Q(A)$, $E := Q(B)$, K be the normal closure of E over F , and C be the integral closure of A in K . Notice that $A \subseteq B \subseteq C$ is a tower of integral extensions, so we have surjections $\text{Spec } C \twoheadrightarrow \text{Spec } B \twoheadrightarrow \text{Spec } A$. By lying-over, find $\mathfrak{P}_0, \mathfrak{P}_1 \in \text{Spec } C$ such that $\mathfrak{P}_0 \cap A = \mathfrak{q}_0$ and $\mathfrak{P}_1 \cap B = \mathfrak{q}_1$. Now by Going-up, there exists $\mathfrak{P}_0 \subsetneq \mathfrak{P}'_1 \in \text{Spec } C$ such that $\mathfrak{P}'_1 \cap A = \mathfrak{p}_1 = \mathfrak{P}_1 \cap A$. So $\mathfrak{P}_1, \mathfrak{P}'_1 \in (f^*)^{-1}(\mathfrak{p}_1)$, and by the lemma, there exists $\sigma \in \text{Aut}_F(K)$ such that $\sigma(\mathfrak{P}'_1) = \mathfrak{P}_1$. But $\mathfrak{P}_0 \subsetneq \mathfrak{P}'_1$, so then $\sigma(\mathfrak{P}_0) \subsetneq \mathfrak{P}_1$ where $\sigma(\mathfrak{P}_0) \in (f^*)^{-1}(\mathfrak{p}_0)$. Consider $\mathfrak{q}_0 := \sigma(\mathfrak{P}_0) \cap B \in \text{Spec } B$. Then $\mathfrak{q}_0 \cap A = \mathfrak{p}_0$ and $\mathfrak{q}_0 \subsetneq \mathfrak{P}_1 \cap B = \mathfrak{q}_1$, so we are done. $\square$

- (6). *Noetherianity (Corollary 3.5.2): if B/A is an integral domain, B is an integral domain, and A is integrally closed, then if A is Noetherian, so is B .*

The main strategy is to realize B as the submodule of a finitely-generated A -module which is Noetherian as A is. To this end we need to go to the field of fraction $E := Q(B)$ where we can write $B = \sum_{i=1}^n Ae_i$ for some $e_i \in E$. This is again a highlight of the importance of considering the ambient fields and their extensions, since the effect of their automorphisms is easier to see and describe.

Nullstellensatz and algebraic geometry

- (1). *Nullstellensatz I (Theorem 4.1.6): if A is a finitely generated k -algebra and also a field, then A/k is a finite extension.*
- (2). *Nullstellensatz II (Theorem 4.1.7): if k is algebraically closed and $A \simeq k[x_1, \dots, x_n]/\mathfrak{a}$ is a finitely generated k -algebra, then there is a bijection $\text{Max } A \leftrightarrow V(\mathfrak{a})$.*
- (3). *Nullstellensatz III (Theorem 4.1.8): if k is algebraically closed and $\mathfrak{a} \triangleleft k[x_1, \dots, x_n]$ is a proper ideal, then $V(\mathfrak{a}) \neq \emptyset$.*
- (4). *Nullstellensatz IV (Theorem 4.1.9): if k is algebraically closed and $\mathfrak{a} \leq k[x_1, \dots, x_n]$, then $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.*

Since $V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}})$ for every $\mathfrak{a} \triangleleft k[x_1, \dots, x_n]$, (4) shows that for an algebraically closed field k , there is a bijection

$$\text{Rad } k[x_1, \dots, x_n] \xrightleftharpoons[I(-)]{V(-)} \{\text{closed subsets of } \mathbb{A}_k^n\}.$$

If we consider a finitely generated k -algebra, say $A \simeq k[x_1, \dots, x_n]/\mathfrak{a}$, then we obtain a bijection

$$\text{Rad } A \leftrightarrow \{\text{closed subsets of } V(\mathfrak{a})\}$$

where $V(\mathfrak{a}) \subseteq \mathbb{A}_k^n$ carries the subspace topology.

Dedekind domains and DVR

A Dedekind domain is an integrally closed Noetherian domain of dimension 1. There are other equivalent characterizations which are more common and useful in other fields.

- (1). *Characterization of DVR (Theorem 5.3.2): if D is a local Noetherian domain of dimension 1 with $\text{Max } D = \{\mathfrak{m}\}$, then TFAE*
 - (i). D is integrally closed;
 - (ii). $\mathfrak{m}$ is principal;
 - (iii). $\dim_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) = 1$;
 - (iv). if $0 \neq \mathfrak{a} \triangleleft D$ is a proper ideal, then $\mathfrak{a} = \mathfrak{m}^k$ for some unique $k \in \mathbb{N}$.

In particular, every DVR is a local PID and its ideals are very rigid: they are all of the form $\langle x^k \rangle$ for some $k \in \mathbb{N}$ where x is *any* generator of $\mathfrak{m}$; indeed, $\mathfrak{m}$ is generated by *any* element in $\mathfrak{m} \setminus \mathfrak{m}^2$ by (iv).

For a local Noetherian ring D with $\dim D \geq 1$, we say D is regular local ring if $\dim D = \dim_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2)$, which is analogous to (iii); in fact, one can show that any regular local ring is necessarily a domain by a result similar to Lemma 5.3.1. The geometric interpretation of regularity is *smoothness*, or no singularity. Recall that a manifold $\mathcal{M}$ is said to be smooth at p if $\dim \mathcal{M} = \dim T_p(\mathcal{M})$. The notion of the tangent space $T_p(\mathcal{M})$ is captured by $\mathfrak{m}_p/\mathfrak{m}_p^2$ which represents the k -span of all linear factors in regular functions that vanish around p . In the context of variety, if $A(X)$ is the coordinate ring, then $A(X)_{\mathfrak{m}_p}$ is the Noetherian local ring in question with unique maximal ideal $\tilde{\mathfrak{m}}_p$; notice that $A(X)/\mathfrak{m}_p \simeq k$ for every $p \in X$.

- (2). *Characterization of Dedekind domains (Theorem 5.3.3): if D is a Noetherian domain of dimension 1, then D is integrally closed $\iff D_{\mathfrak{m}}$ is a DVR for every $\mathfrak{m} \in \text{Max } D$.*

If $X \subseteq \mathbb{A}_k^2$ is an irreducible curve, then $A(X)$ is a Dedekind domain precisely when X is non-singular; i.e. the localization of $A(X)$ at every maximal ideal is a regular local ring, or simply a DVR since it is one-dimensional. Indeed, $A(X) \simeq k[x, y]/\mathfrak{p}$ is a Noetherian domain which is one-dimensional.

Miscellaneous

- (1). *Fiber of contraction (Theorem 1.3.8): if $f : A \rightarrow B$ is a ring homomorphism and $\mathfrak{p} \in \text{Spec } A$, then $(f^*)^{-1}(\mathfrak{p}) \leftrightarrow \text{Spec}(B \otimes_A k(\mathfrak{p}))$ where $B \otimes_A k(\mathfrak{p}) \simeq f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e$; in particular, $(f^*)^{-1}(\mathfrak{p}) \neq \emptyset \iff \mathfrak{p}^{ec} = \mathfrak{p}$.*

The proof starts with the bijection $(f^*)^{-1}(\mathfrak{p}) \leftrightarrow \text{Spec}(f(S_{\mathfrak{p}})^{-1}B/f(S_{\mathfrak{p}})^{-1}\mathfrak{p}^e)$ which follows from the correspondences between the ideals in a ring and those in its quotient or localization. The mentioned isomorphism follows from manipulating tensor product and A -algebra. The in-particular part is the most useful: it works with any ring homomorphism whatsoever (need not be a ring extension even!).

- (2). *Extension of morphism (Theorem 4.1.5): let B be a finitely generated A -algebra, B/A be an extension of domains, and Ω be an algebraically closed field; suppose $b_0 \in B \setminus \{0\}$; then for some $a_0 \in A$, if $\varphi \in \text{Hom}_{\text{Ring}}(A, \Omega)$ satisfies $\varphi(a_0) \neq 0$, then there exists $\theta \in \text{Hom}_{\text{Alg}_A}(B, \Omega)$ such that $\theta(b_0) \neq 0$ and $\theta|_A = \varphi$.*

This result is particularly useful when φ is an embedding; in that case for any $b_0 \in B \setminus \{0\}$ we can find an extension θ that does not vanish at b_0 . We use it to prove Nullstellensatz I (Theorem 4.1.6). In the remark of Theorem 4.1.3 we also note that if $X \subseteq \mathbb{A}_k^n$ is an affine variety, then there is a bijection $X \leftrightarrow \text{Hom}_{\text{Alg}_k}(A(X), k)$. If we pick $f \in A(X) \setminus \{0\}$, then the existence of $\theta \in \text{Hom}_{\text{Alg}_k}(A(X), k)$ with $\theta(f) \neq 0$ is just a paraphrase of the fact that f does not vanish for some $p \in X$.

- (3). *Krull's HT (Theorem 6.1.6): let A be a Noetherian ring and $\mathfrak{a} := \langle a_1, \dots, a_n \rangle \triangleleft A$; if $\mathfrak{p} \in \text{Ass}(\mathfrak{a})$ is a minimal element, then $\text{ht}(\mathfrak{p}) \leq n$.*

Aside from being very useful on its own, this result motivates the definition of a regular local ring. If A is a local Noetherian ring with $\text{Max } A = \{\mathfrak{m}\}$, then by Nakayama's lemma

$$\dim A = \text{ht}(\mathfrak{m}) \leq d(\mathfrak{m}) = \dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2).$$

When equality holds, we say A is a regular local ring.

- (4). *Dimension of finitely generated k -domains (Theorem 6.3.2): if A is a finitely generated k -domain, then for every $\mathfrak{p} \in \text{Spec } A$, $\text{ht}(\mathfrak{p}) + \dim A/\mathfrak{p} = \dim A$.*

One immediate consequence of this result is that $\dim A = \text{ht}(\mathfrak{m})$ for every $\mathfrak{m} \in \text{Max } A$.

- (5) *Finiteness of integral extension (Generalization of Theorem 3.5.1): let A be a finitely generated k -domain, $F := Q(A)$, E/F be a finite extension, and B be the integral closure of A in E ; then B is a finitely generated A -module and also a finitely generated k -algebra.*

Compared with Theorem 3.5.1, the following improvements are made: we don't need the assumption $E = Q(B)$ (since a posteriori this is always the case because of Lemma 3.5.9); and A need not be integrally closed...